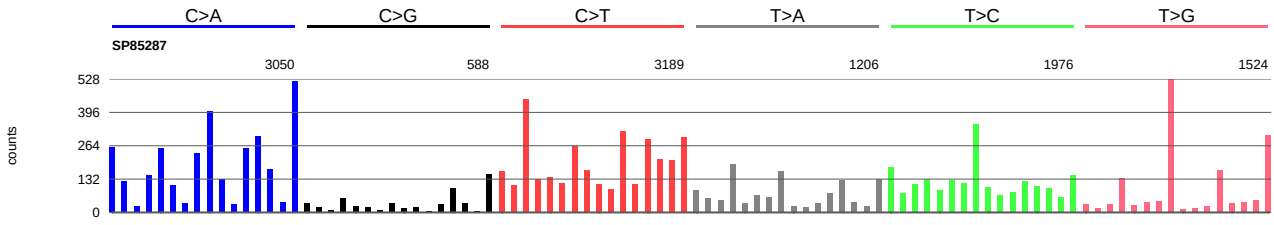
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C>G
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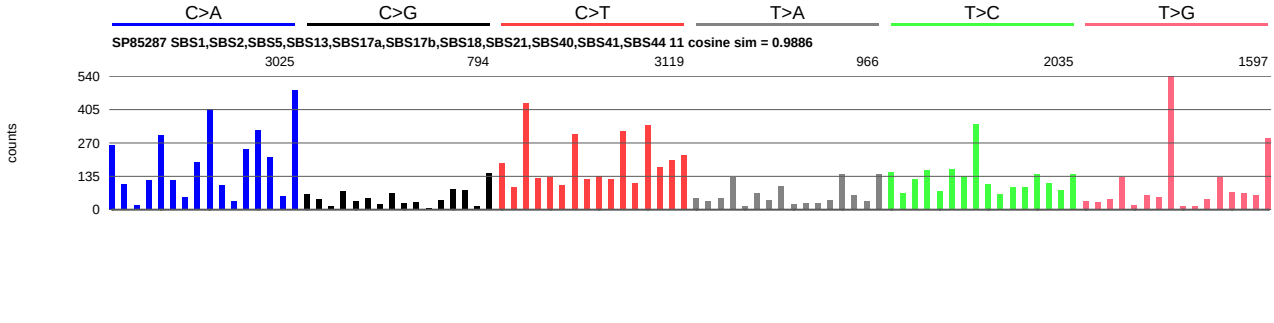
SP85287
3050
588
3189
1206
1976
1524
counts
528
396
264
132
0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T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.9886
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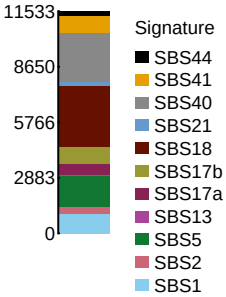
794
3119
966
2035
1597
counts
540
405
270
135
0


11533
8650
5766
2883
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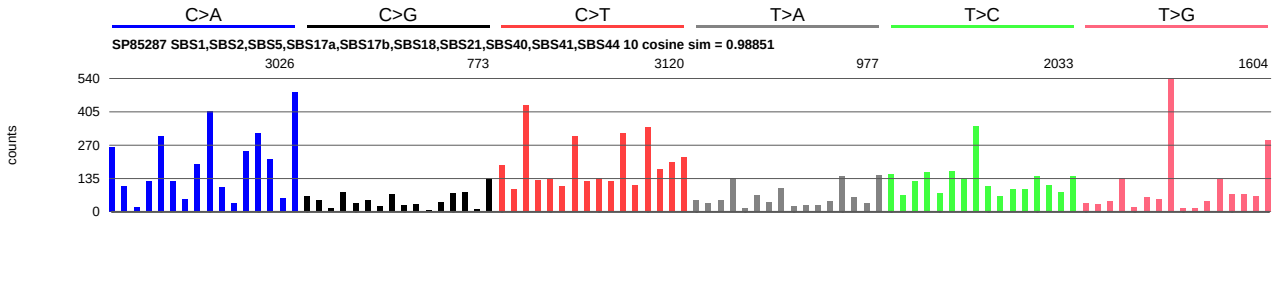
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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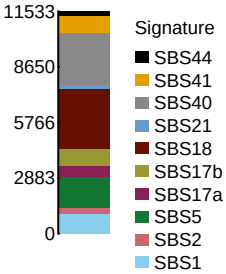
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98851
3026
773
3120
977
2033
1604
counts
540
405
270
135
0


11533
8650
5766
2883
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Signature
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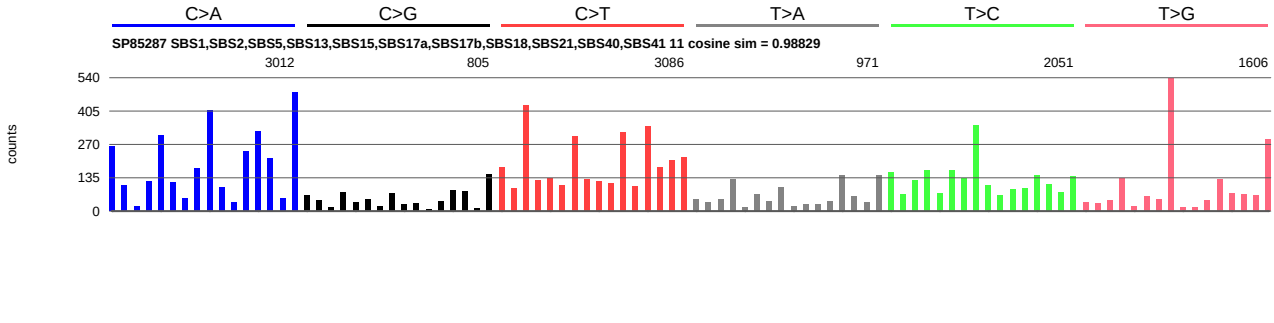
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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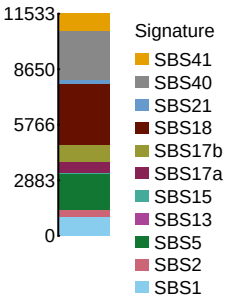
SP85287 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98829
3012
805
3086
971
2051
1606
counts
540
405
270
135
0


11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
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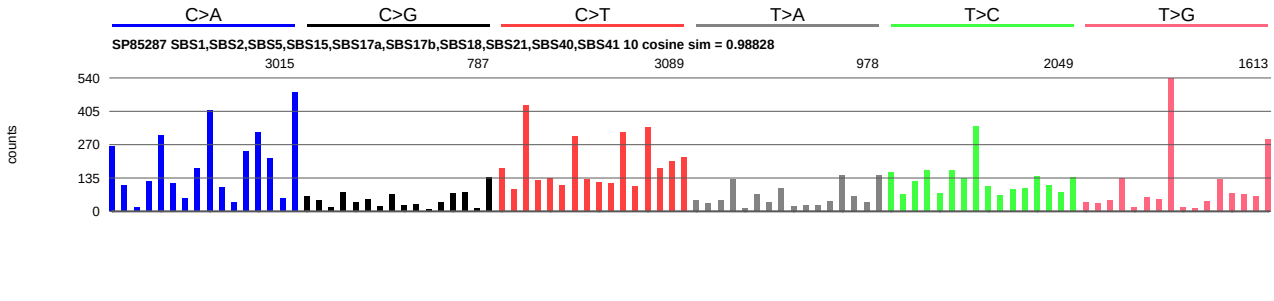
SBS2
SBS1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C>G
C>T
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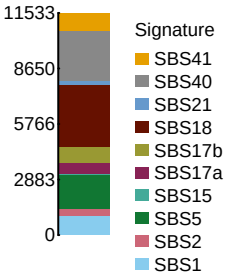
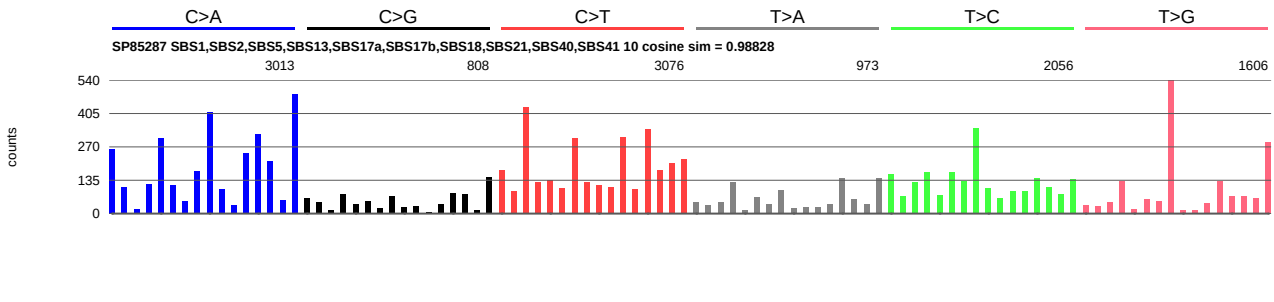
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98828
3015
787
3089
978
2049
1613
counts
540
405
270
135
0


11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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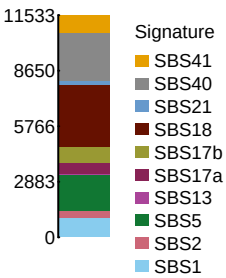
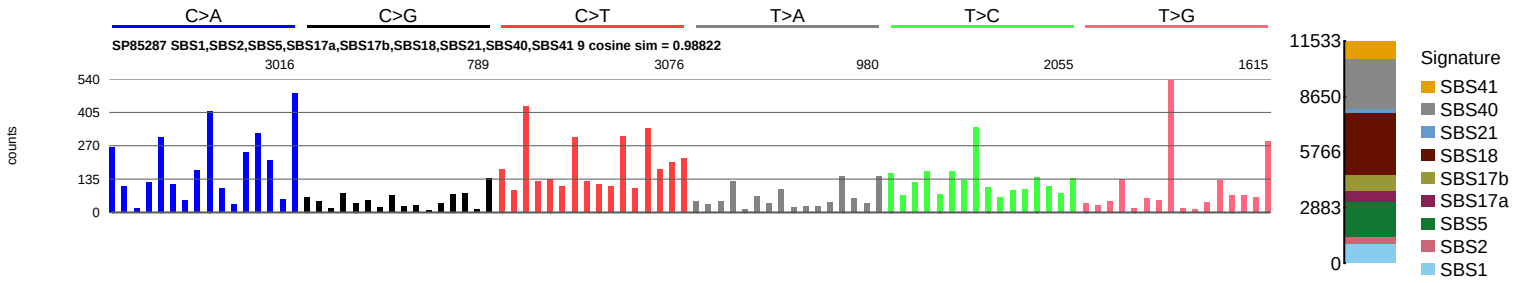
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98828
3013
808
3076
973
2056
1606
counts
540
405
270
135
0


11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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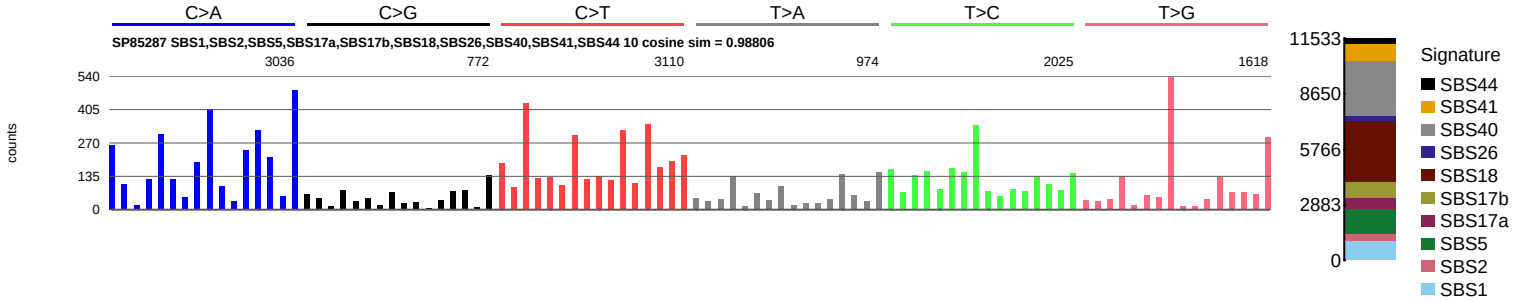
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98822
3016
789
3076
980
2055
1615
counts
540
405
270
135
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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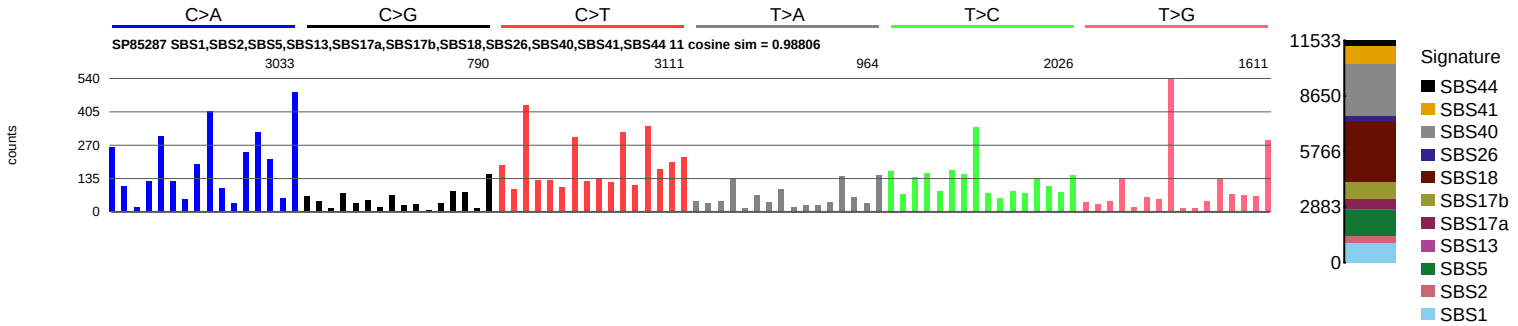
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98806
3036
772
3110
974
2025
1618
counts
540
405
270
135
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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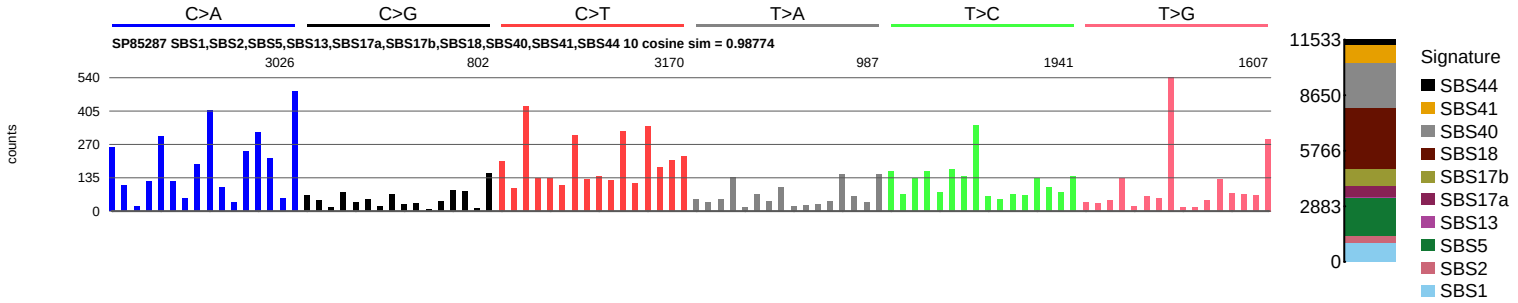
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98806
3033
790
3111
964
2026
1611
counts
540
405
270
135
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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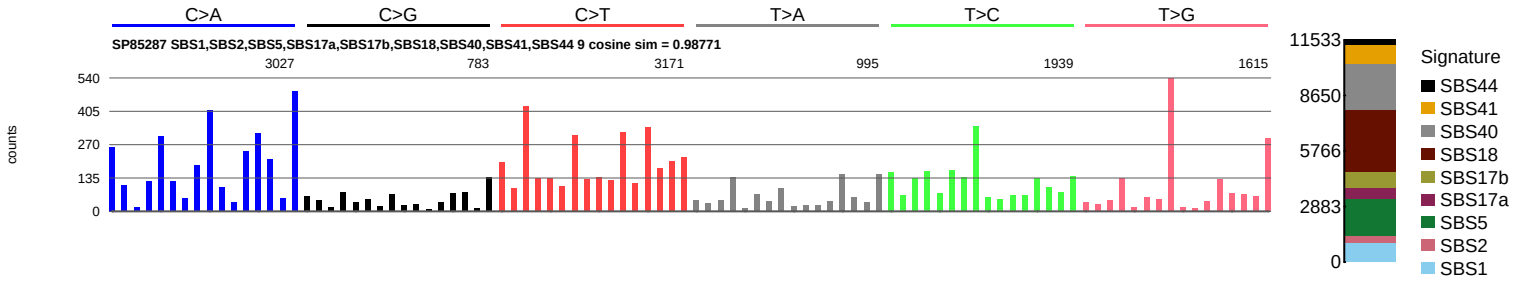
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98774
3026
802
3170
987
1941
1607
counts
540
405
270
135
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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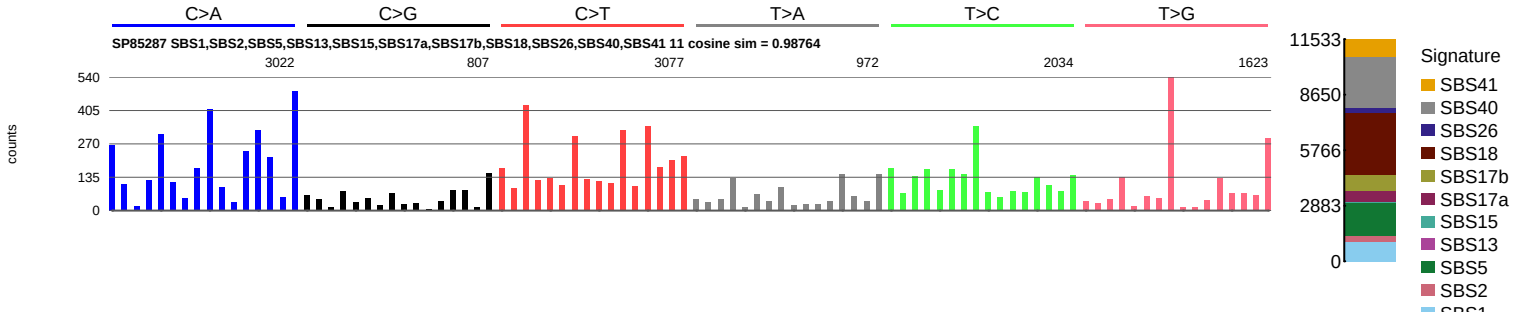
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98771
3027
783
3171
995
1939
1615
counts
540
405
270
135
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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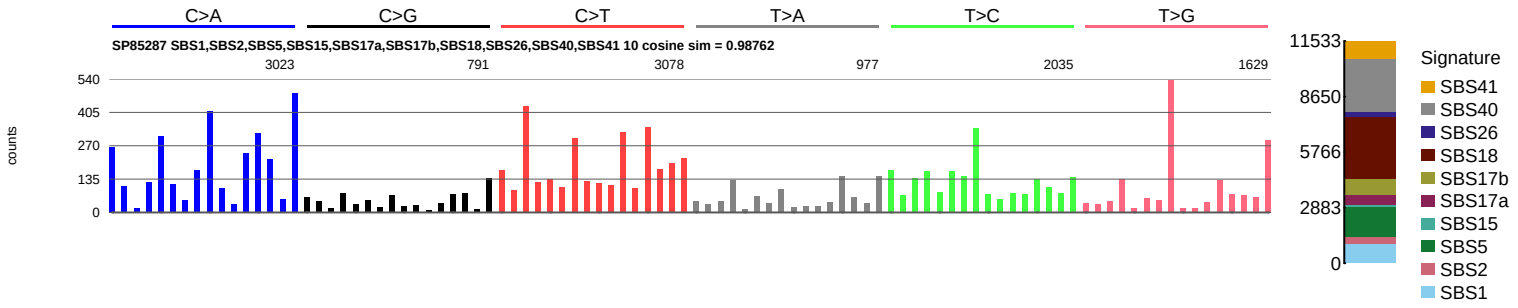
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.98764
3022
807
3077
972
2034
1623
counts
540
405
270
135
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2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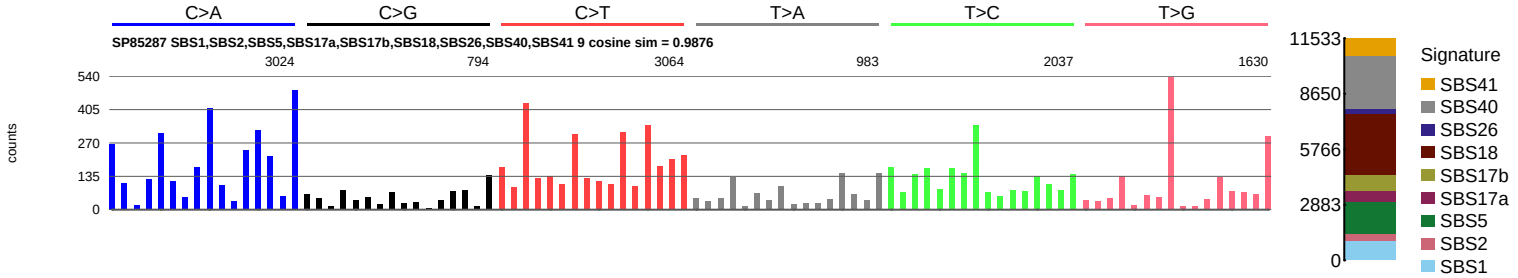
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98762
3023
791
3078
977
2035
1629
counts
540
405
270
135
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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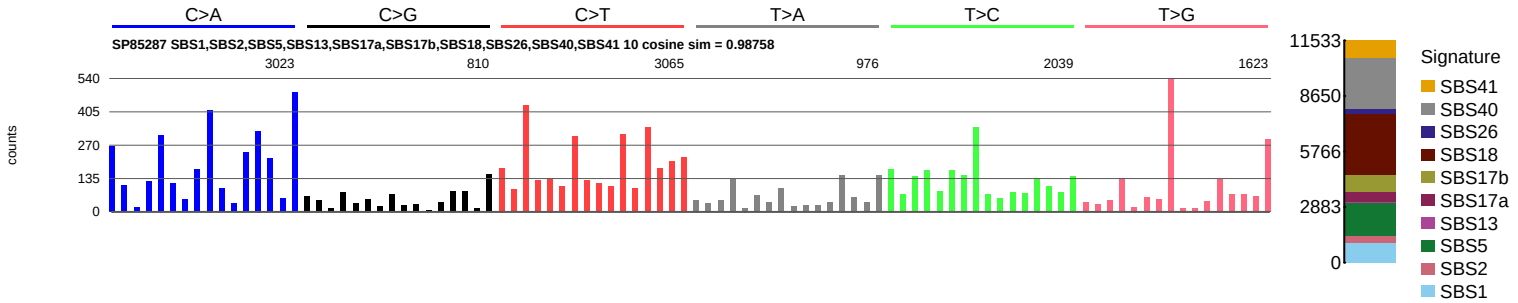
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.9876
3024
794
3064
983
2037
1630
counts
540
405
270
135
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
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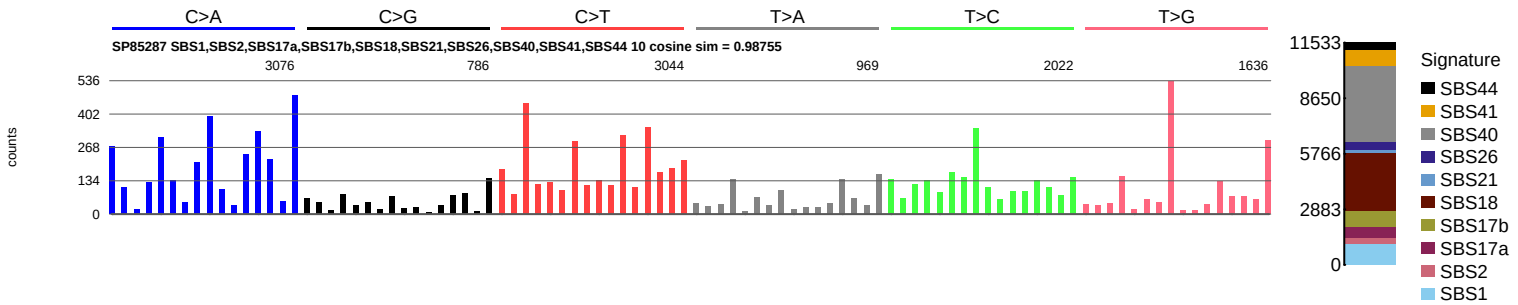
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98758
3023
810
3065
976
2039
1623
counts
540
405
270
135
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
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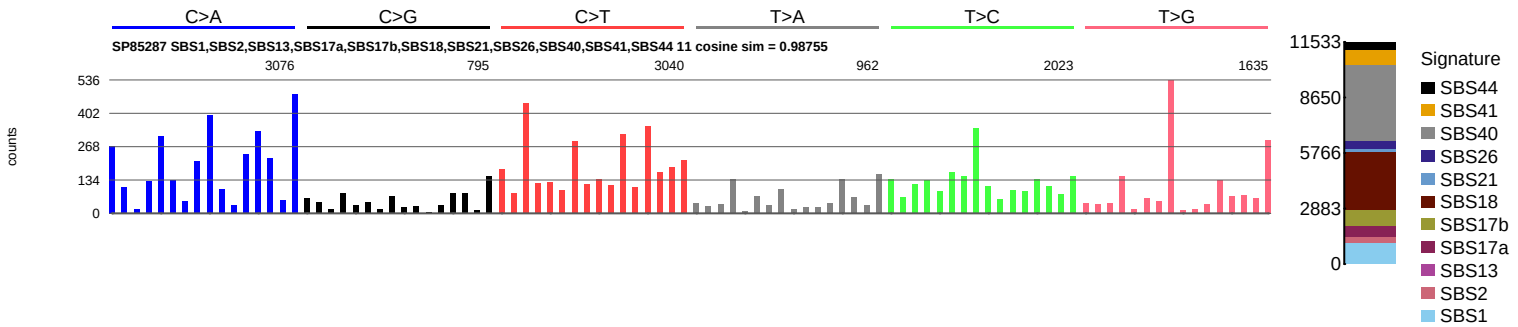
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98755
3076
786
3044
969
2022
1636
counts
536
402
268
134
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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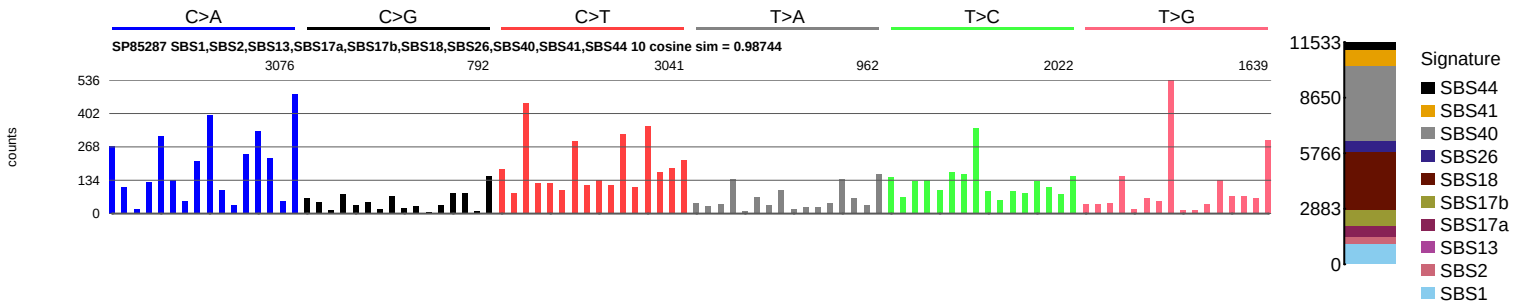
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98755
3076
795
3040
962
2023
1635
counts
536
402
268
134
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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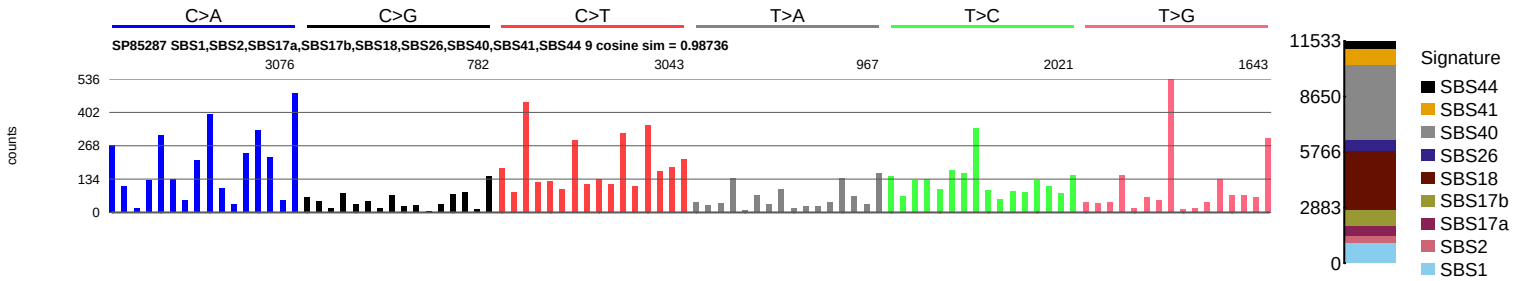
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98744
3076
792
3041
962
2022
1639
counts
536
402
268
134
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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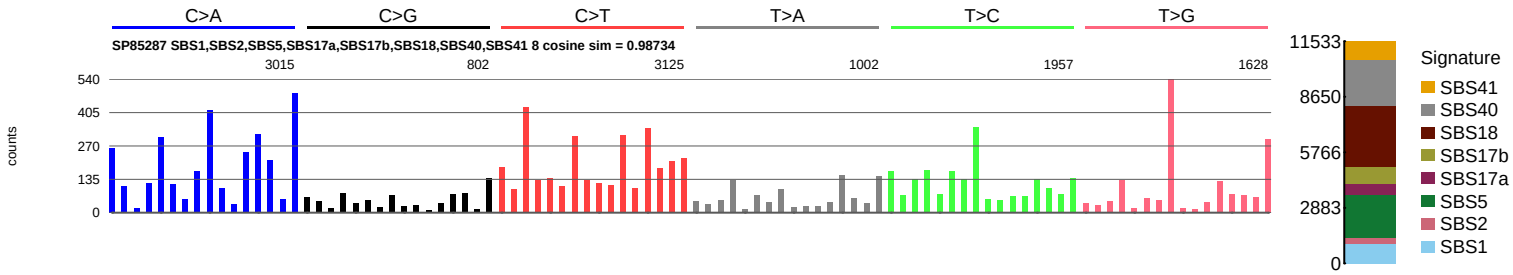
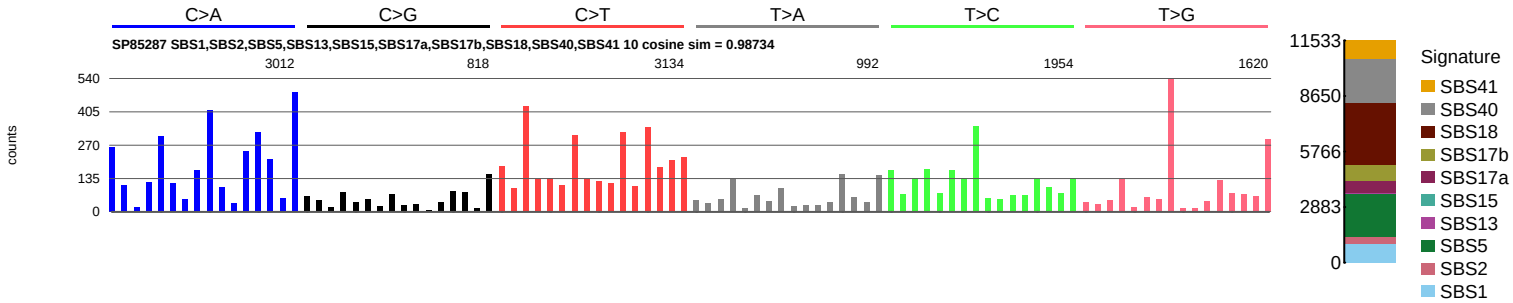
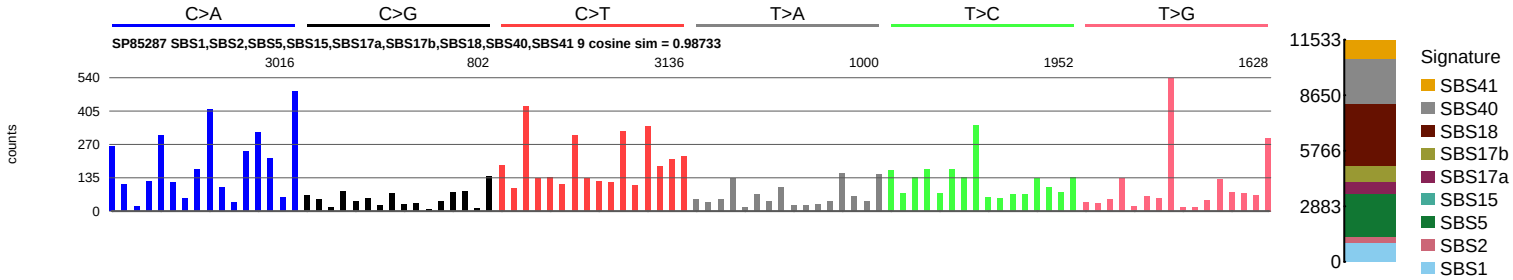
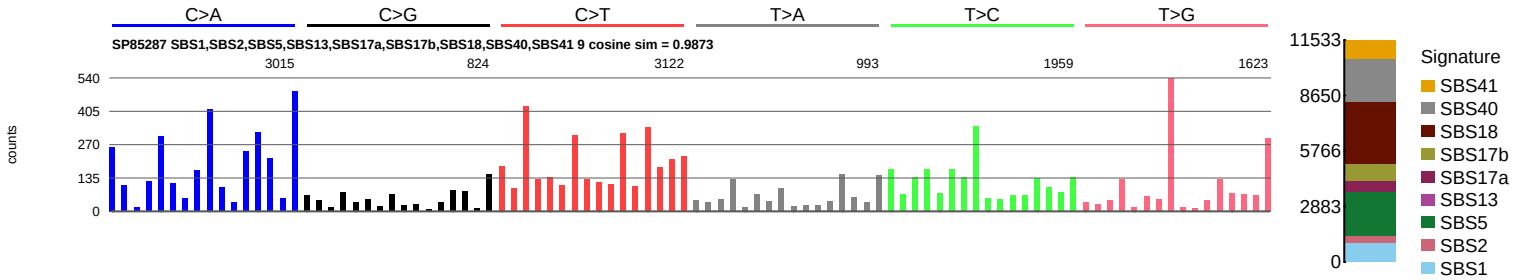
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98736
3076
782
3043
967
2021
1643
counts
536
402
268
134
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98734
3015
802
3125
1002
1957
1628
counts
540
405
270
135
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98734
3012
818
3134
992
1954
1620
counts
540
405
270
135
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98733
3016
802
3136
1000
1952
1628
counts
540
405
270
135
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>T
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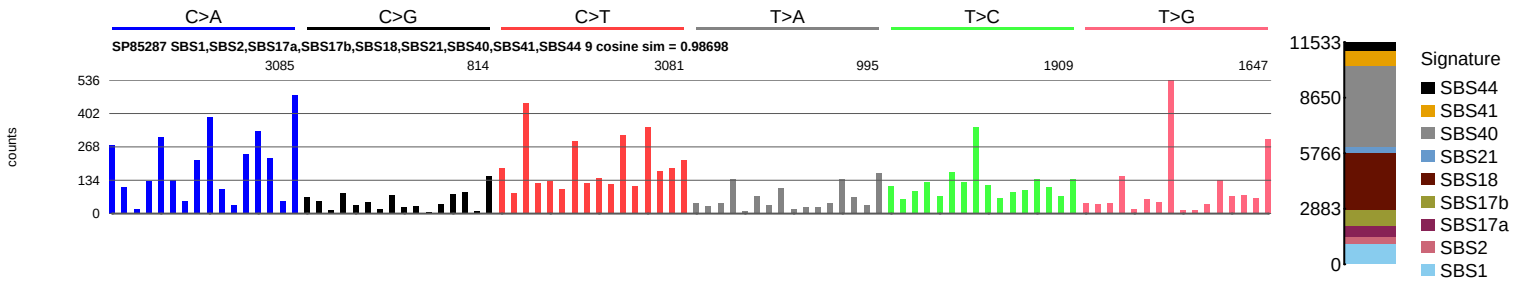
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.9873
3015
824
3122
993
1959
1623
counts
540
405
270
135
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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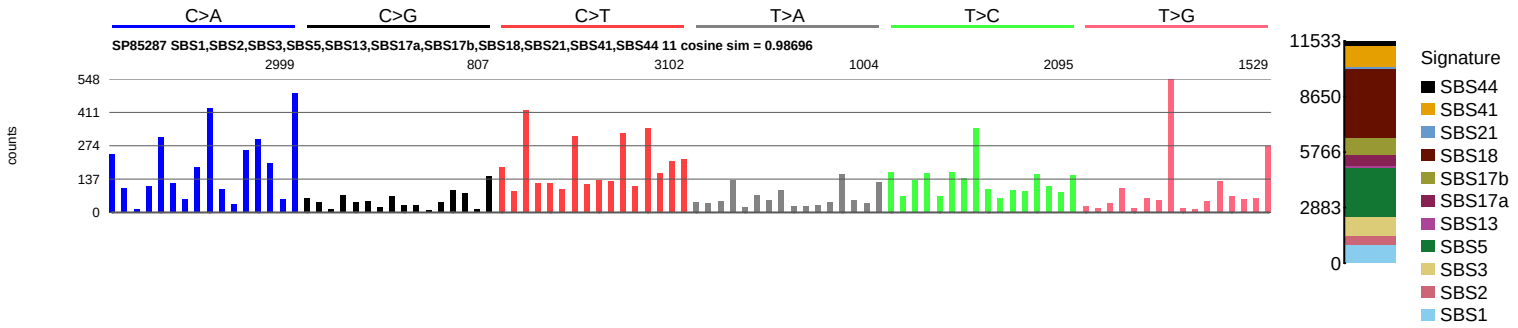
T>G
SP85287 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98698
3085
814
3081
995
1909
1647
counts
536
402
268
134
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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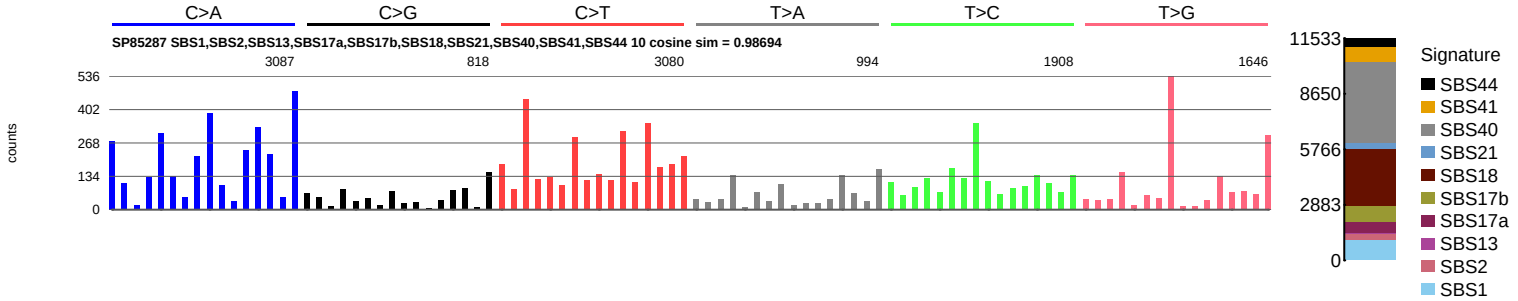
SP85287 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98696
2999
807
3102
1004
2095
1529
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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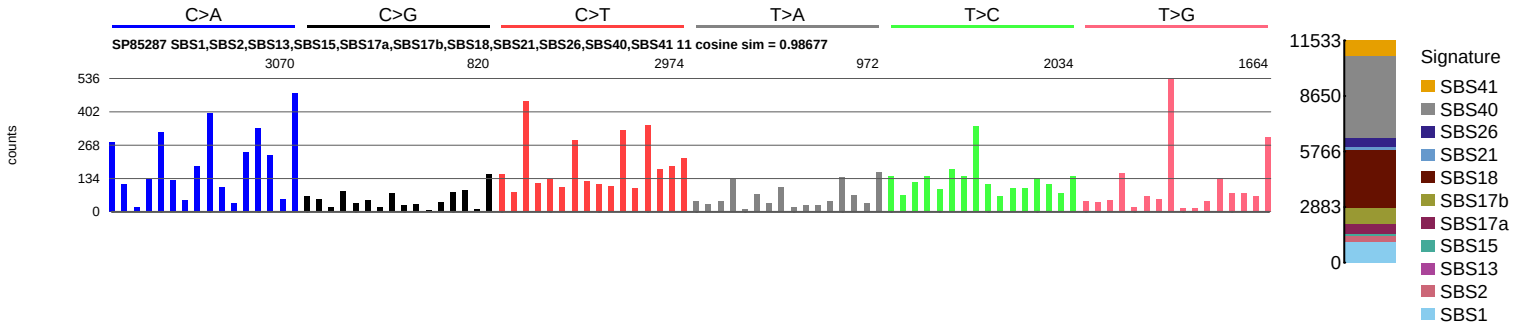
T>G
SP85287 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98694
3087
818
3080
994
1908
1646
counts
536
402
268
134
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
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T>C
T>G
SP85287 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98677
3070
820
2974
972
2034
1664
counts
536
402
268
134
0
11533
8650
5766
2883
0
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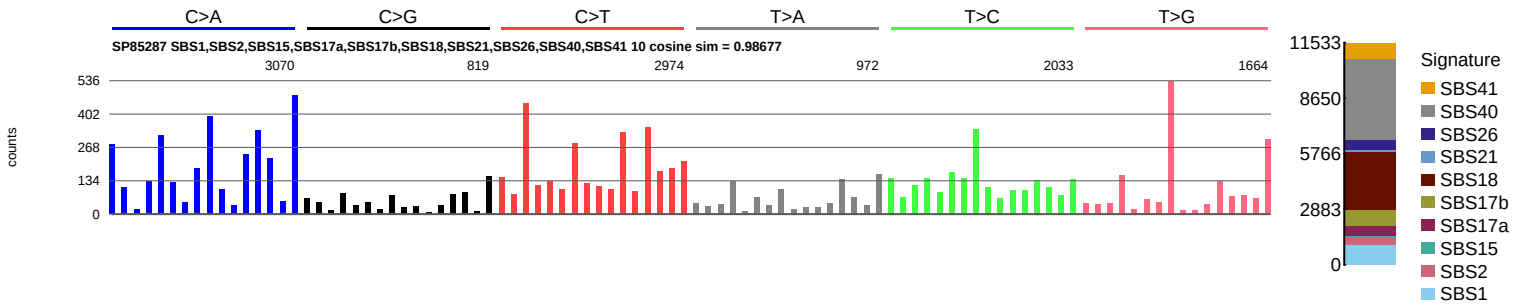
SBS2
SBS1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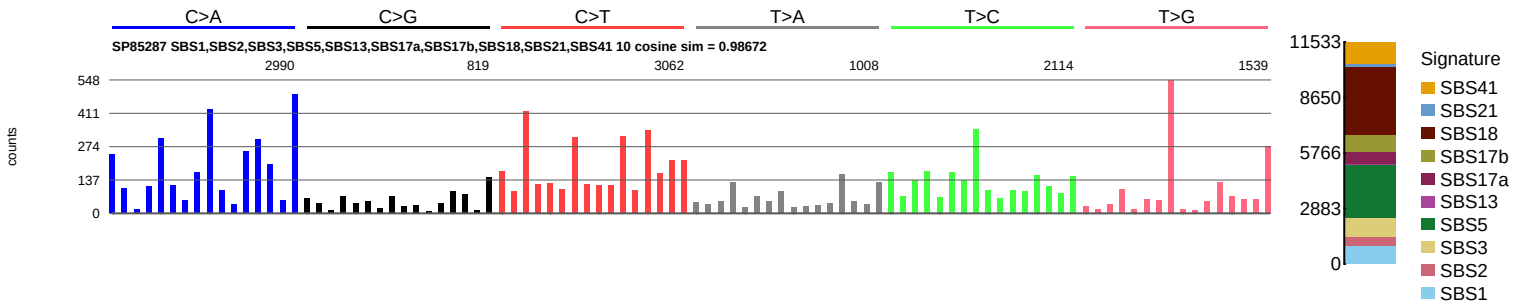
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98677
3070
819
2974
972
2033
1664
counts
536
402
268
134
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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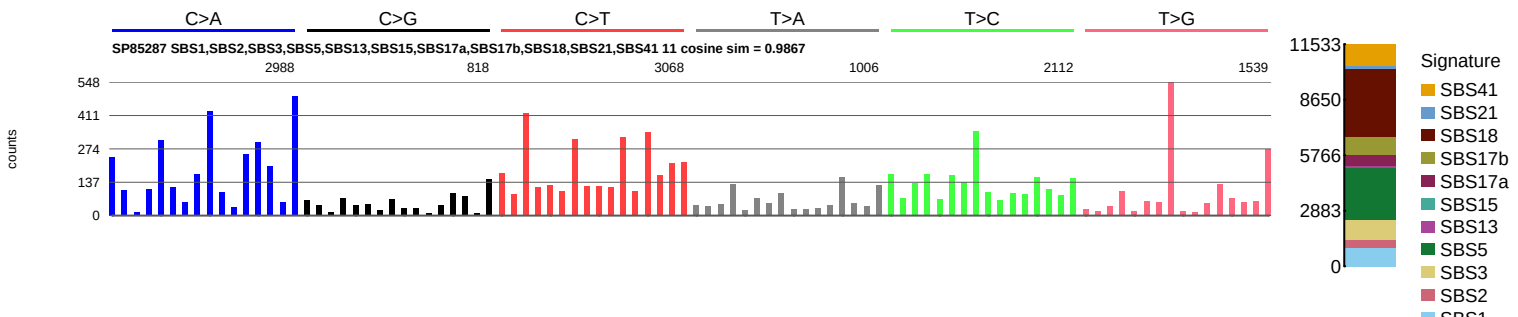
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98672
2990
819
3062
1008
2114
1539
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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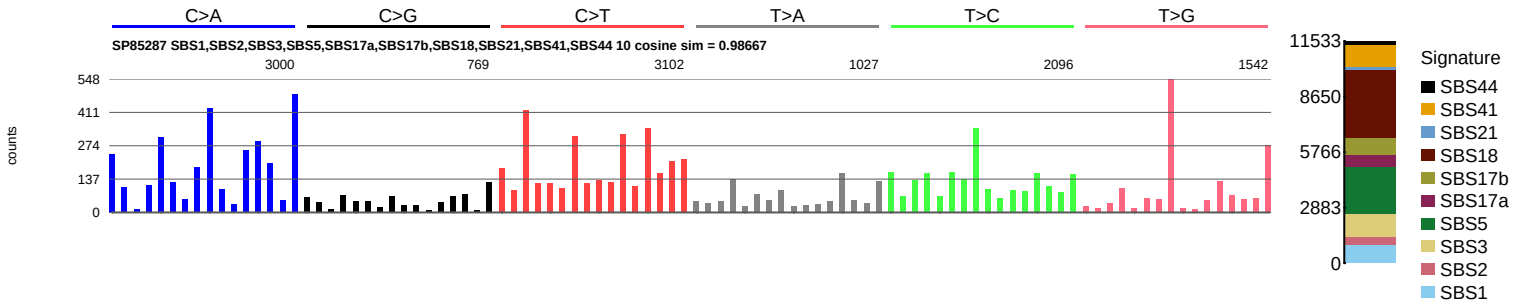
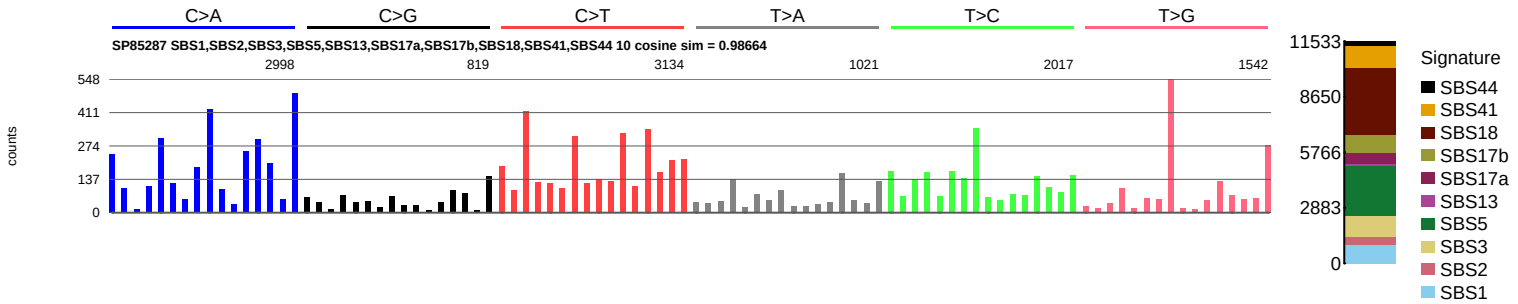
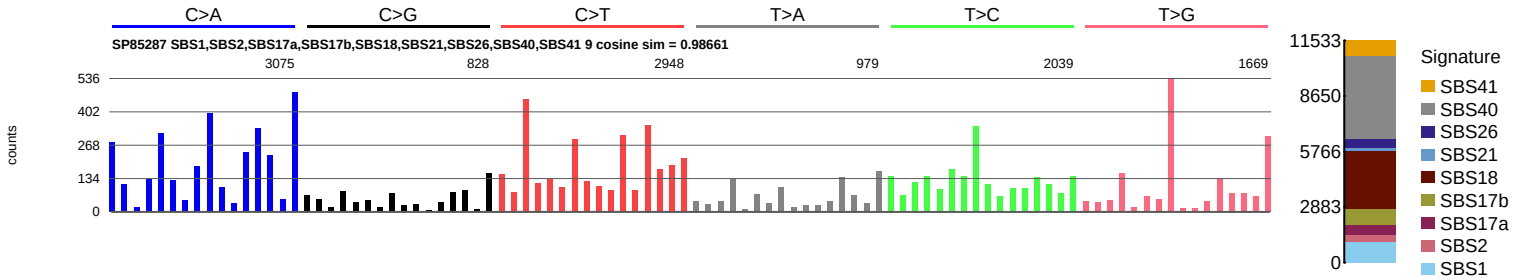
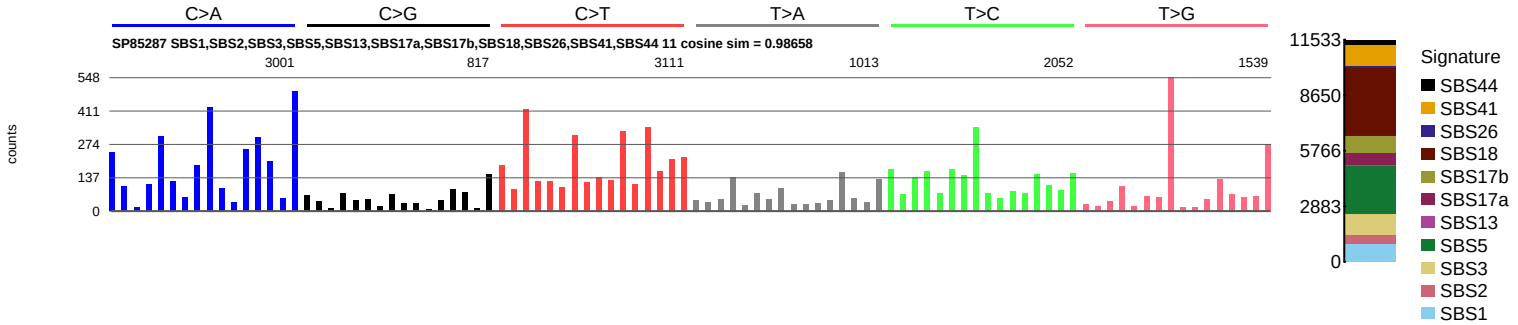
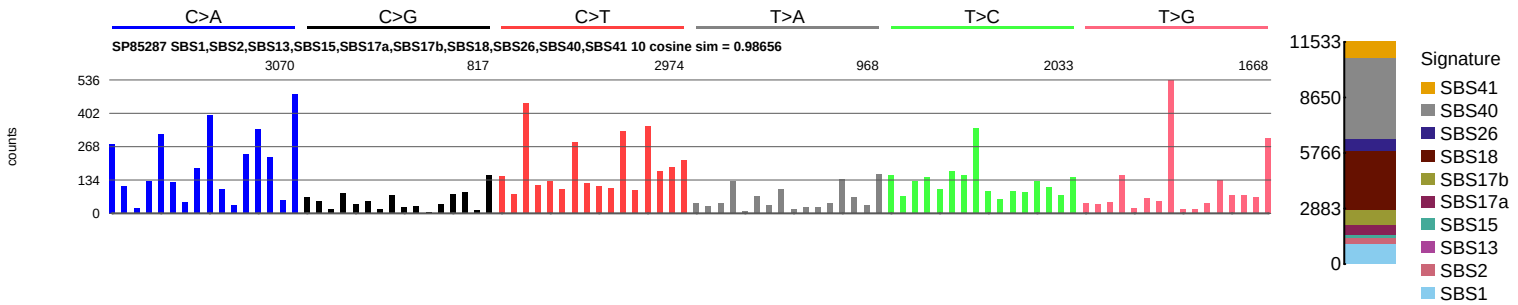
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.9867
2988
818
3068
1006
2112
1539
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98667
3000
769
3102
1027
2096
1542
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98664
2998
819
3134
1021
2017
1542
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98661
3075
828
2948
979
2039
1669
counts
536
402
268
134
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.98658
3001
817
3111
1013
2052
1539
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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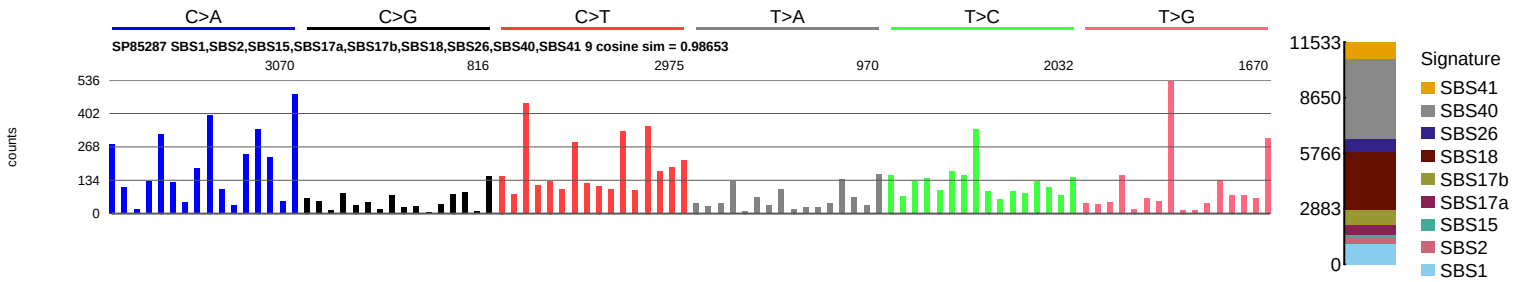
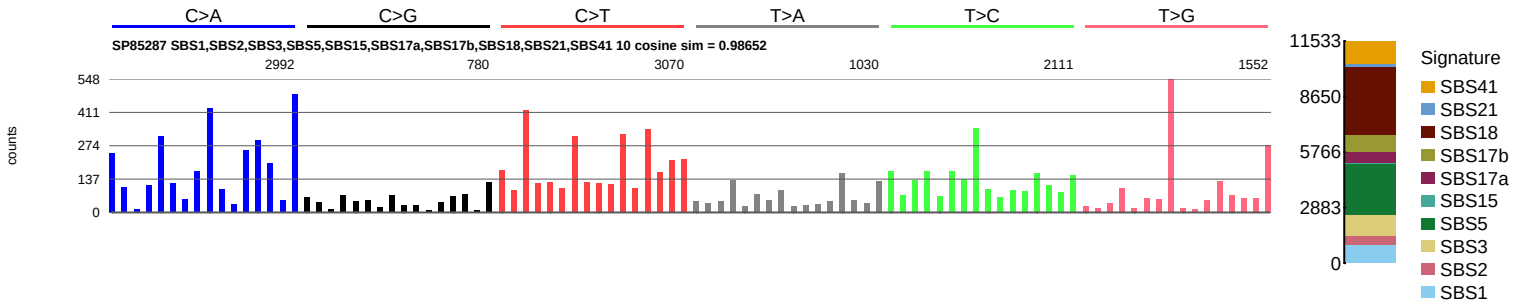
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98656
3070
817
2974
968
2033
1668
counts
536
402
268
134
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98653
3070
816
2975
970
2032
1670
counts
536
402
268
134
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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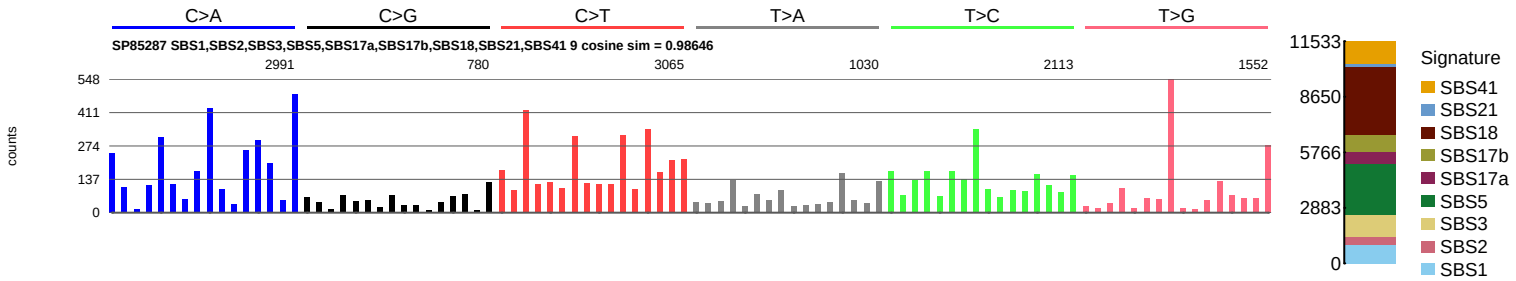
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98652
2992
780
3070
1030
2111
1552
548
411
274
137
0
counts
11533
8650
5766
2883
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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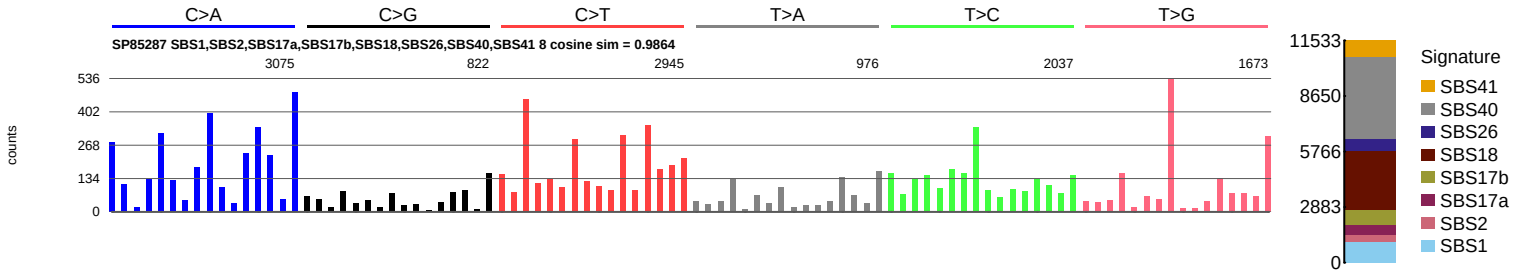
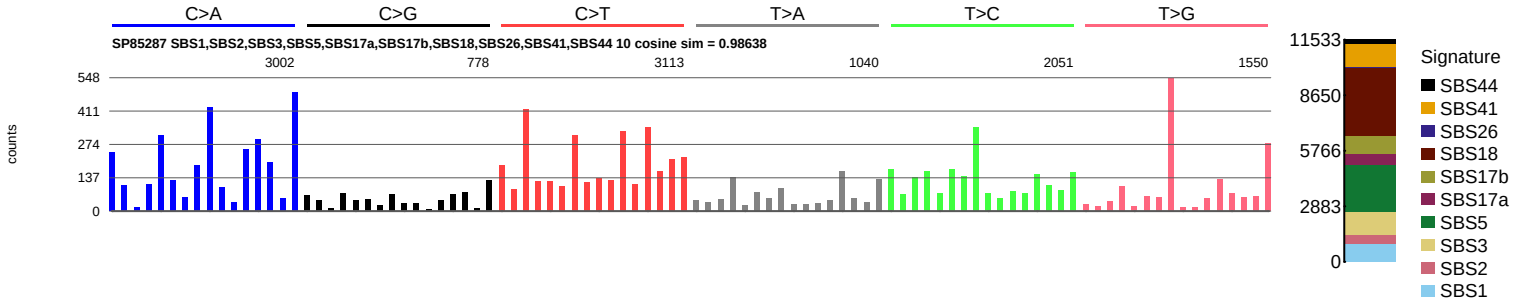
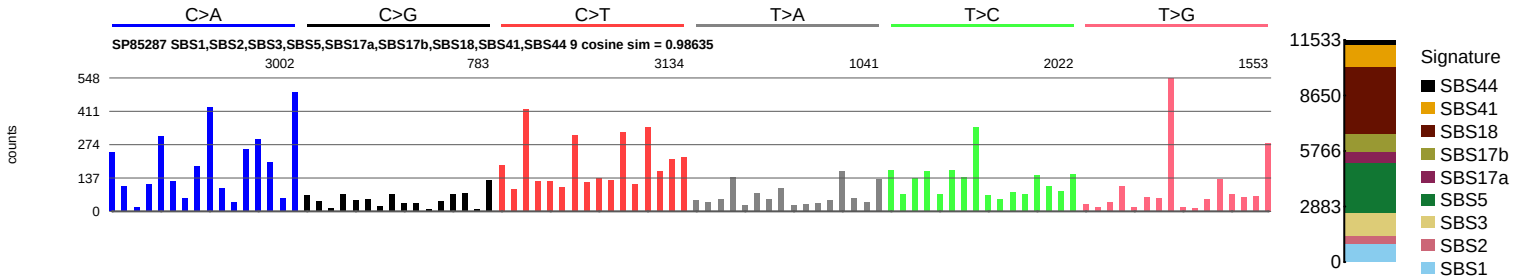
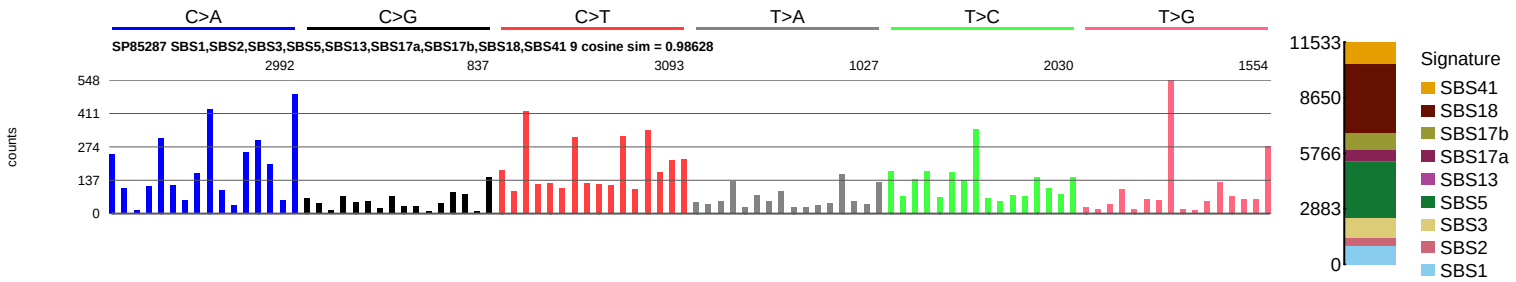
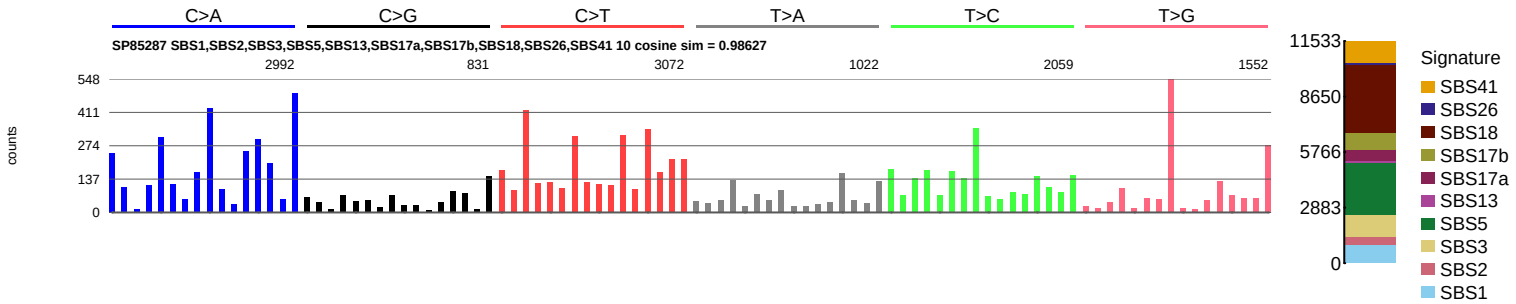
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98646
2991
780
3065
1030
2113
1552
548
411
274
137
0
counts
11533
8650
5766
2883
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

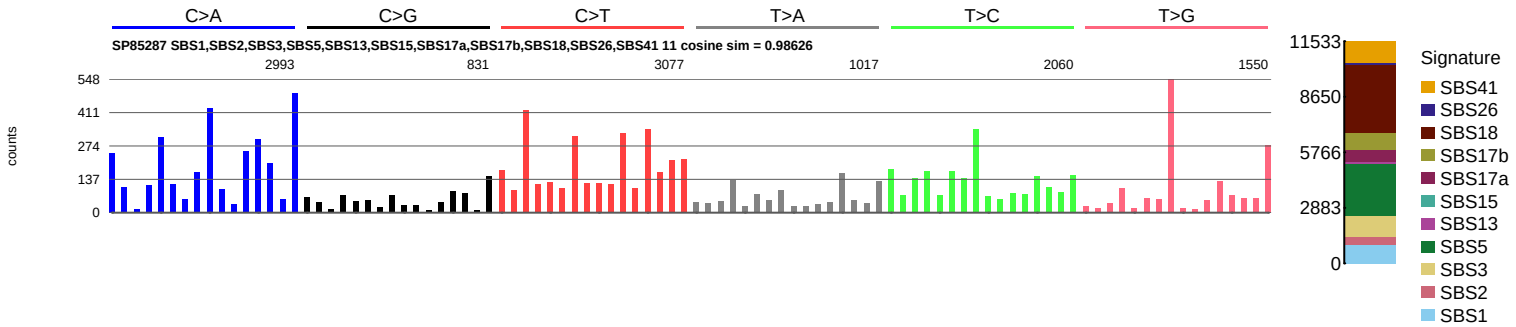

C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.9864
3075
822
2945
976
2037
1673
536
402
268
134
0
counts
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

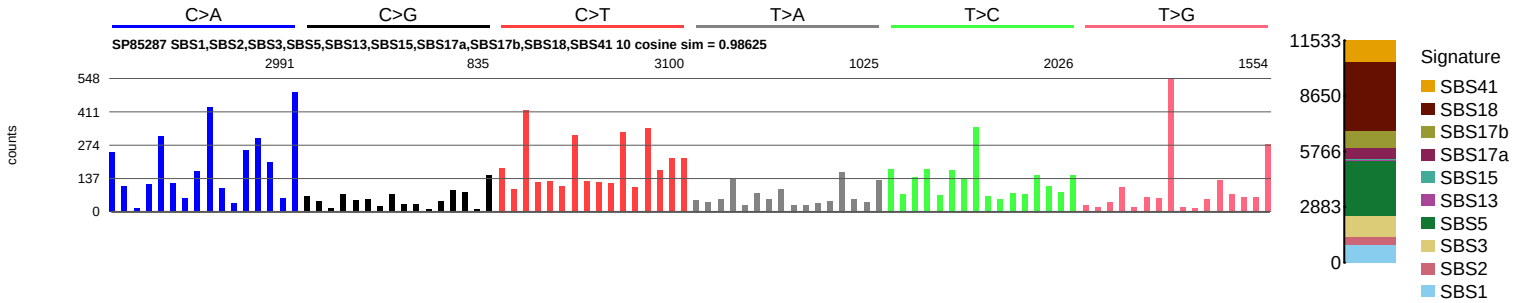

C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98638
3002
778
3113
1040
2051
1550
548
411
274
137
0
counts
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

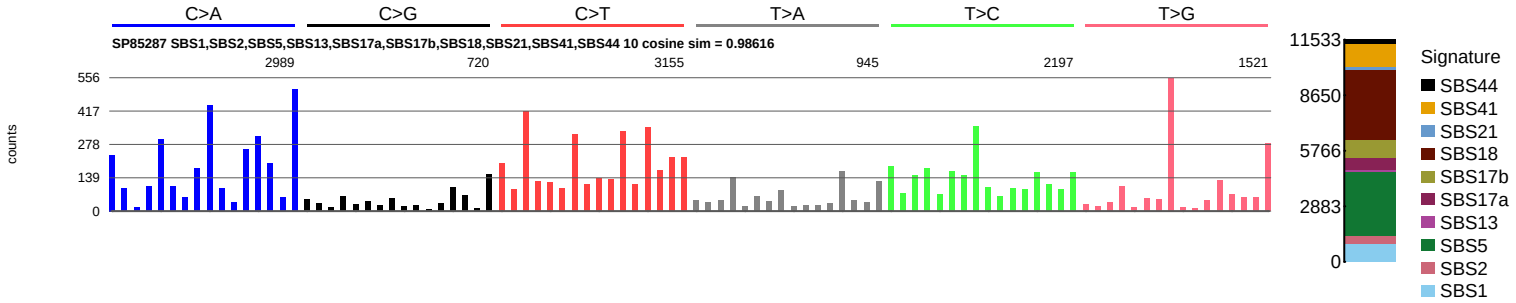

C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98635
3002
783
3134
1041
2022
1553
548
411
274
137
0
counts
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98628
2992
837
3093
1027
2030
1554
548
411
274
137
0
counts
11533
8650
5766
2883
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98627
2992
831
3072
1022
2059
1552
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 11 cosine sim = 0.98626
2993
831
3077
1017
2060
1550
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.98625
2991
835
3100
1025
2026
1554
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98616
2989
720
3155
945
2197
1521
counts
556
417
278
139
0
11533
8650
5766
2883
0
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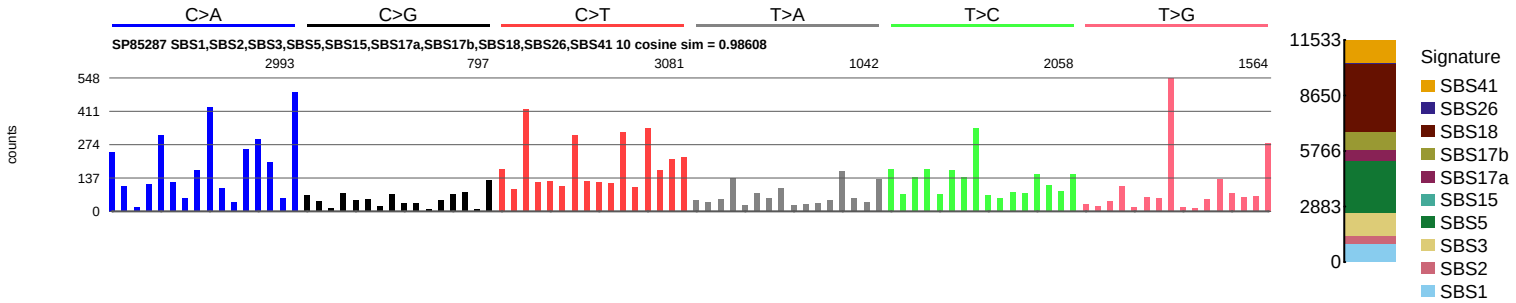
SBS1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SP85287 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98608
2993
797
3081
1042
2058
1564
counts
548
411
274
137
0
11533
8650
5766
2883
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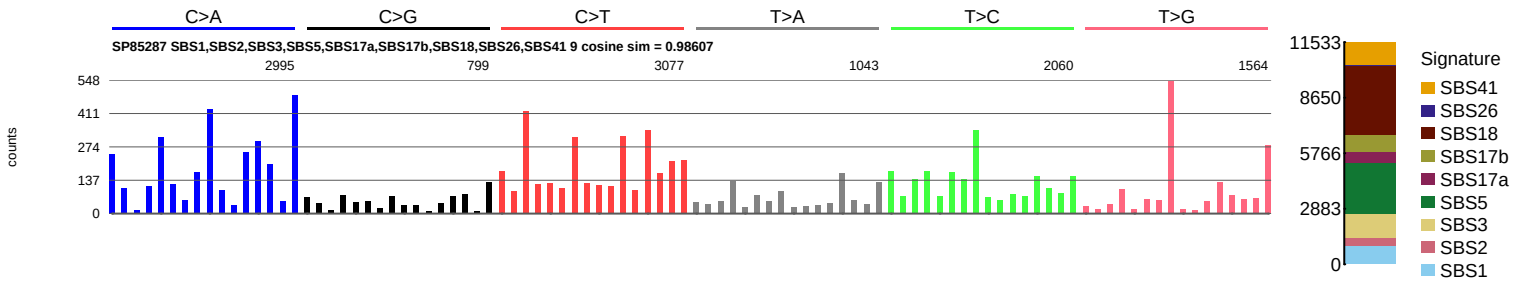
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SP85287 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98607
2995
799
3077
1043
2060
1564
counts
548
411
274
137
0
11533
8650
5766
2883
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SBS17b
SBS17a
SBS5
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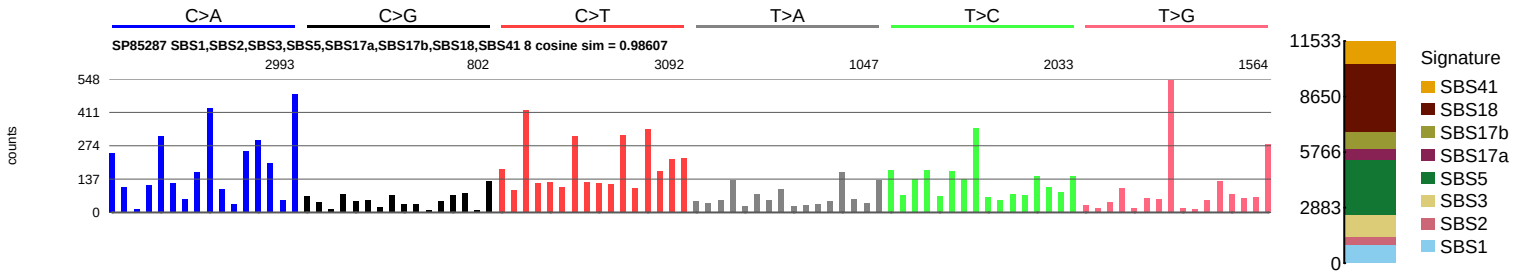
SBS2
SBS1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C>T
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SP85287 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98607
2993
802
3092
1047
2033
1564
548
411
274
137
0
counts
11533
8650
5766
2883
0
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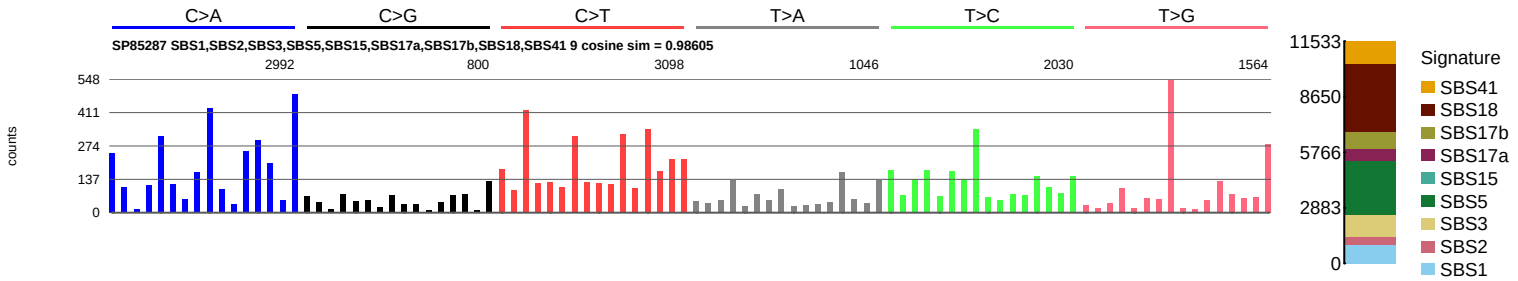
SBS2
SBS1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SP85287 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98605
2992
800
3098
1046
2030
1564
548
411
274
137
0
counts
11533
8650
5766
2883
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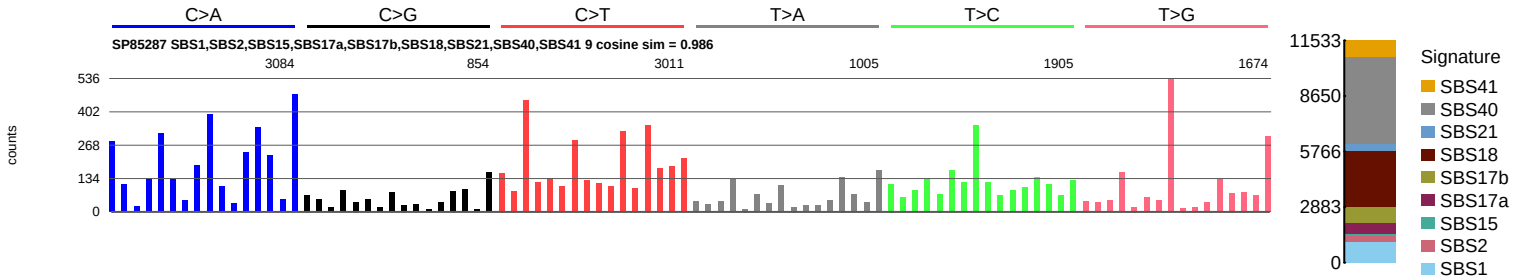
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
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SP85287 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.986
3084
854
3011
1005
1905
1674
536
402
268
134
0
counts
11533
8650
5766
2883
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Signature
SBS41
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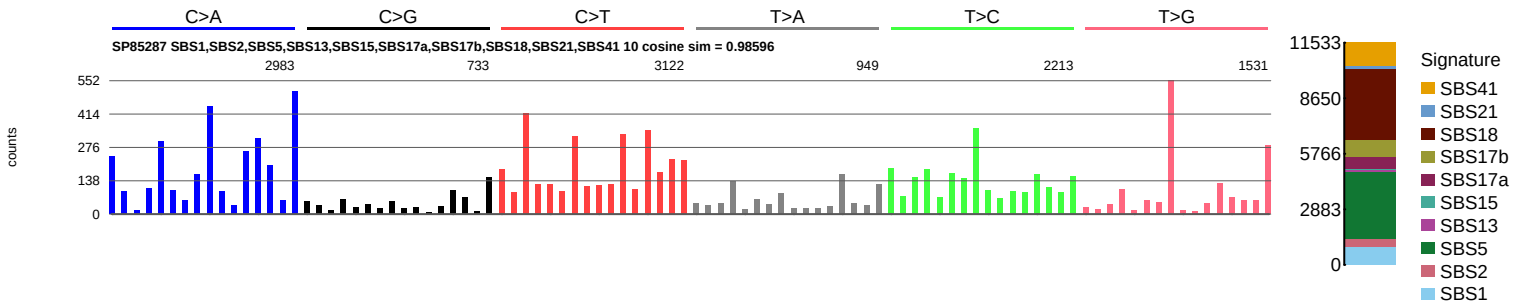
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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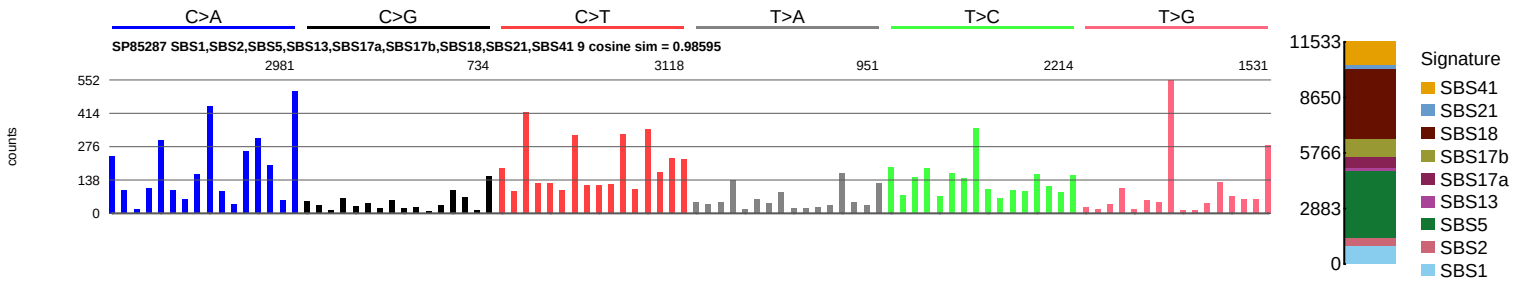
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98596
2983
733
3122
949
2213
1531
552
414
276
138
0
counts
11533
8650
5766
2883
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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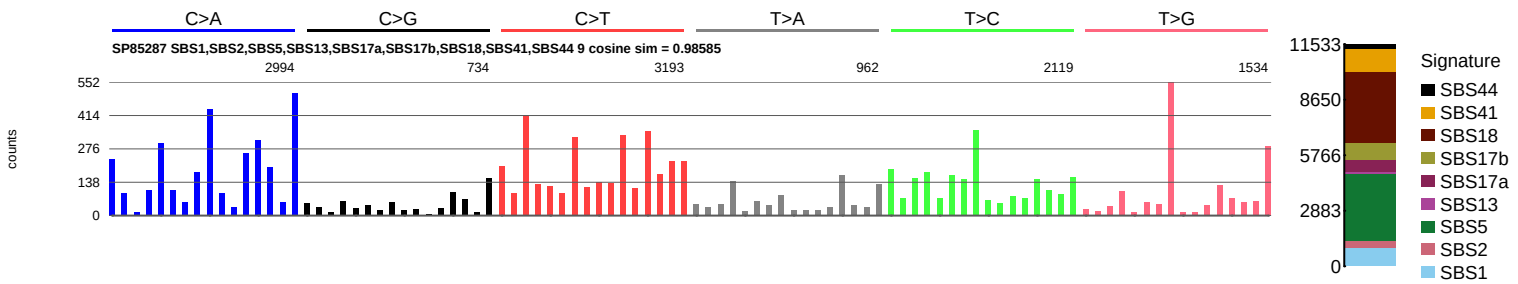
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98595
2981
734
3118
951
2214
1531
552
414
276
138
0
counts
11533
8650
5766
2883
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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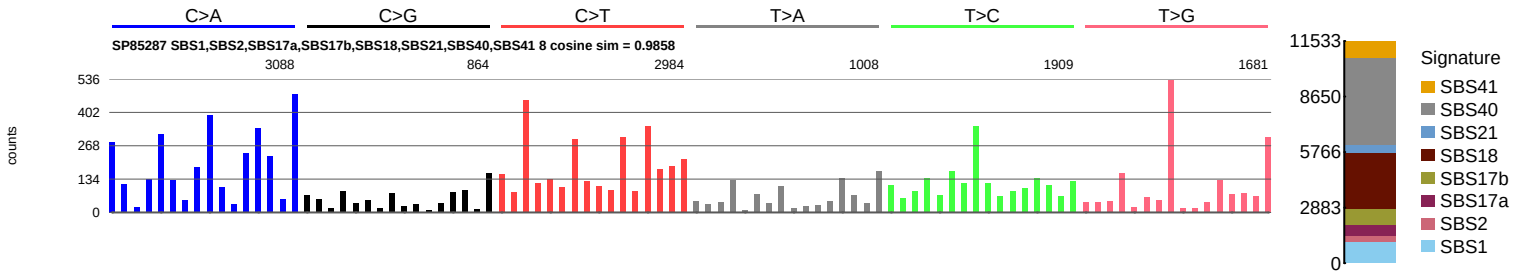
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98585
2994
734
3193
962
2119
1534
552
414
276
138
0
counts
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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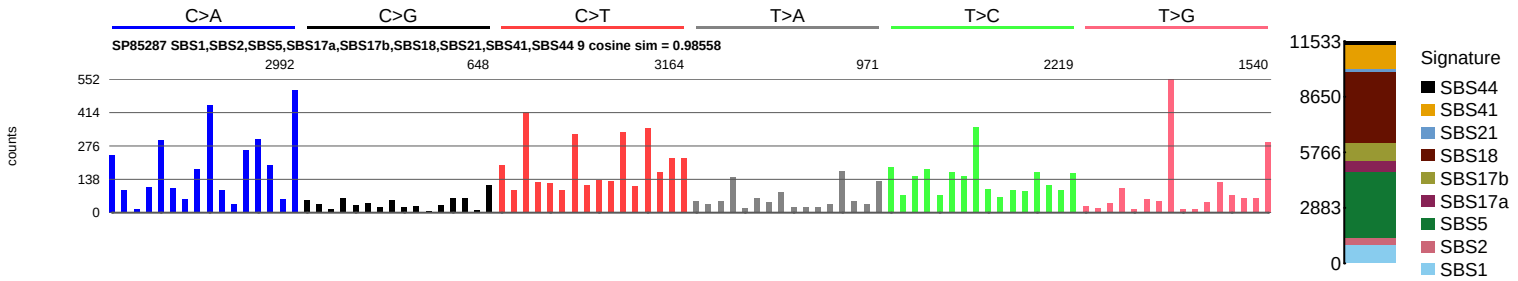
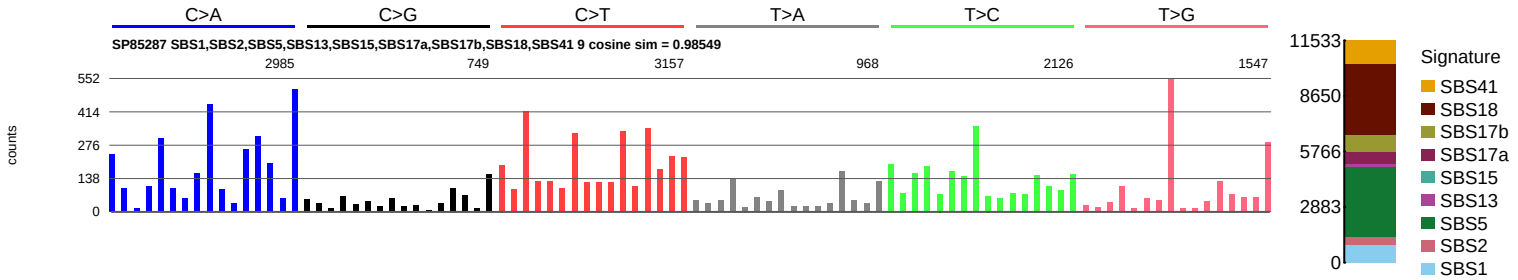
T>C
T>G
SP85287 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.9858
3088
864
2984
1008
1909
1681
counts
536
402
268
134
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98558
2992
648
3164
971
2219
1540
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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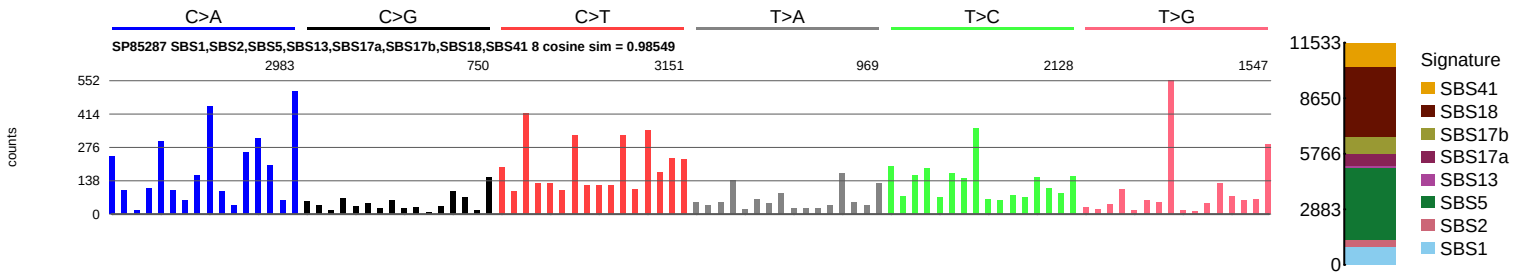
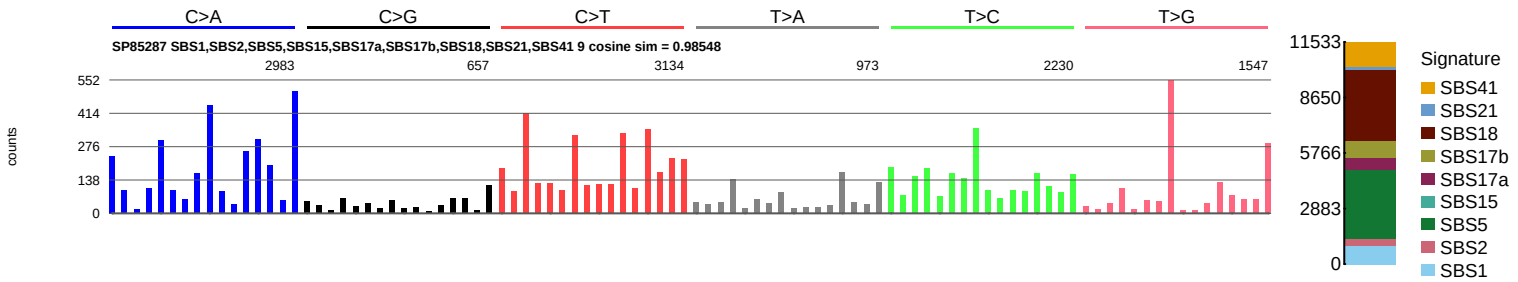
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98549
2985
749
3157
968
2126
1547
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98549
2983
750
3151
969
2128
1547
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98548
2983
657
3134
973
2230
1547
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
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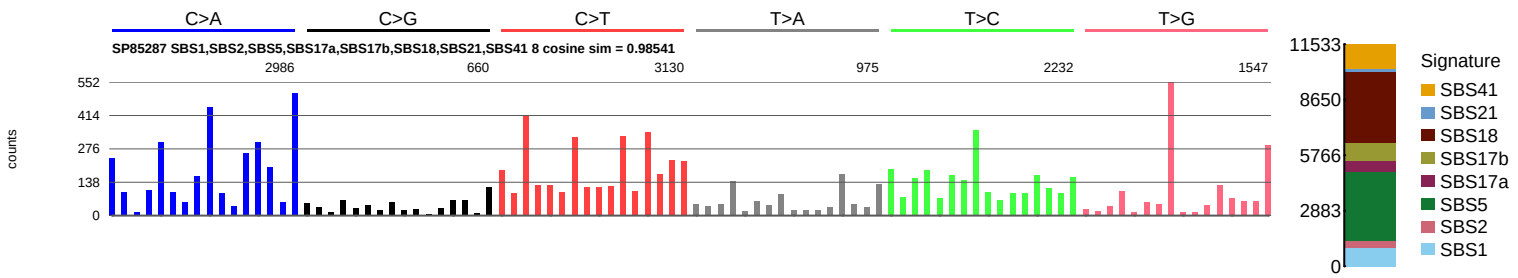
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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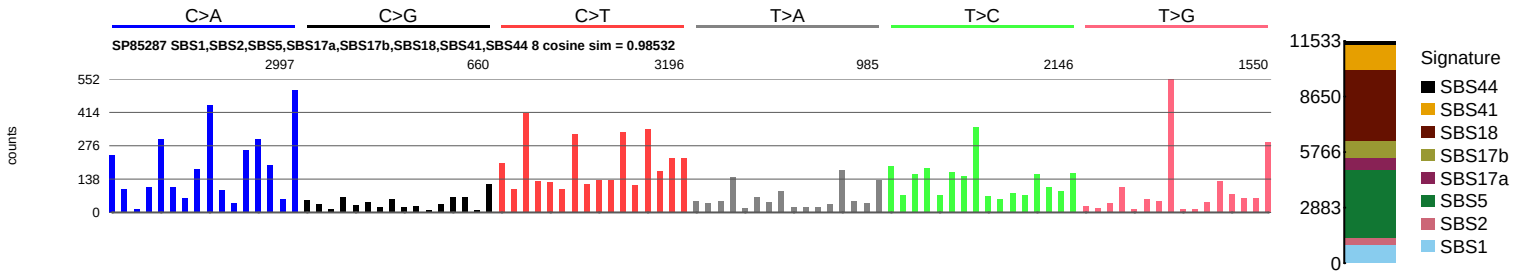
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98541
2986
660
3130
975
2232
1547
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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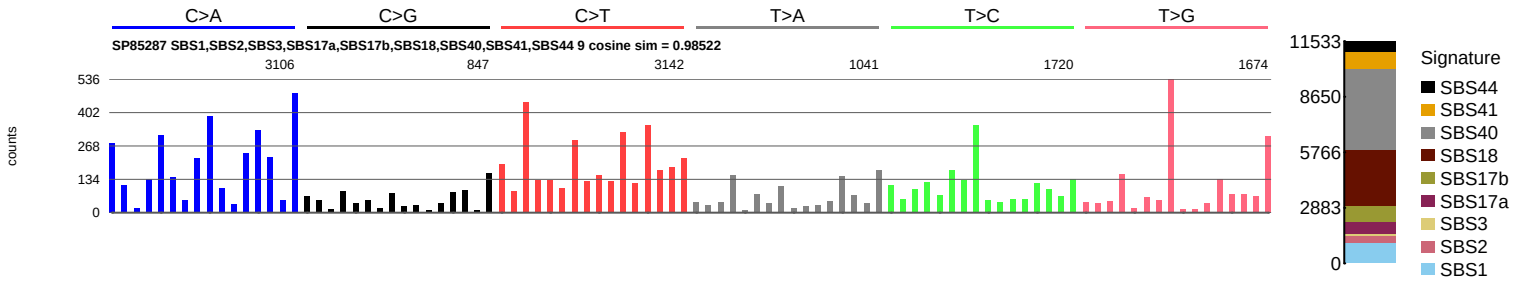
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98532
2997
660
3196
985
2146
1550
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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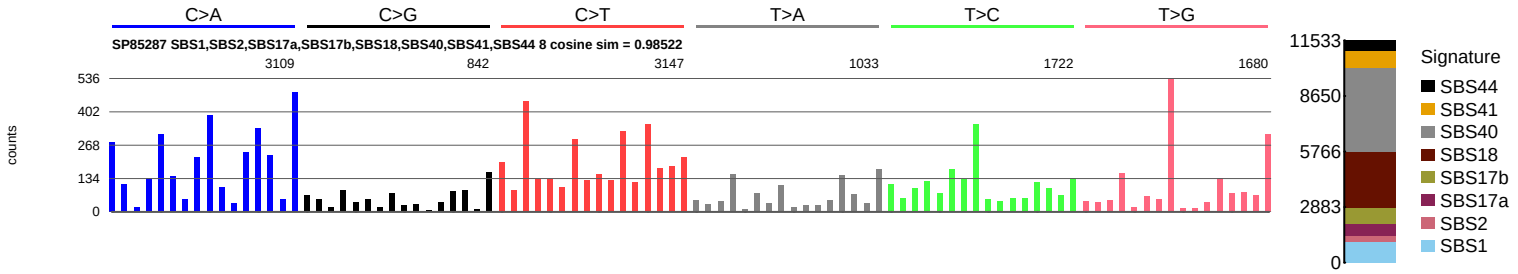
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98522
3106
847
3142
1041
1720
1674
counts
536
402
268
134
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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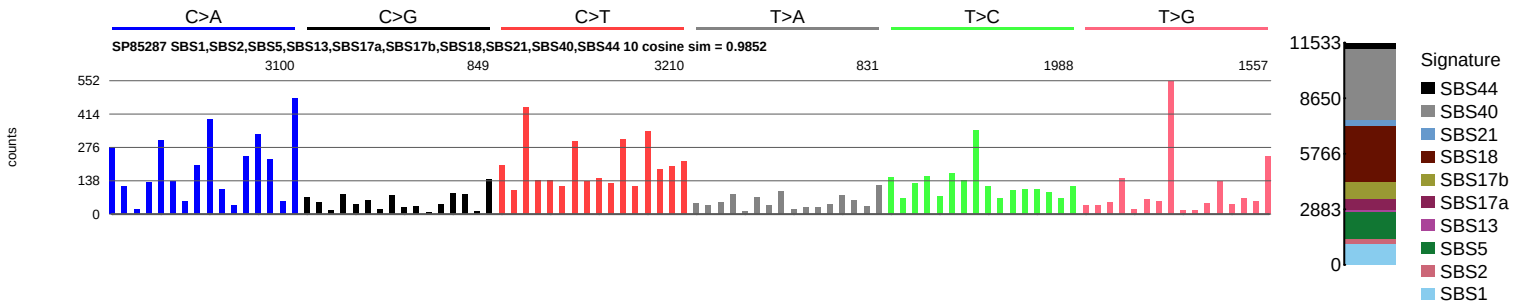
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98522
3109
842
3147
1033
1722
1680
counts
536
402
268
134
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
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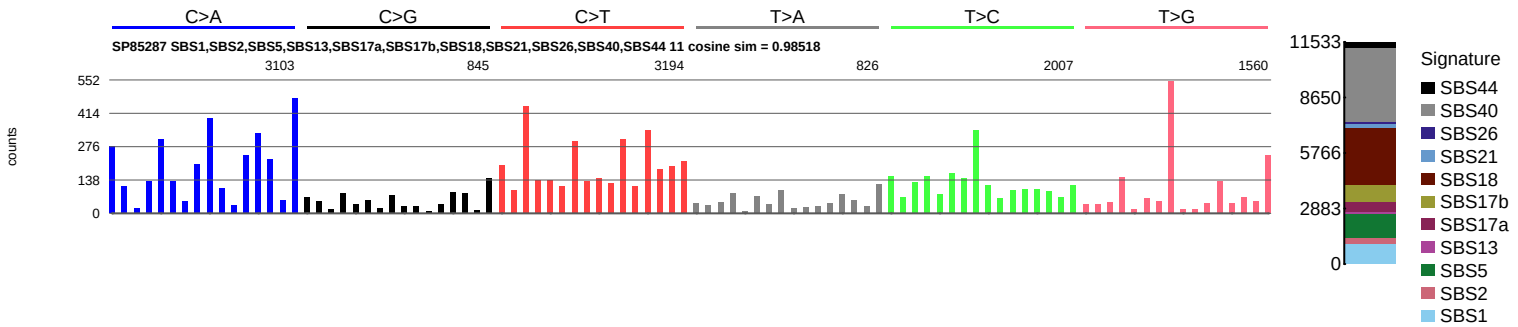
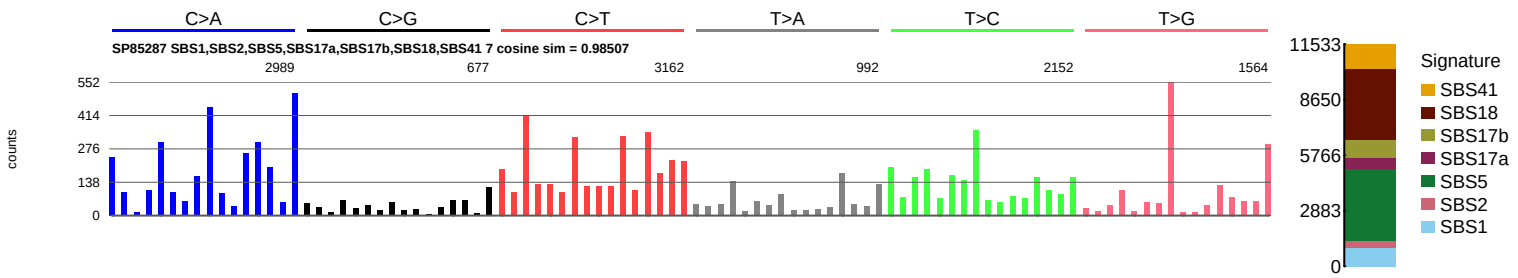
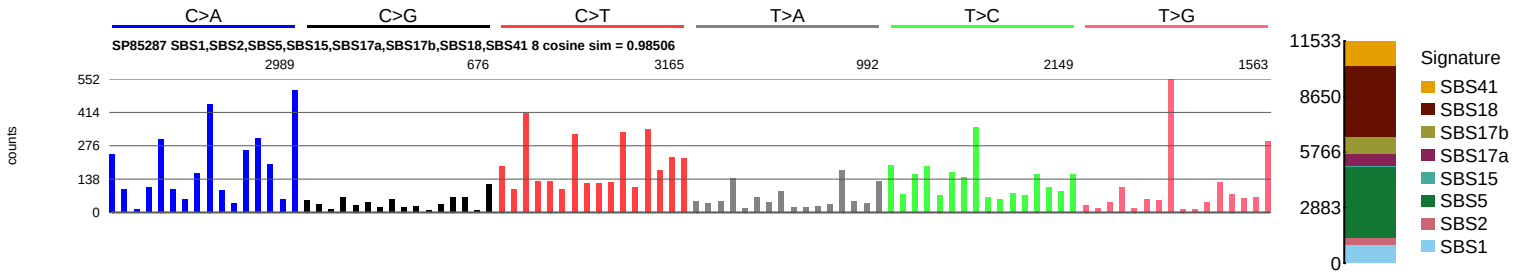
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.9852
3100
849
3210
831
1988
1557
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98518
3103
845
3194
826
2007
1560
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98507
2989
677
3162
992
2152
1564
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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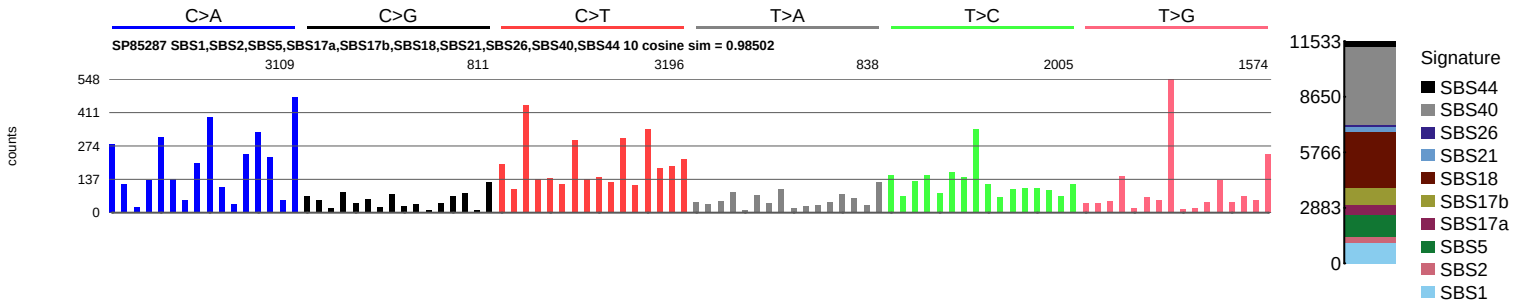
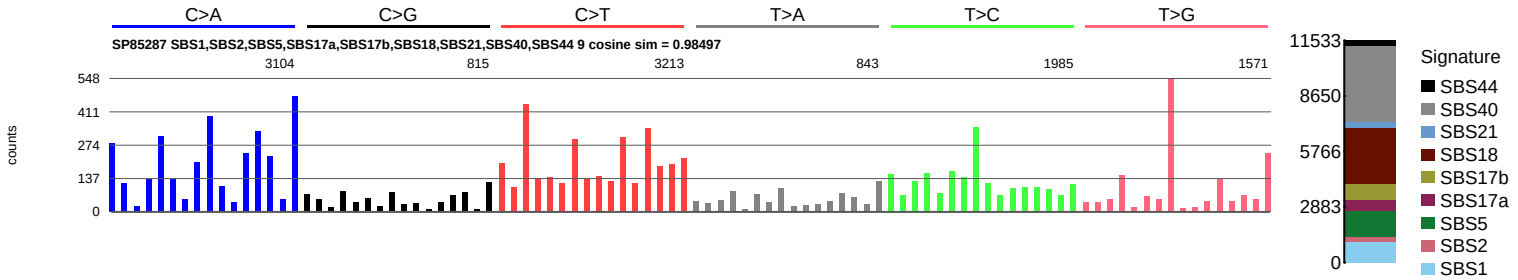
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98506
2989
676
3165
992
2149
1563
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98502
3109
811
3196
838
2005
1574
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98497
3104
815
3213
843
1985
1571
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
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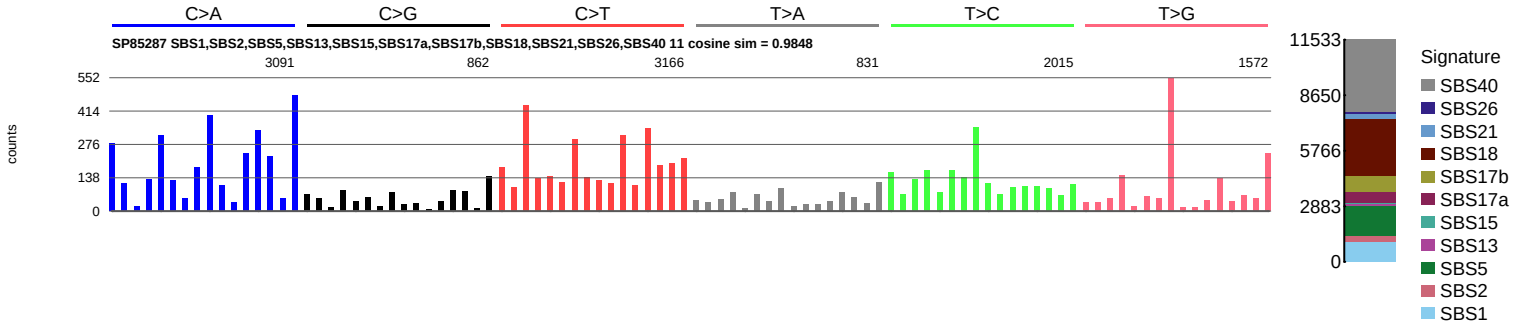
SBS2
SBS1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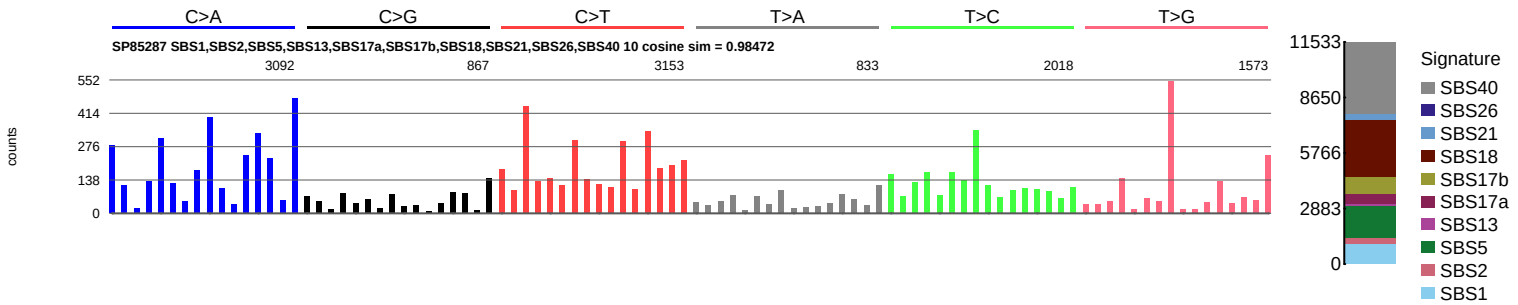
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 11 cosine sim = 0.9848
3091
862
3166
831
2015
1572
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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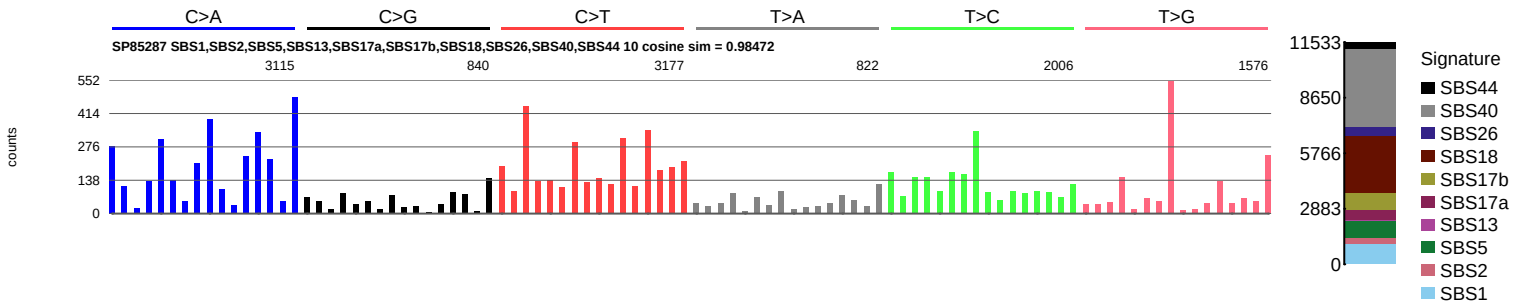
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98472
3092
867
3153
833
2018
1573
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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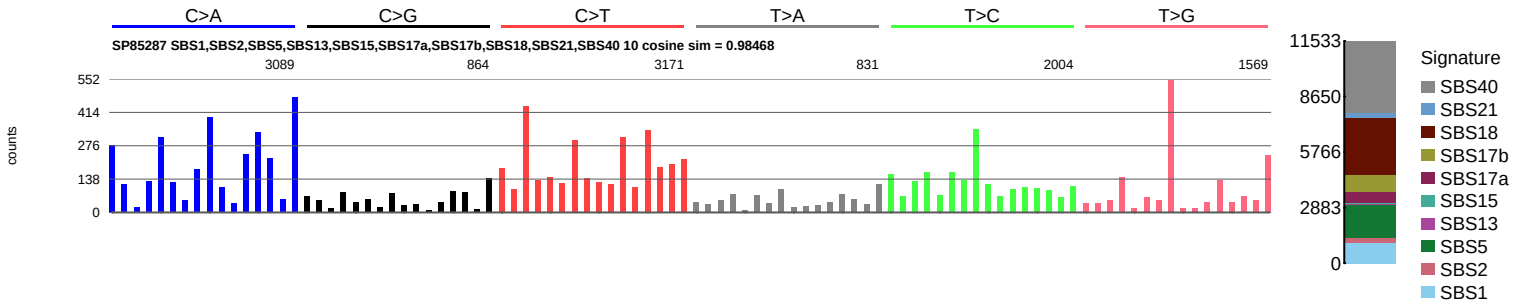
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98472
3115
840
3177
822
2006
1576
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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SP85287 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98468
3089
864
3171
831
2004
1569
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
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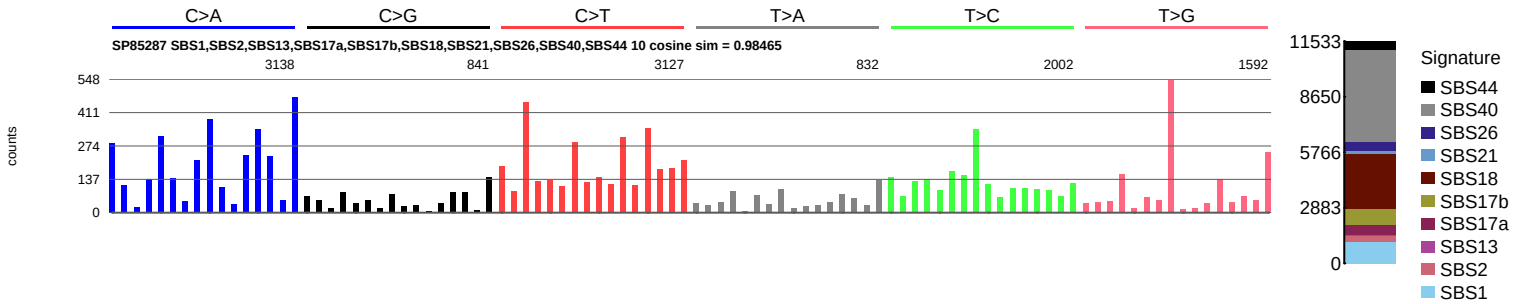
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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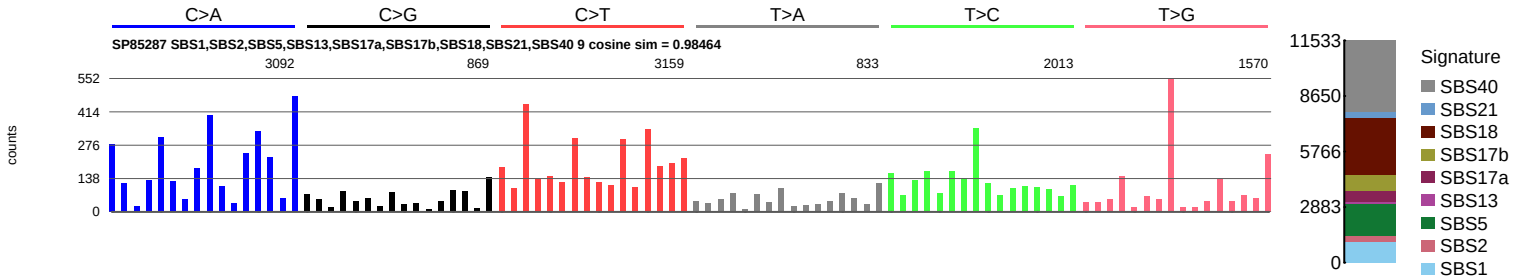
T>C
T>G
SP85287 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98465
3138
841
3127
832
2002
1592
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>T
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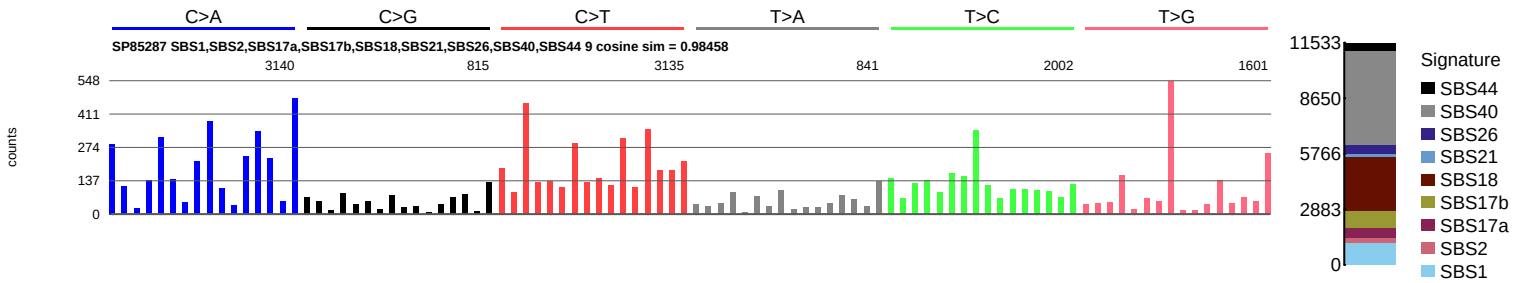
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98464
3092
869
3159
833
2013
1570
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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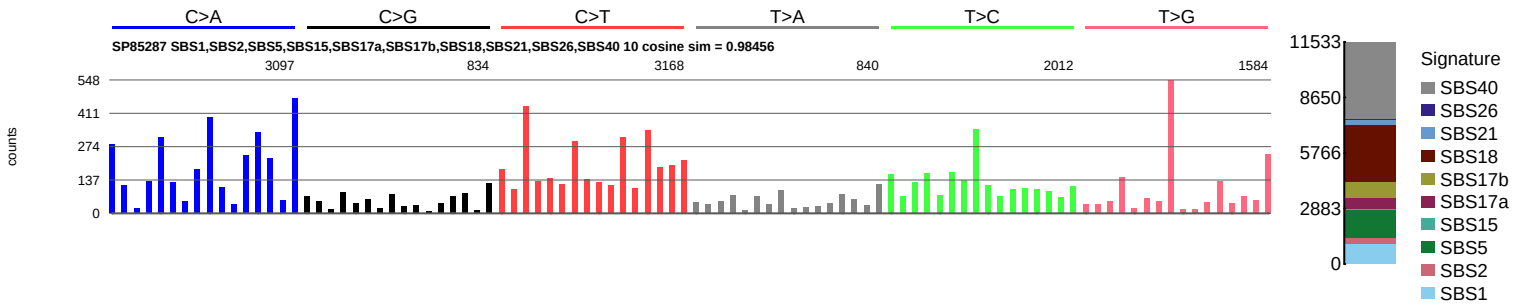
T>C
T>G
SP85287 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98458
3140
815
3135
841
2002
1601
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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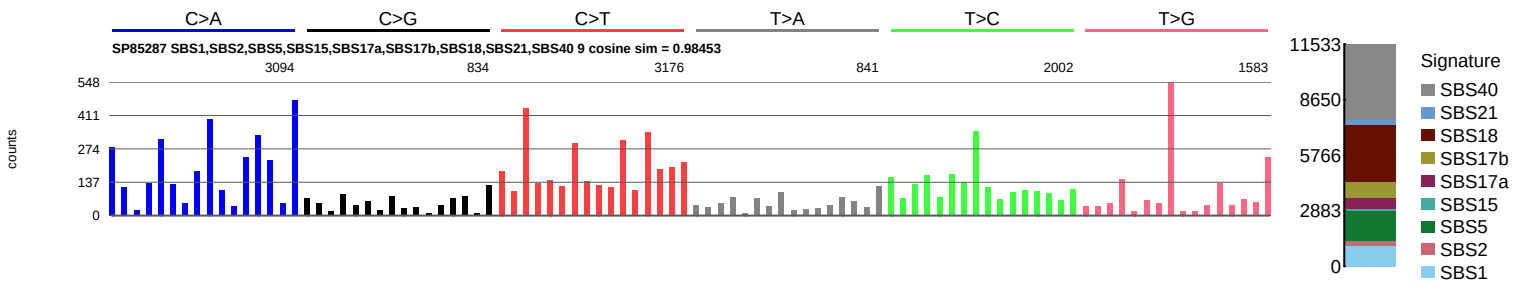
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98456
3097
834
3168
840
2012
1584
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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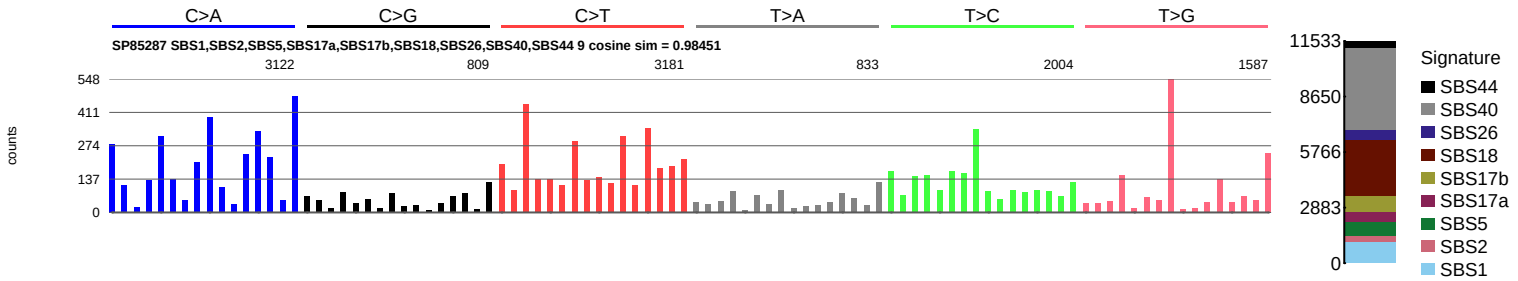
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98453
3094
834
3176
841
2002
1583
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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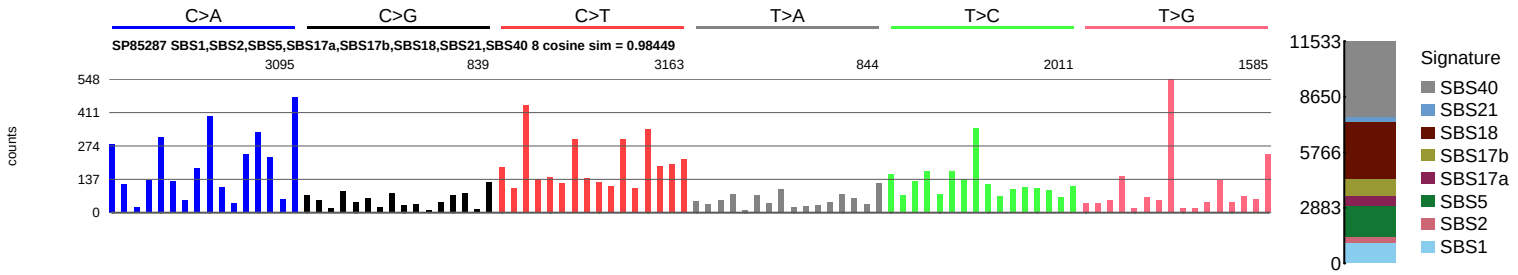
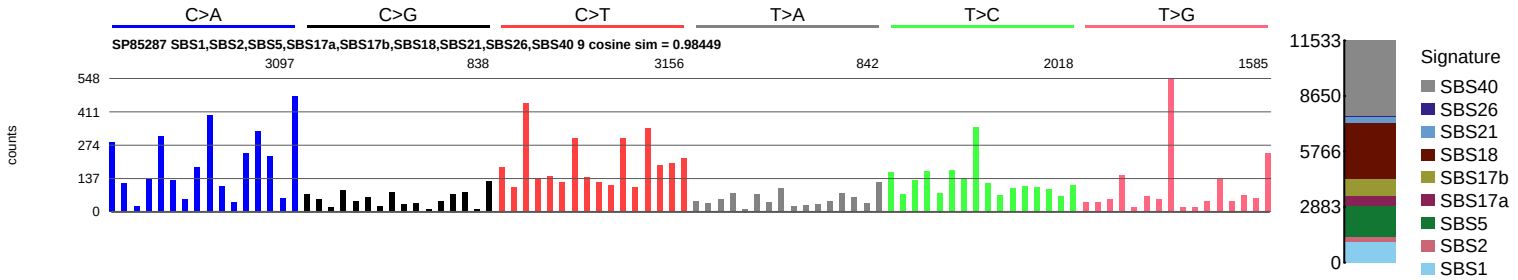
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98451
3122
809
3181
833
2004
1587
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98449
3095
839
3163
844
2011
1585
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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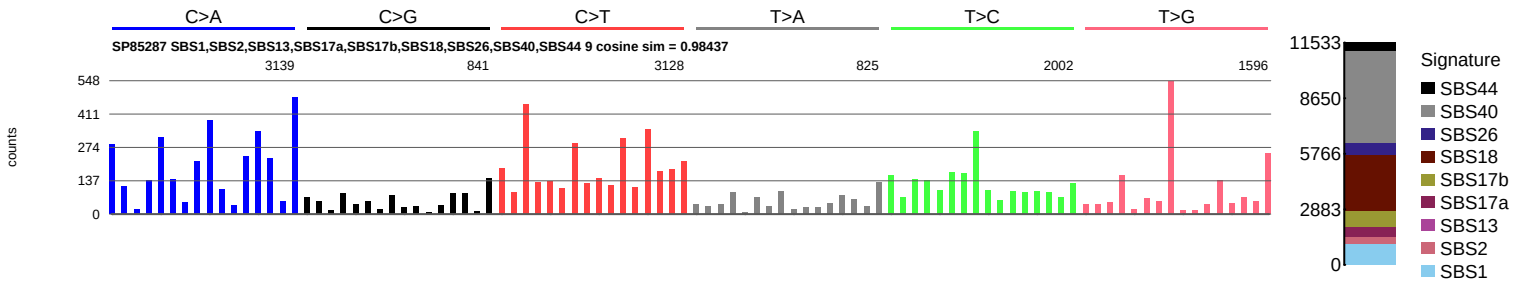
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98449
3097
838
3156
842
2018
1585
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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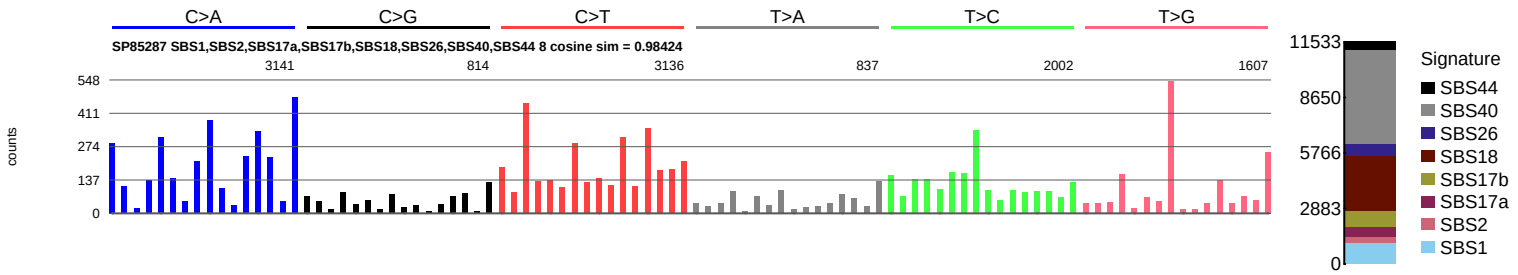
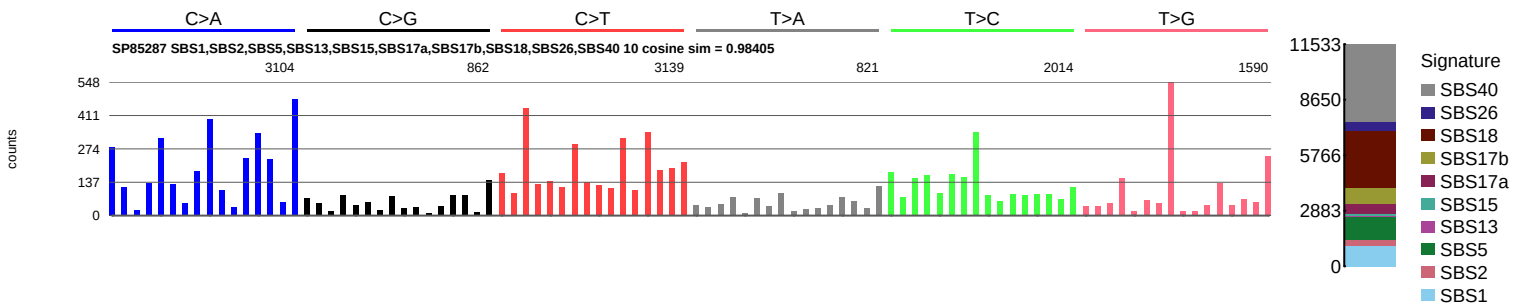
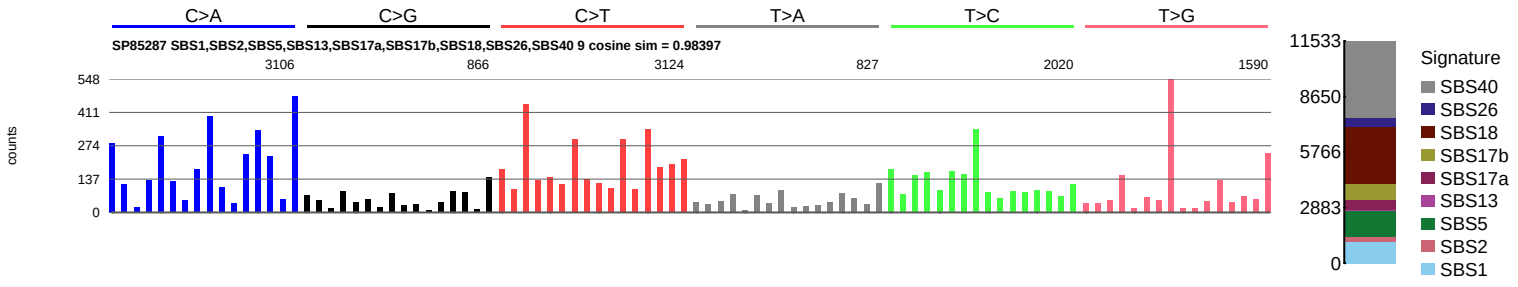
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98437
3139
841
3128
825
2002
1596
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98424
3141
814
3136
837
2002
1607
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98405
3104
862
3139
821
2014
1590
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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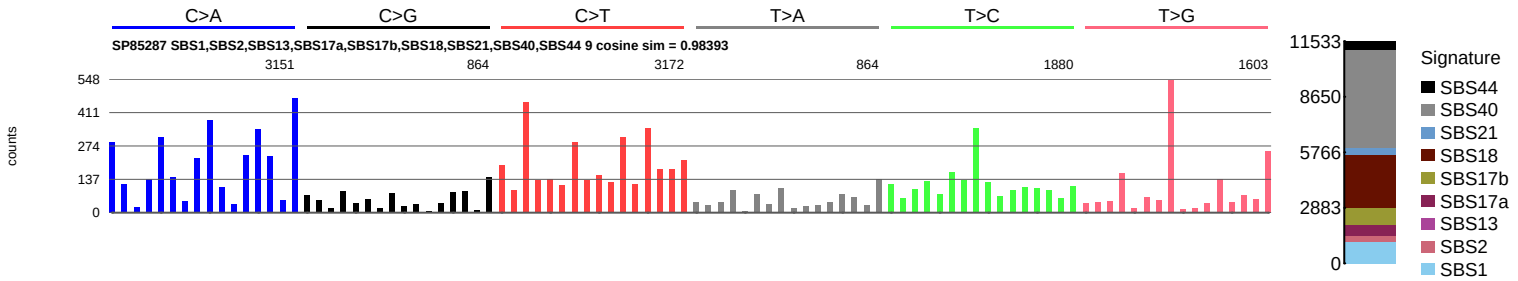
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98397
3106
866
3124
827
2020
1590
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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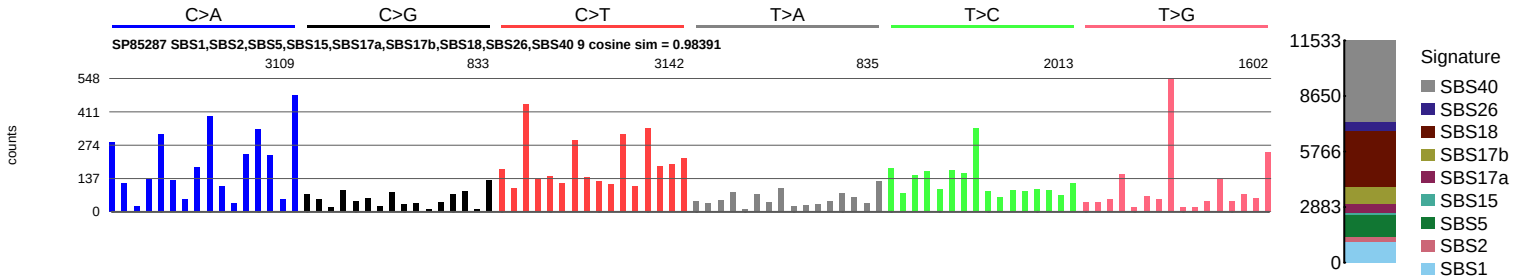
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98393
3151
864
3172
864
1880
1603
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98391
3109
833
3142
835
2013
1602
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS40
SBS26
SBS18
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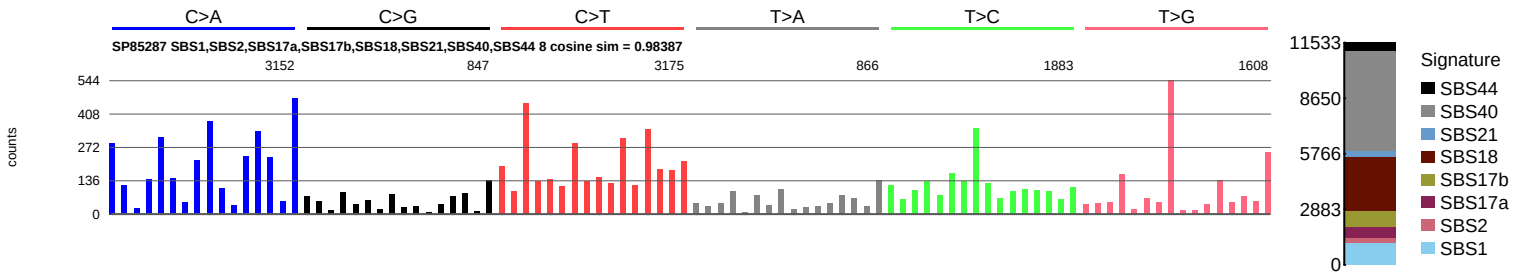
SBS15
SBS5
SBS2
SBS1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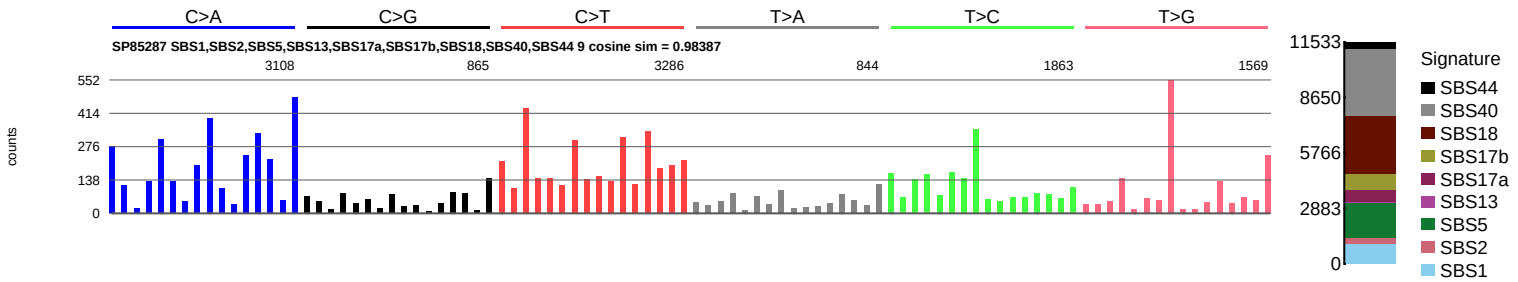
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98387
3152
847
3175
866
1883
1608
counts
544
408
272
136
0
11533
8650
5766
2883
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98387
3108
865
3286
844
1863
1569
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS44
SBS40
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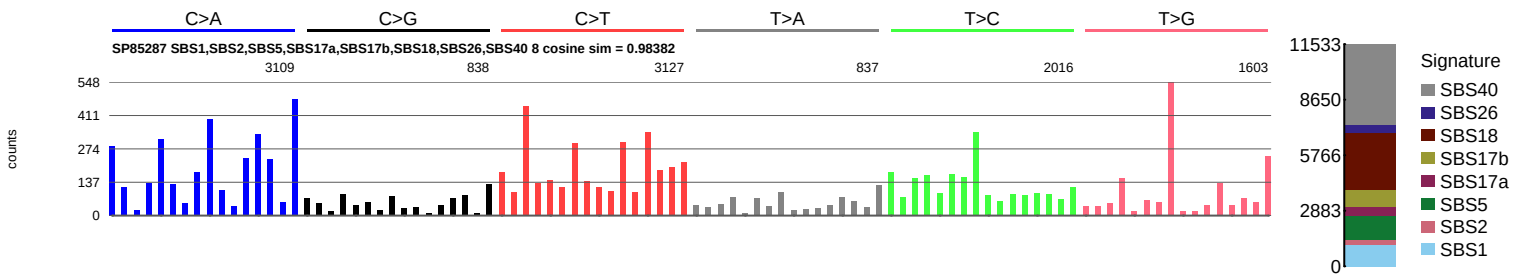
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98382
3109
838
3127
837
2016
1603
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS40
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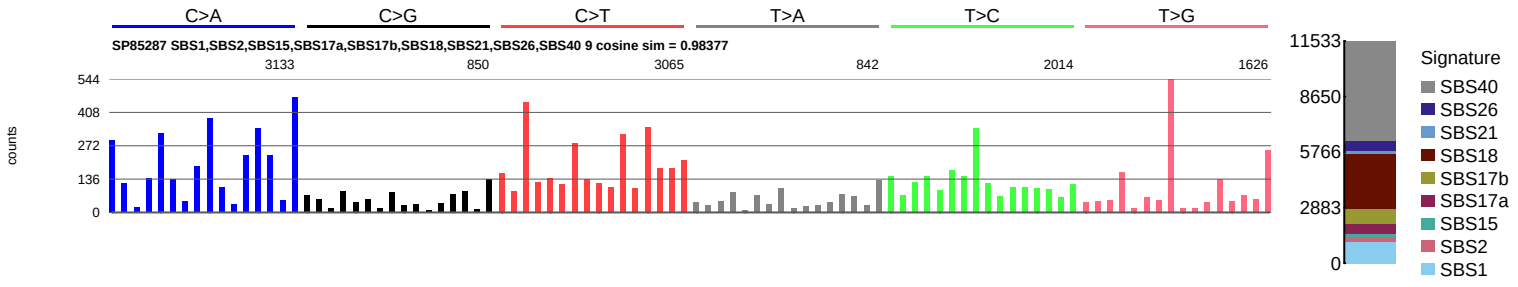
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98377
3133
850
3065
842
2014
1626
counts
544
408
272
136
0
11533
8650
5766
2883
0
Signature
SBS40
SBS26
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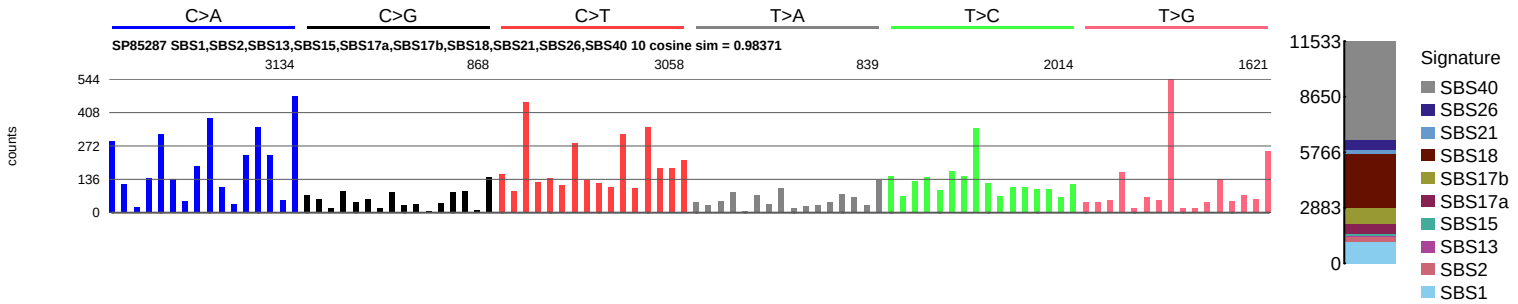
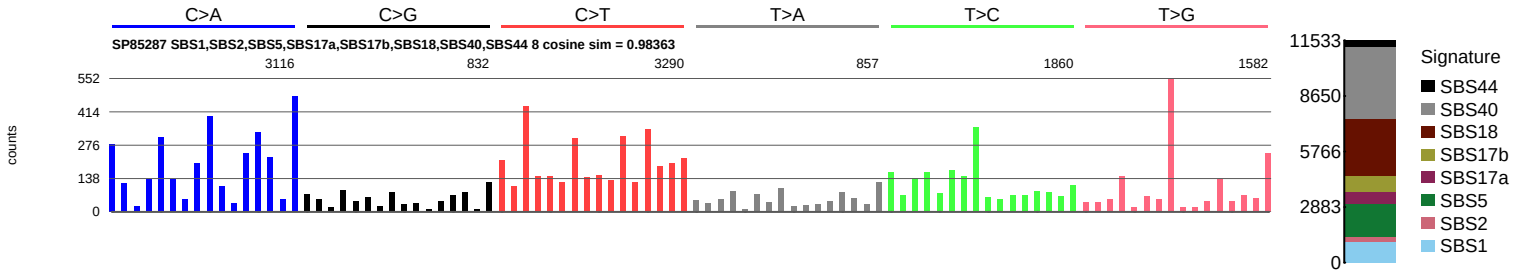
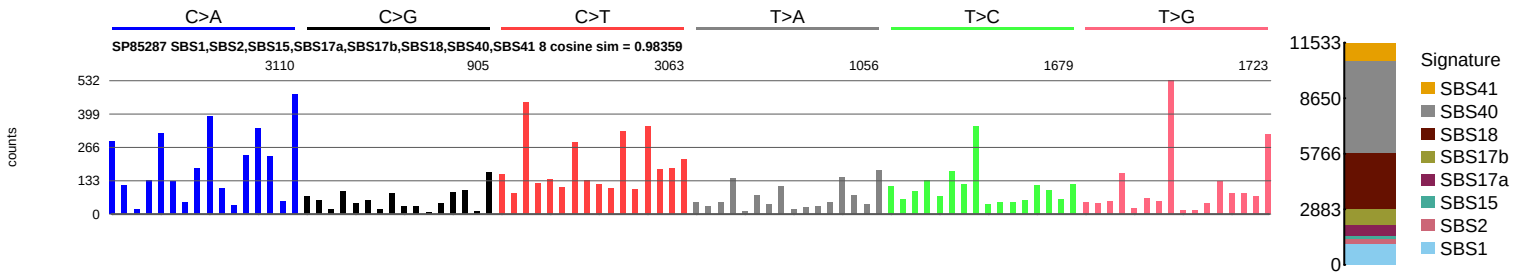
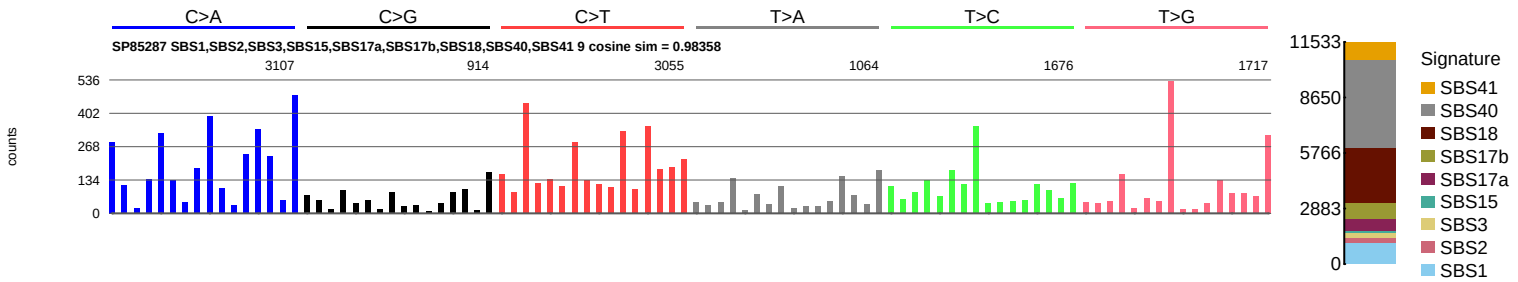
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98371
3134
868
3058
839
2014
1621
counts
544
408
272
136
0
11533
8650
5766
2883
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98363
3116
832
3290
857
1860
1582
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98359
3110
905
3063
1056
1679
1723
counts
532
399
266
133
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>T
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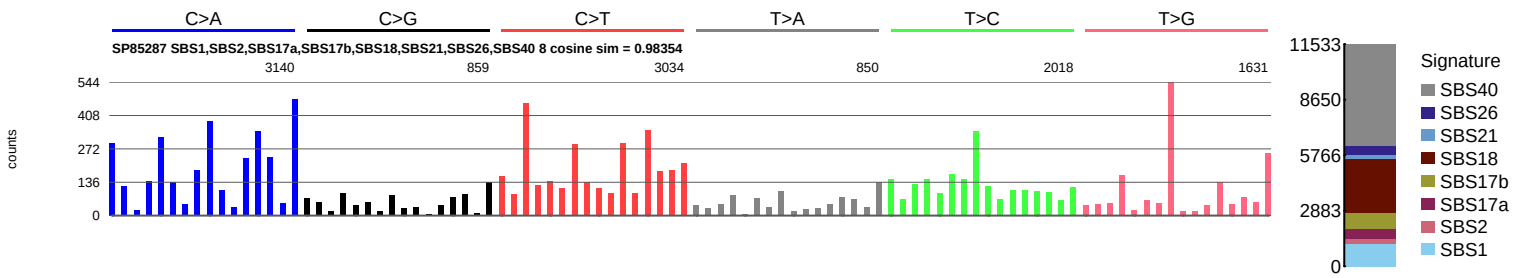
T>G
SP85287 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98358
3107
914
3055
1064
1676
1717
counts
536
402
268
134
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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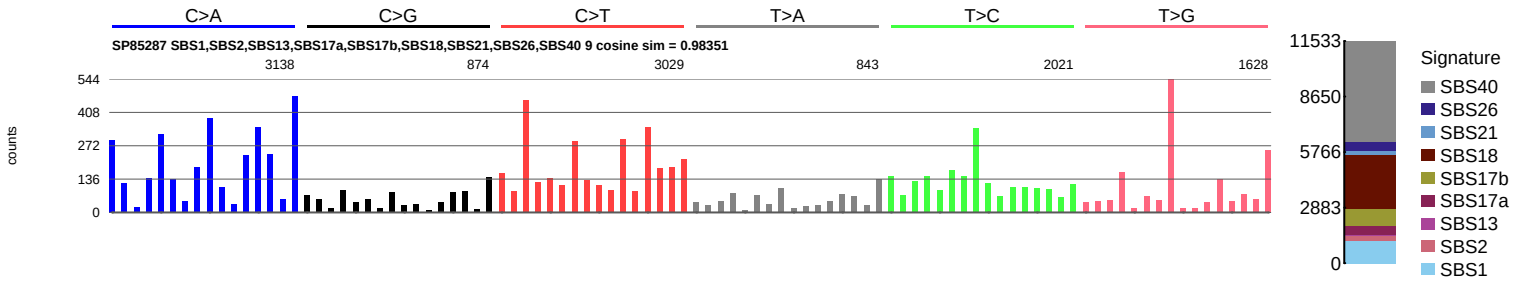
T>C
T>G
SP85287 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98354
3140
859
3034
850
2018
1631
counts
544
408
272
136
0
11533
8650
5766
2883
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>T
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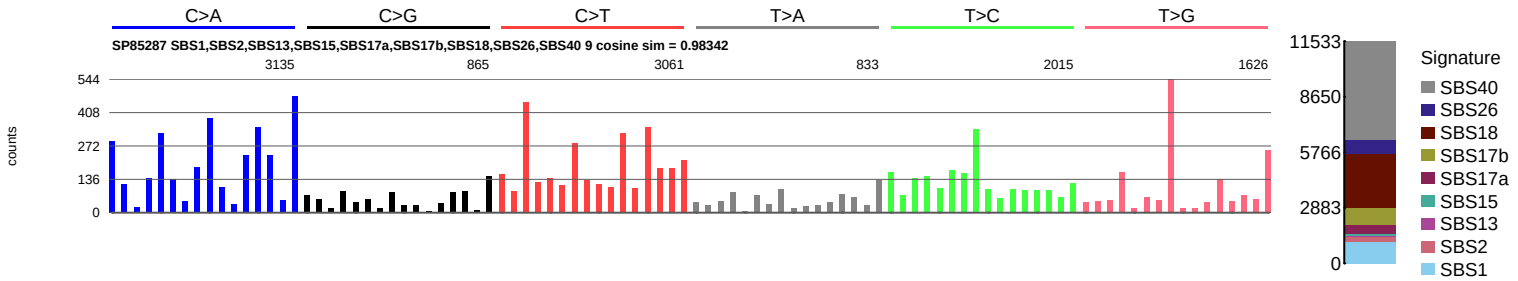
T>G
SP85287 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98351
3138
874
3029
843
2021
1628
counts
544
408
272
136
0
11533
8650
5766
2883
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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SP85287 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98342
3135
865
3061
833
2015
1626
counts
544
408
272
136
0
11533
8650
5766
2883
0
Signature
SBS40
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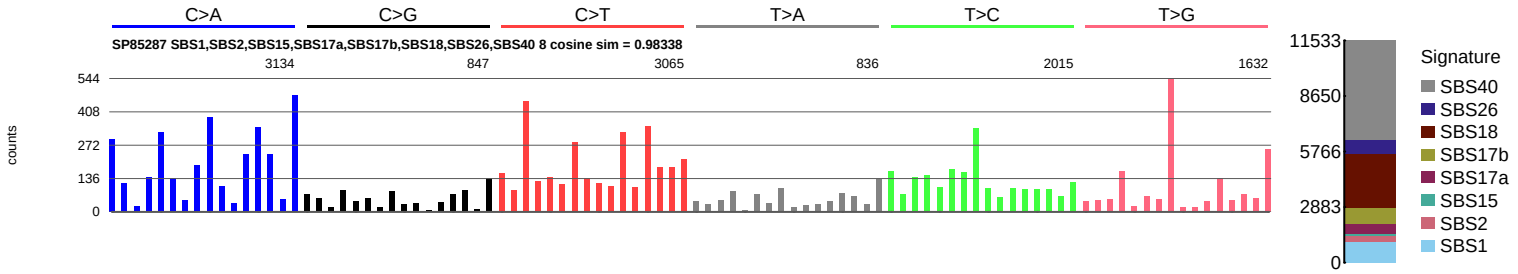
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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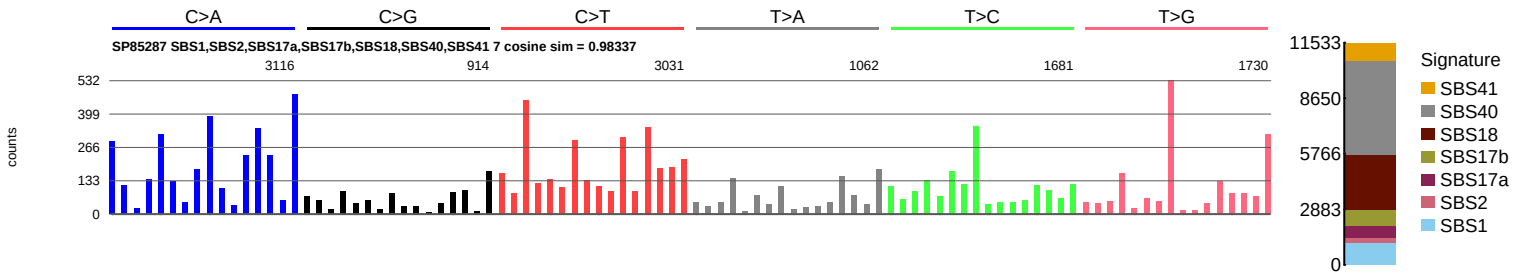
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98338
3134
847
3065
836
2015
1632
counts
544
408
272
136
0
11533
8650
5766
2883
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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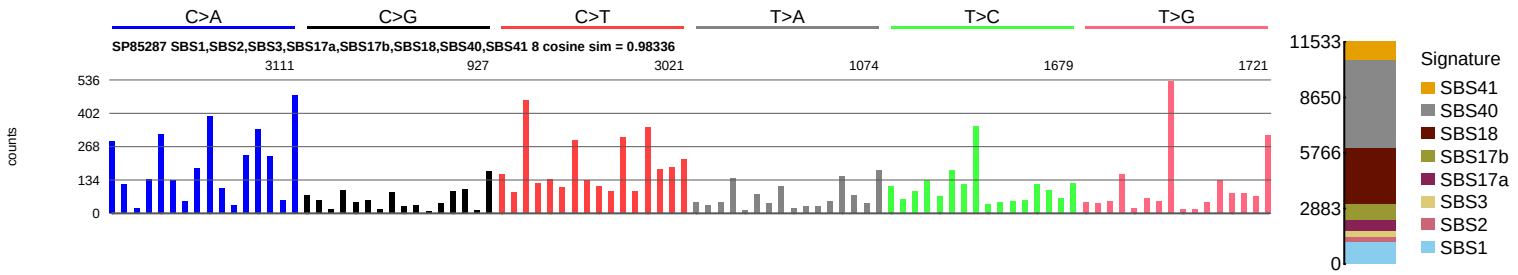
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98337
3116
914
3031
1062
1681
1730
counts
532
399
266
133
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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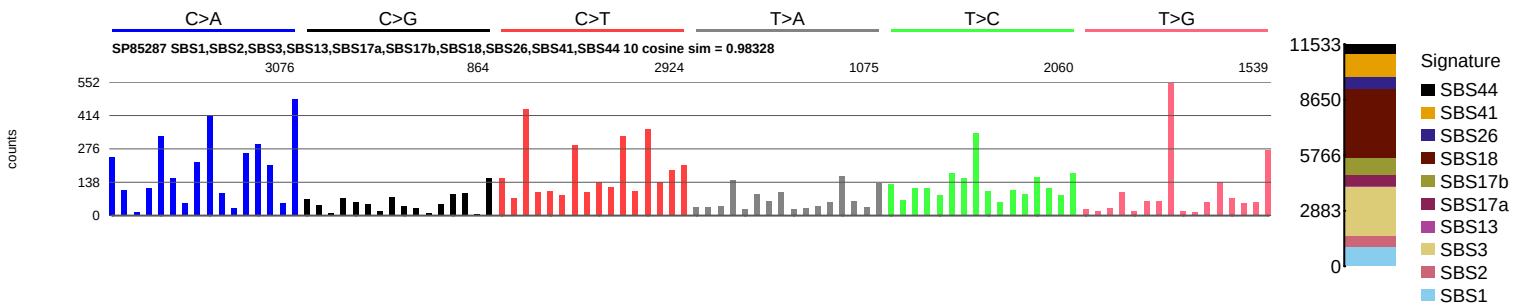
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98336
3111
927
3021
1074
1679
1721
counts
536
402
268
134
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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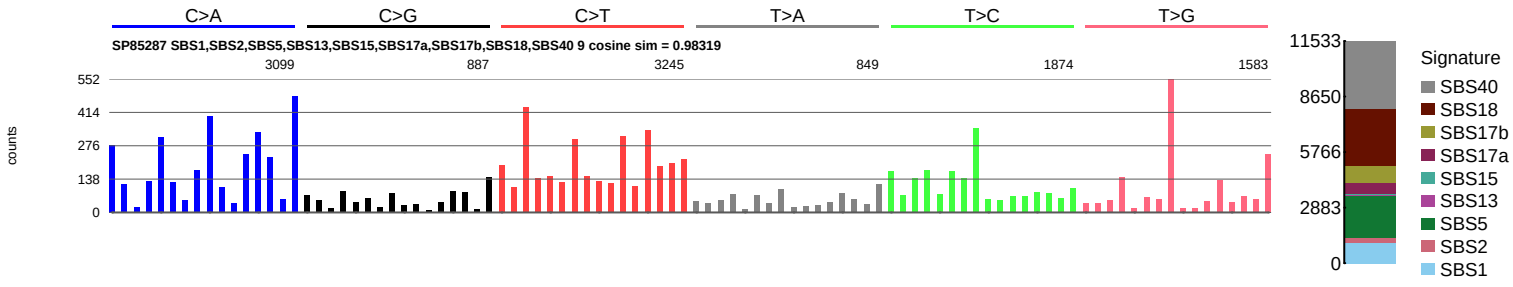
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98328
3076
864
2924
1075
2060
1539
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98319
3099
887
3245
849
1874
1583
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS40
SBS18
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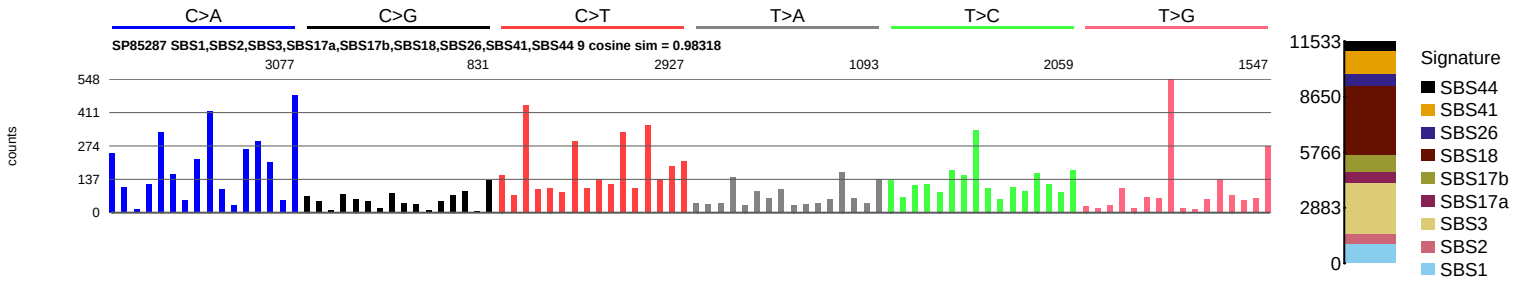
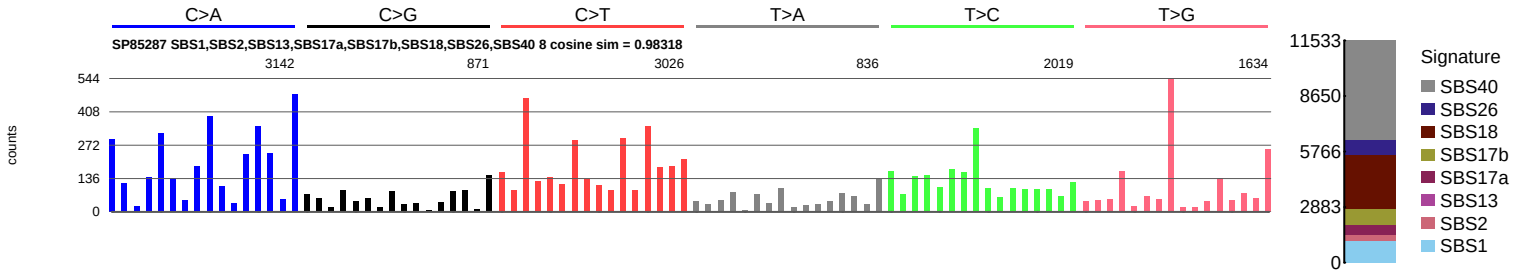
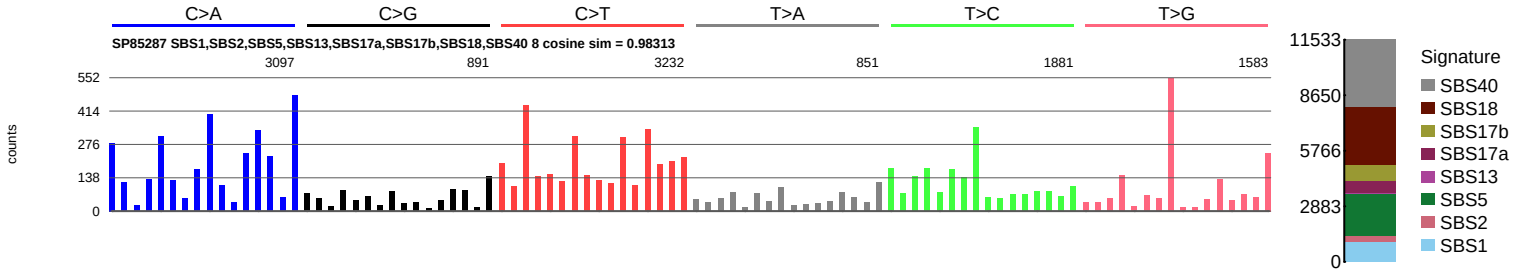
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98318
3077
831
2927
1093
2059
1547
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98318
3142
871
3026
836
2019
1634
counts
544
408
272
136
0
11533
8650
5766
2883
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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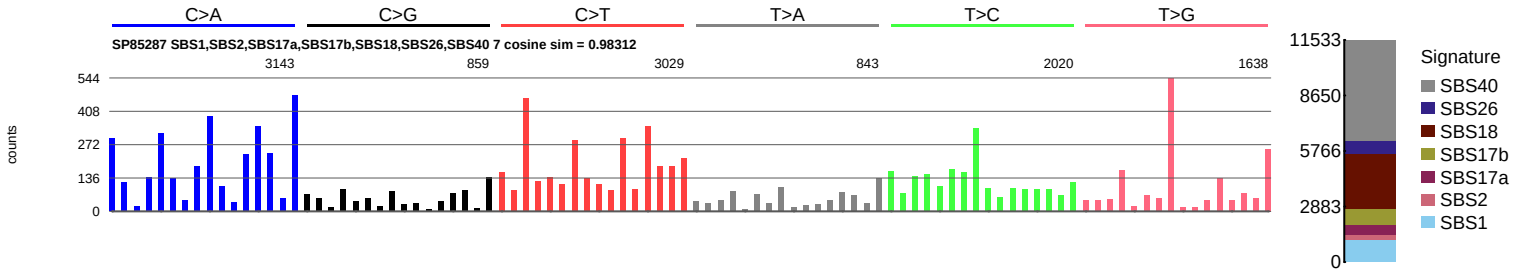
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98313
3097
891
3232
851
1881
1583
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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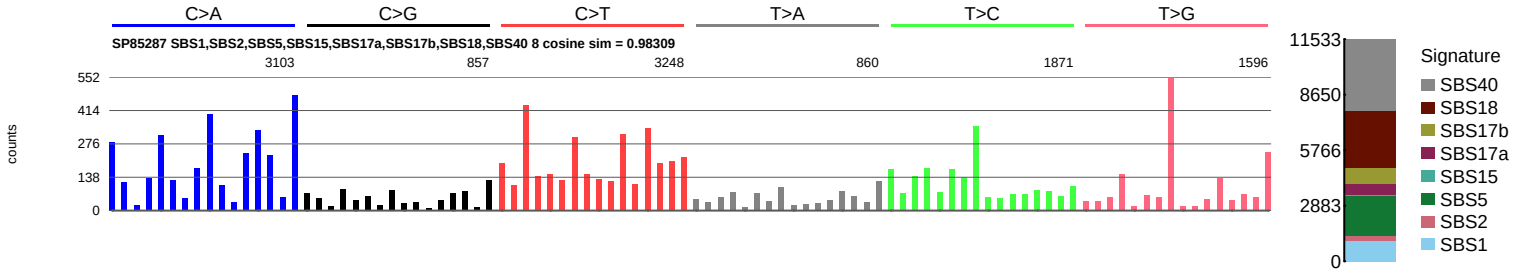
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98312
3143
859
3029
843
2020
1638
counts
544
408
272
136
0
11533
8650
5766
2883
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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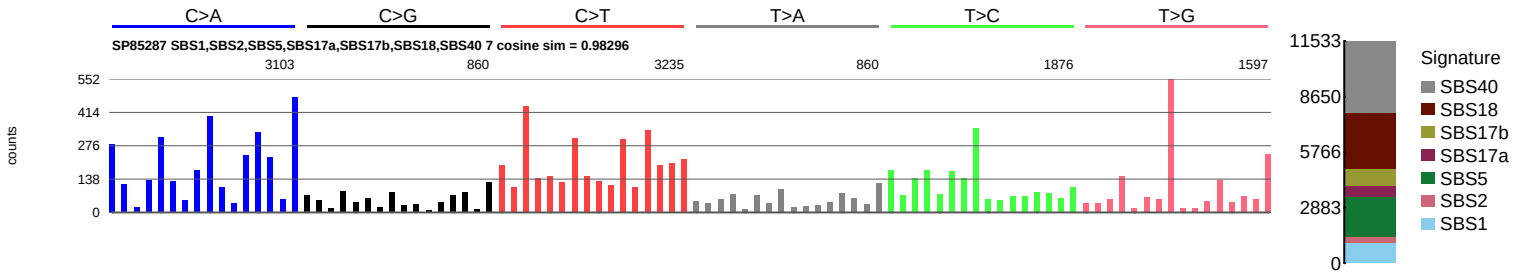
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98309
3103
857
3248
860
1871
1596
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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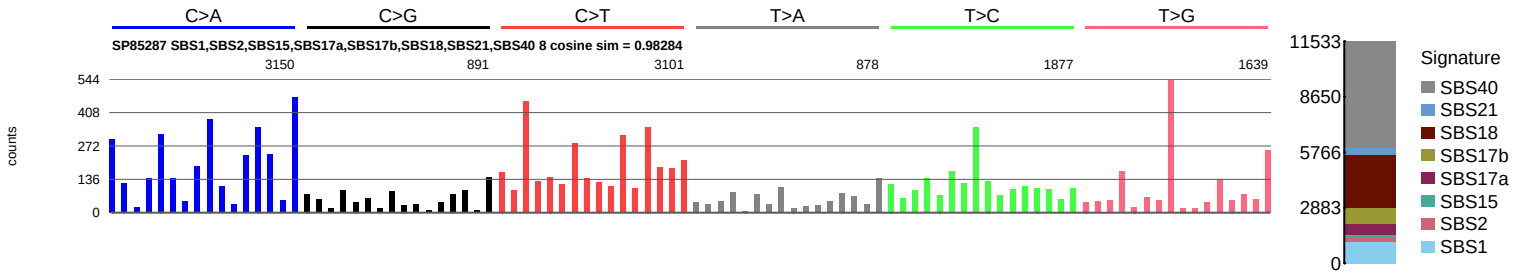
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98296
3103
860
3235
860
1876
1597
552
414
276
138
0
counts
11533
8650
5766
2883
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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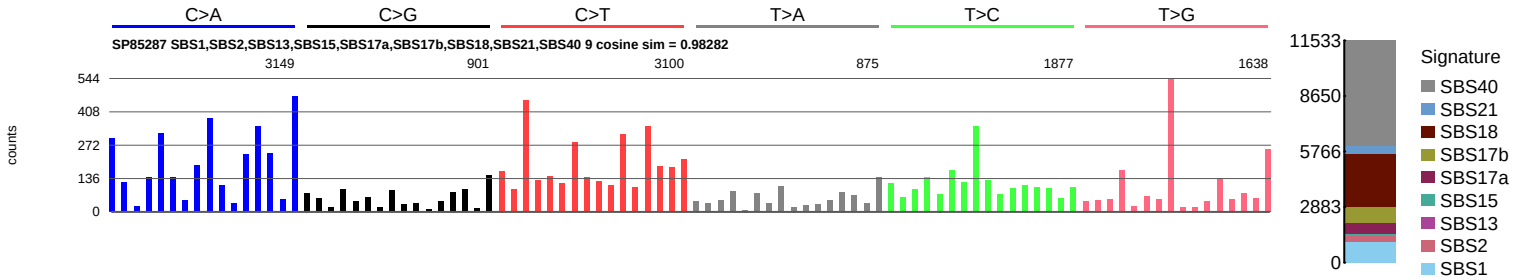
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98284
3150
891
3101
878
1877
1639
544
408
272
136
0
counts
11533
8650
5766
2883
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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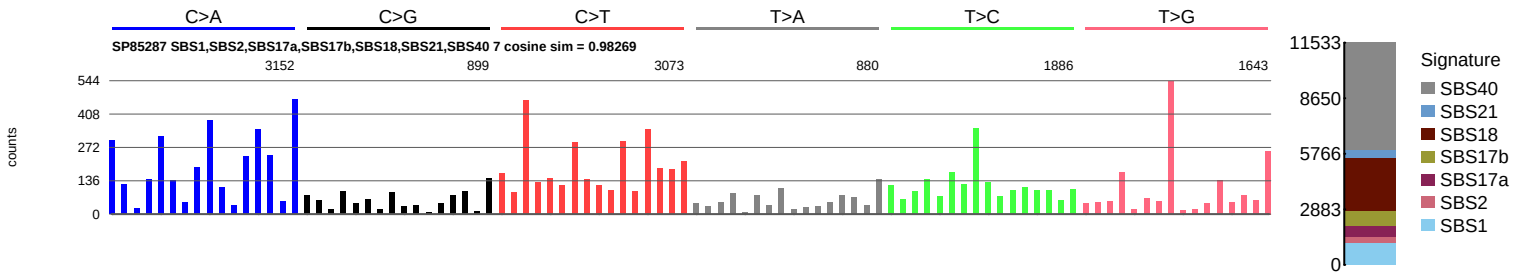
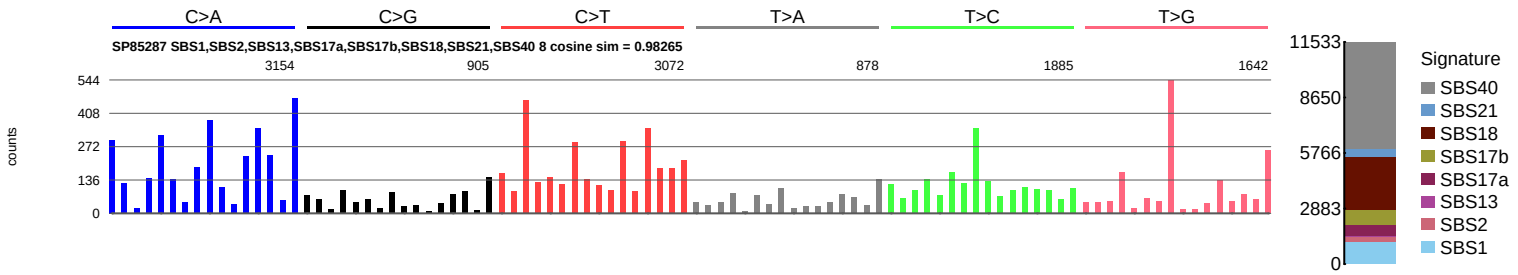
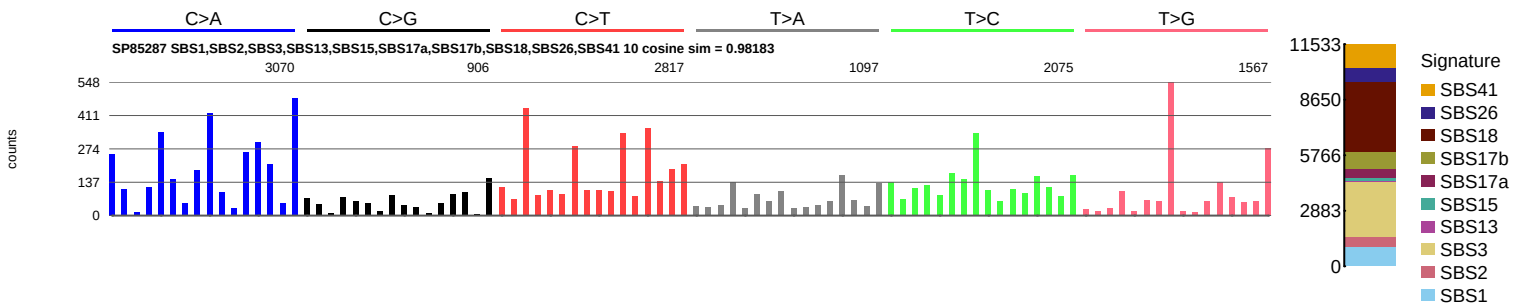
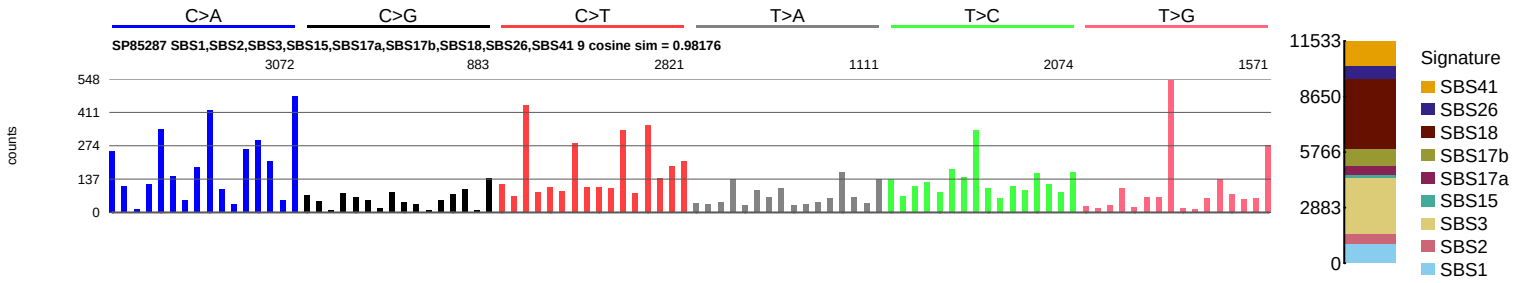
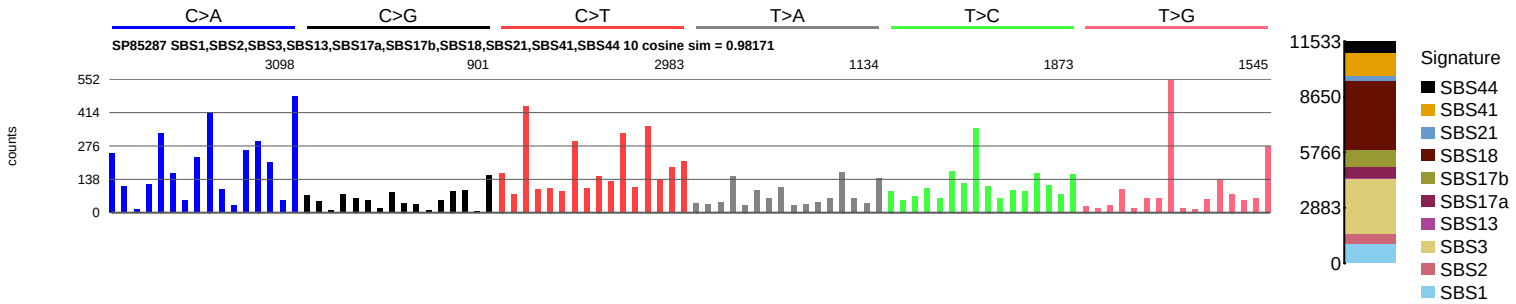
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98282
3149
901
3100
875
1877
1638
544
408
272
136
0
counts
11533
8650
5766
2883
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

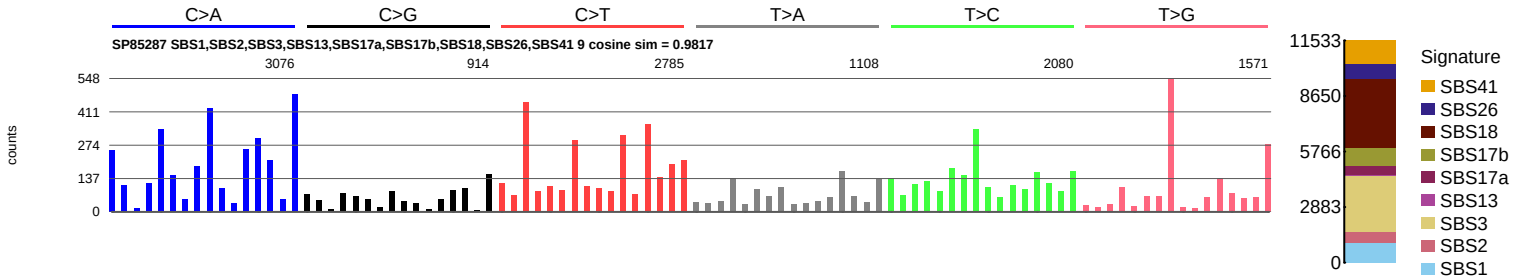

C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98269
3152
899
3073
880
1886
1643
544
408
272
136
0
counts
11533
8650
5766
2883
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

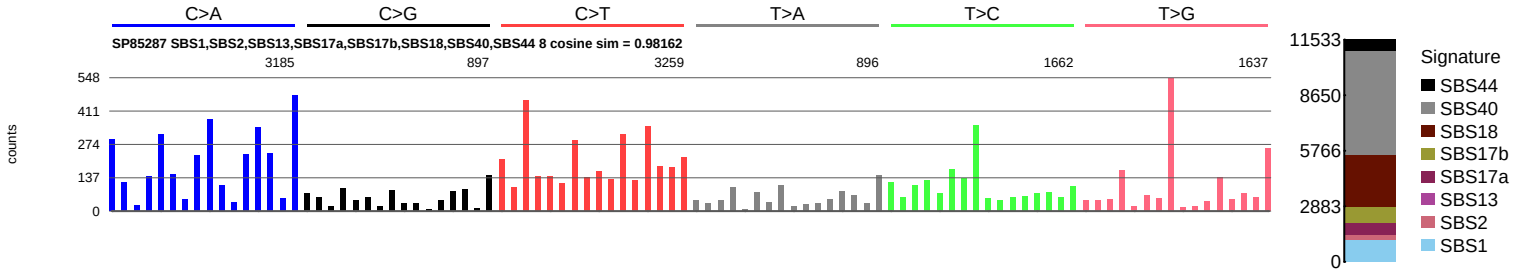

C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98265
3154
905
3072
878
1885
1642
544
408
272
136
0
counts
11533
8650
5766
2883
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

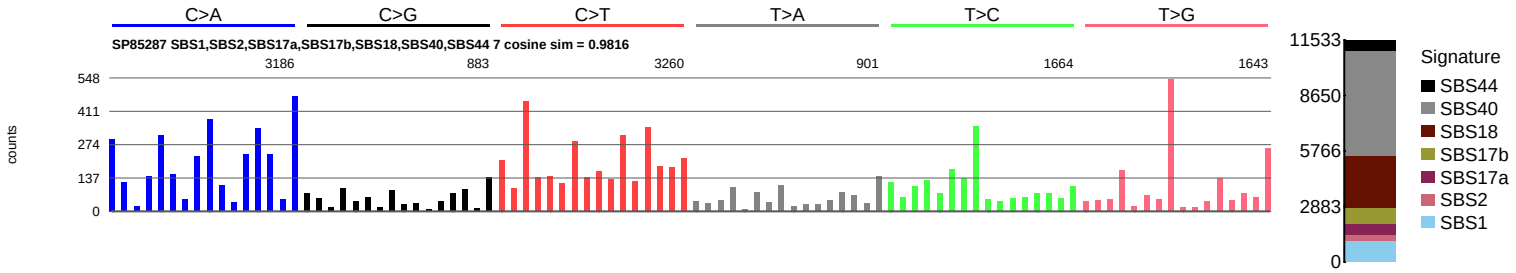

C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98183
3070
906
2817
1097
2075
1567
548
411
274
137
0
counts
11533
8650
5766
2883
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98176
3072
883
2821
1111
2074
1571
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98171
3098
901
2983
1134
1873
1545
counts
552
414
276
138
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.9817
3076
914
2785
1108
2080
1571
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98162
3185
897
3259
896
1662
1637
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.9816
3186
883
3260
901
1664
1643
counts
548
411
274
137
0
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2883
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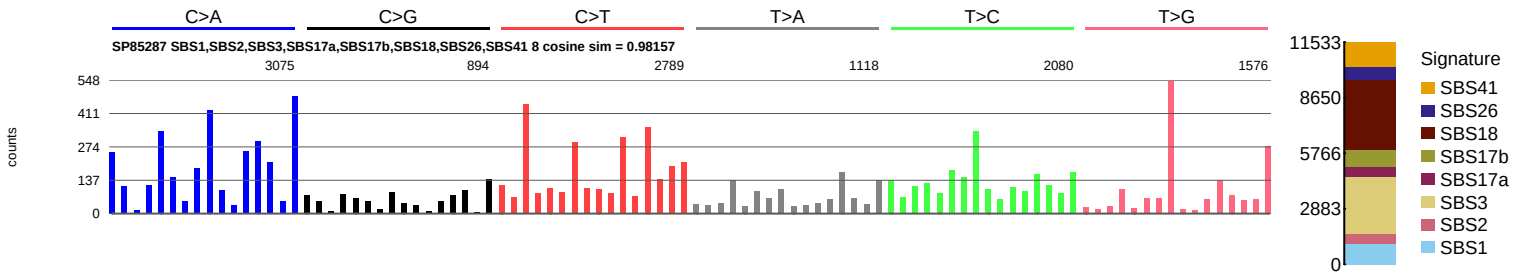
SBS2
SBS1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SP85287 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98157
3075
894
2789
1118
2080
1576
counts
548
411
274
137
0
11533
8650
5766
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SBS41
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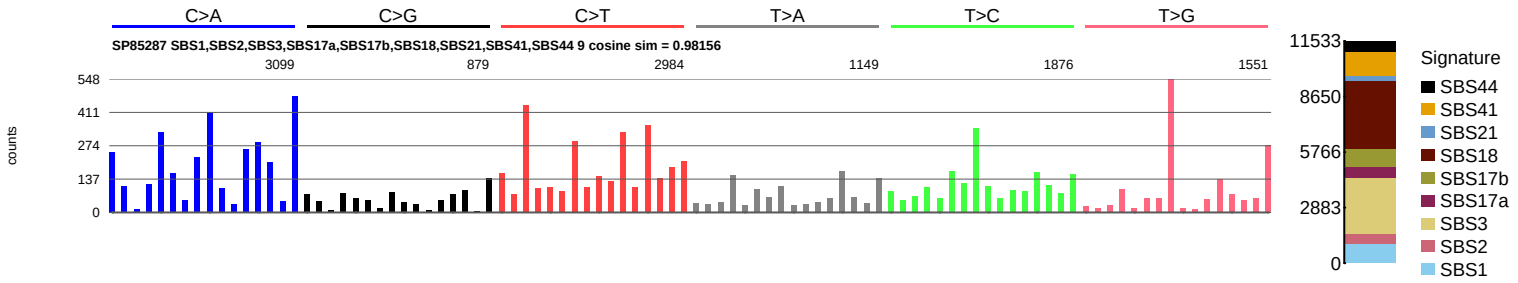
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SP85287 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98156
3099
879
2984
1149
1876
1551
counts
548
411
274
137
0
11533
8650
5766
2883
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SBS21
SBS18
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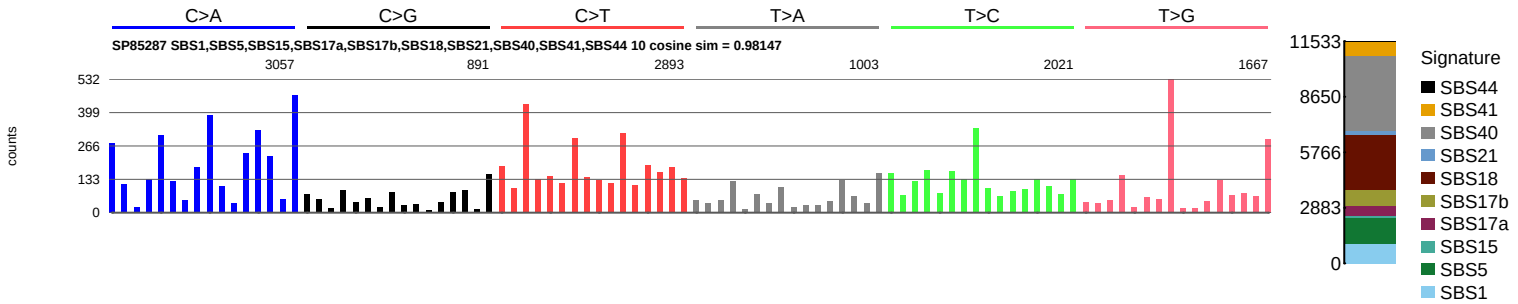
SBS3
SBS2
SBS1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SP85287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98147
3057
891
2893
1003
2021
1667
counts
532
399
266
133
0
11533
8650
5766
2883
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SBS40
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SBS18
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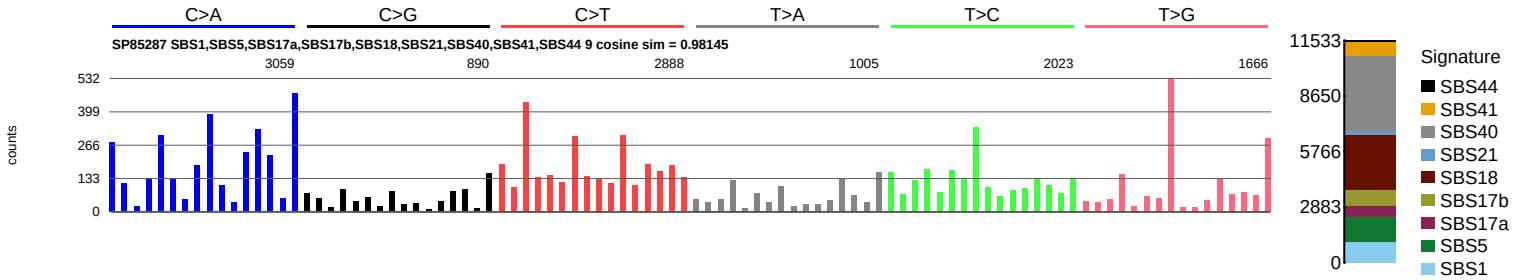
SBS5
SBS1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T>G
SP85287 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98145
3059
890
2888
1005
2023
1666
counts
532
399
266
133
0
11533
8650
5766
2883
0
Signature
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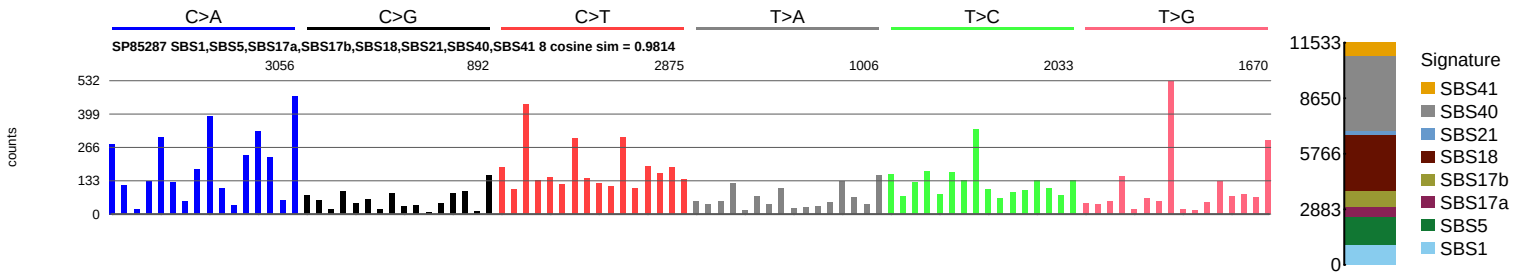
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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C>G
C>T
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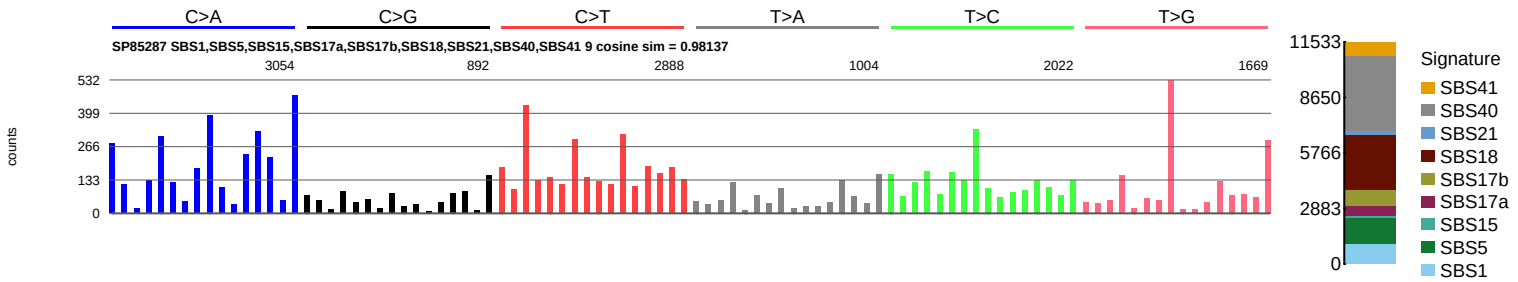
T>C
T>G
SP85287 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.9814
3056
892
2875
1006
2033
1670
counts
532
399
266
133
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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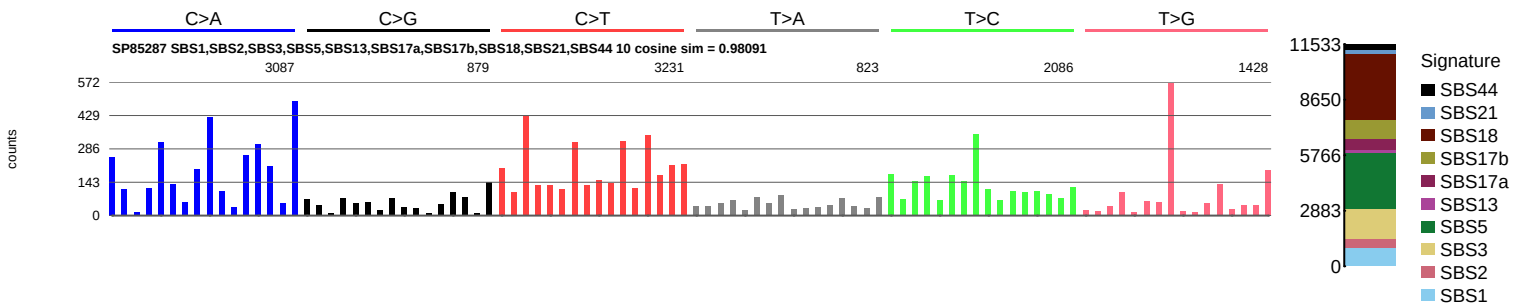
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98137
3054
892
2888
1004
2022
1669
counts
532
399
266
133
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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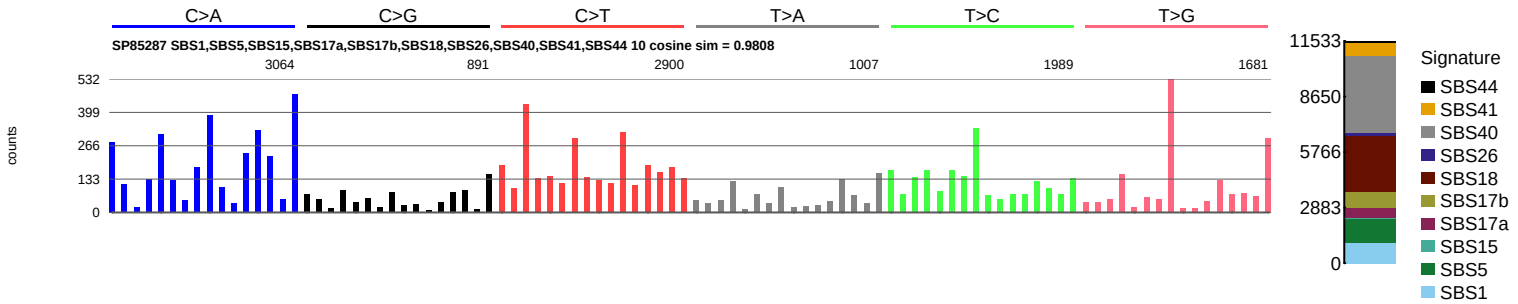
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98091
3087
879
3231
823
2086
1428
counts
572
429
286
143
0
11533
8650
5766
2883
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
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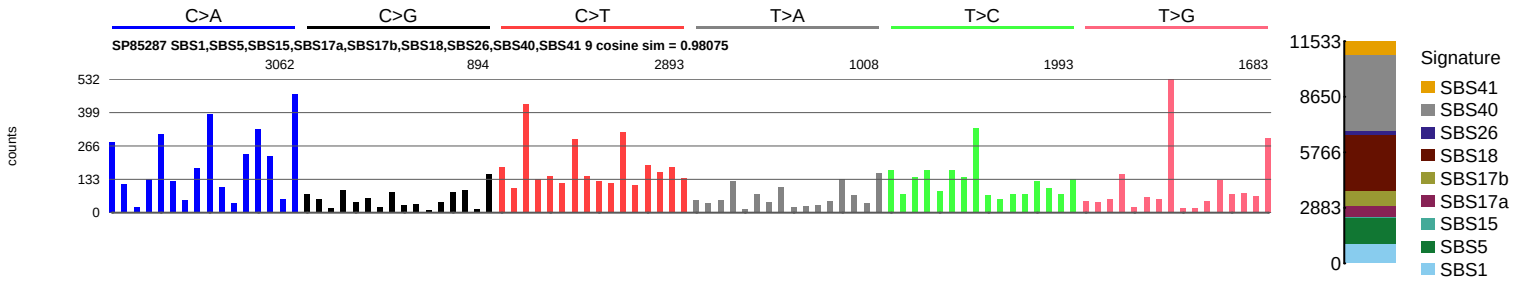
T>C
T>G
SP85287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9808
3064
891
2900
1007
1989
1681
counts
532
399
266
133
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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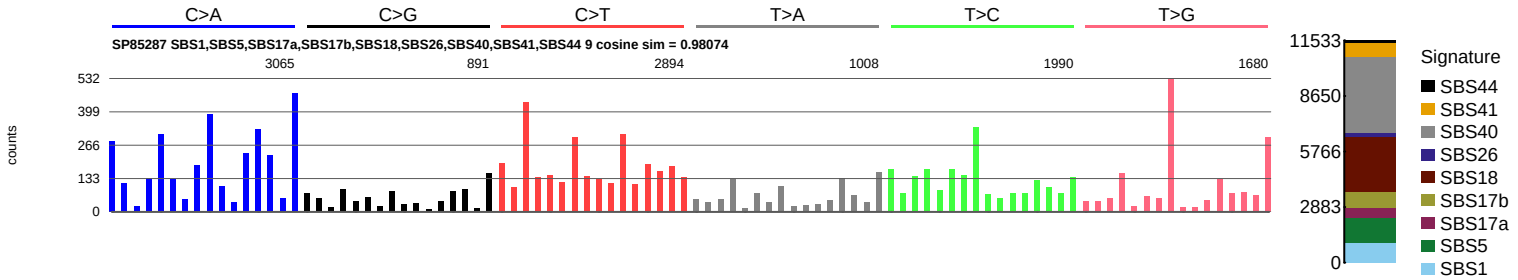
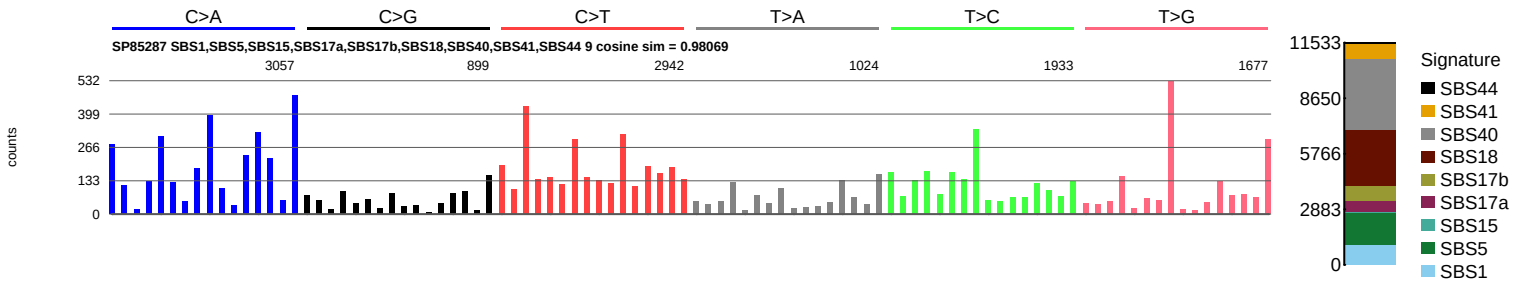
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98075
3062
894
2893
1008
1993
1683
counts
532
399
266
133
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98074
3065
891
2894
1008
1990
1680
counts
532
399
266
133
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1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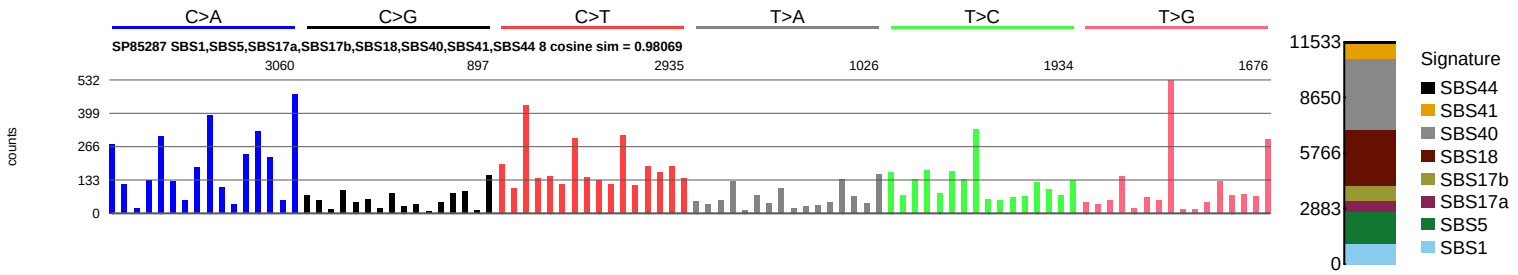
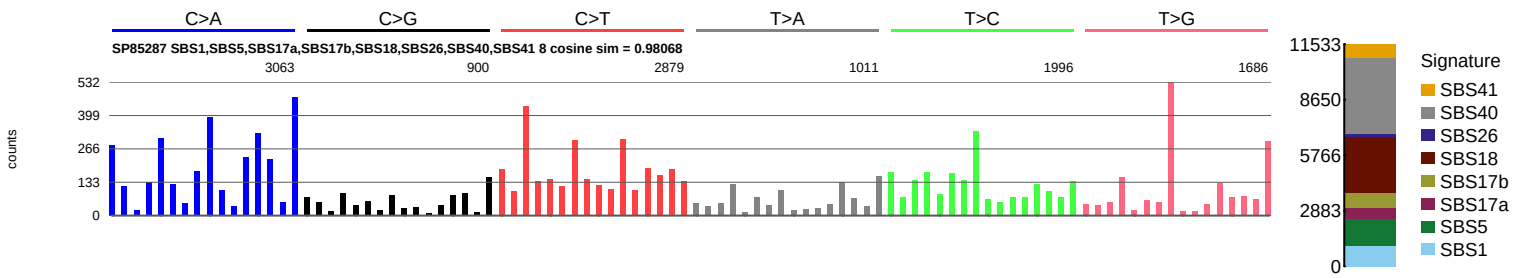
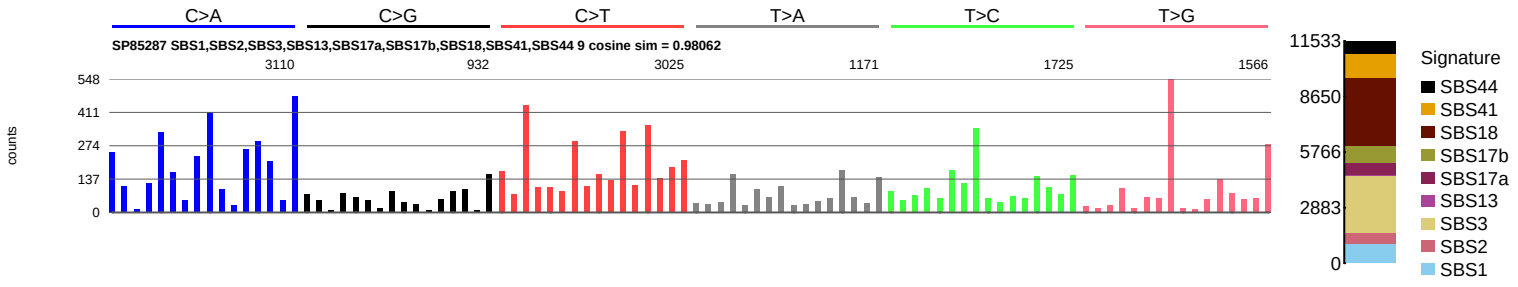
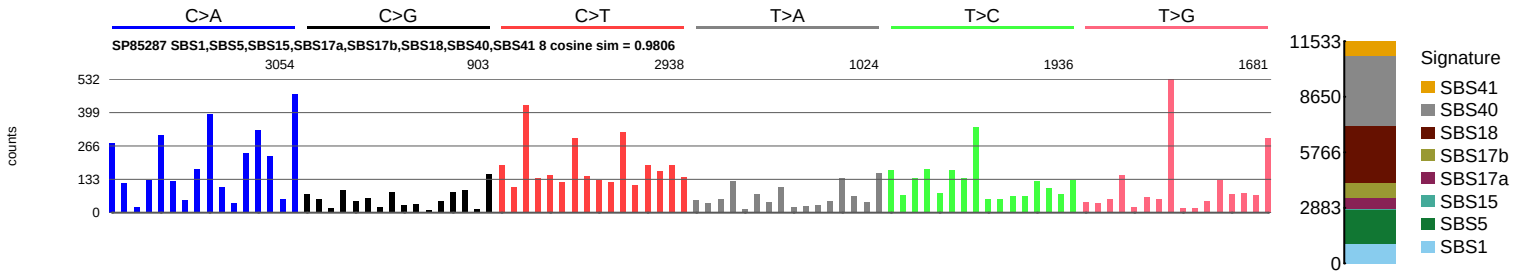
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98069
3057
899
2942
1024
1933
1677
counts
532
399
266
133
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98069
3060
897
2935
1026
1934
1676
counts
532
399
266
133
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98068
3063
900
2879
1011
1996
1686
counts
532
399
266
133
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98062
3110
932
3025
1171
1725
1566
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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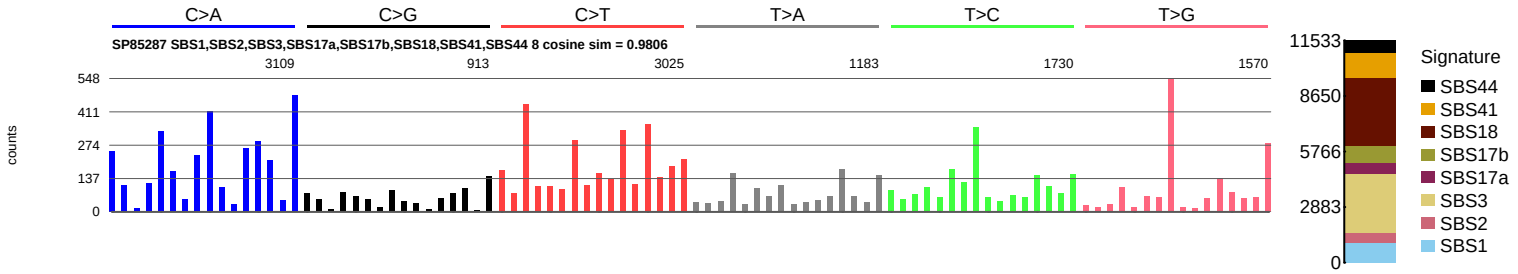
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.9806
3054
903
2938
1024
1936
1681
counts
532
399
266
133
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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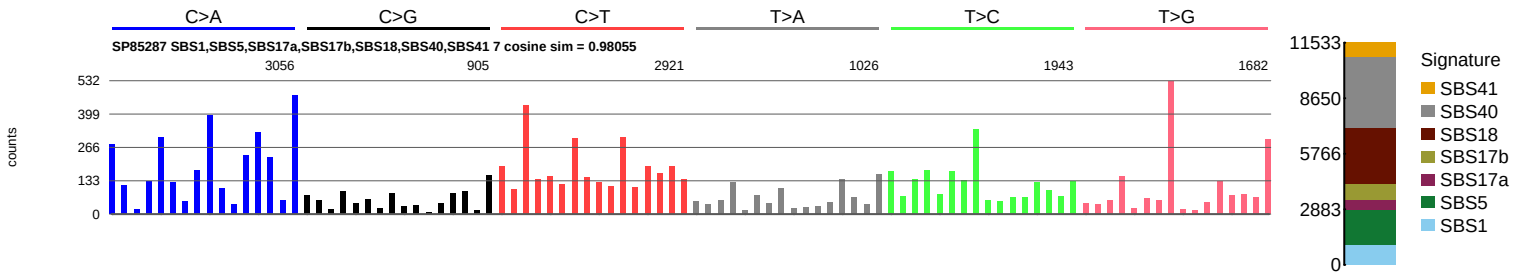
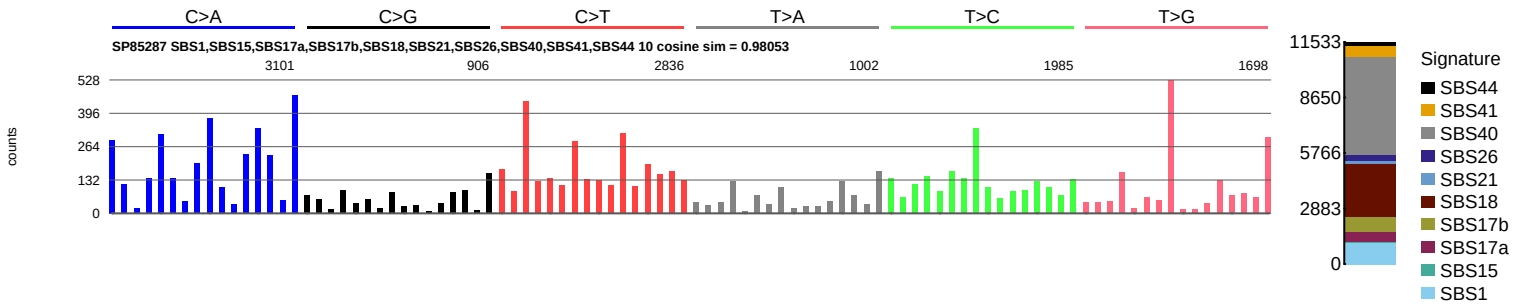
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.9806
3109
913
3025
1183
1730
1570
counts
548
411
274
137
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98055
3056
905
2921
1026
1943
1682
counts
532
399
266
133
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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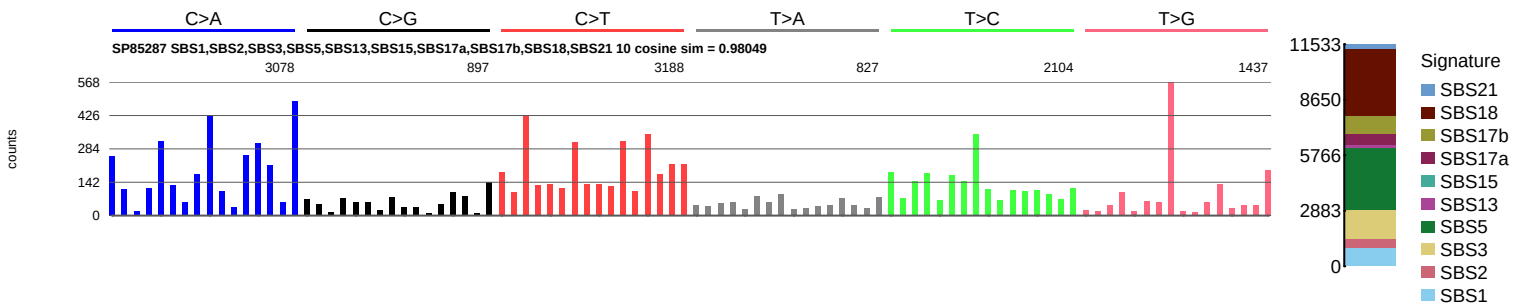
T>A
T>C
T>G
SP85287 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98053
3101
906
2836
1002
1985
1698
counts
528
396
264
132
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
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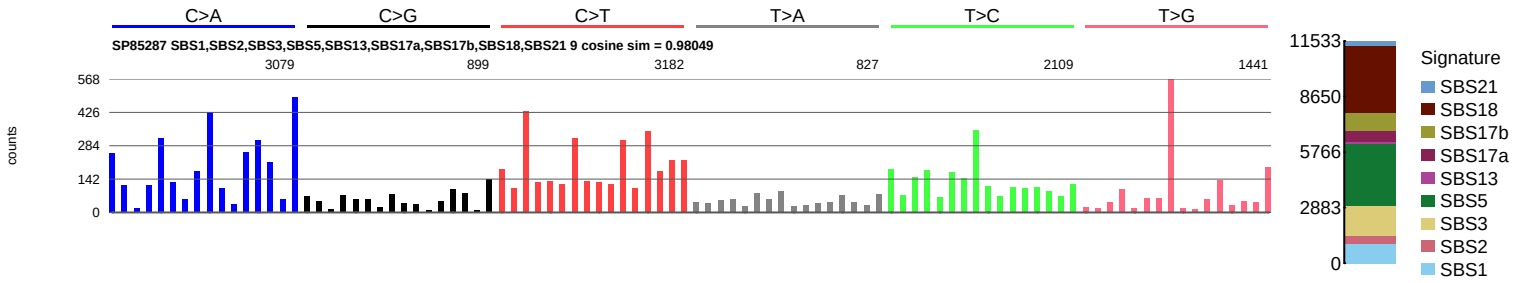
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.98049
3078
897
3188
827
2104
1437
counts
568
426
284
142
0
11533
8650
5766
2883
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98049
3079
899
3182
827
2109
1441
counts
568
426
284
142
0
11533
8650
5766
2883
0
Signature
SBS21
SBS18
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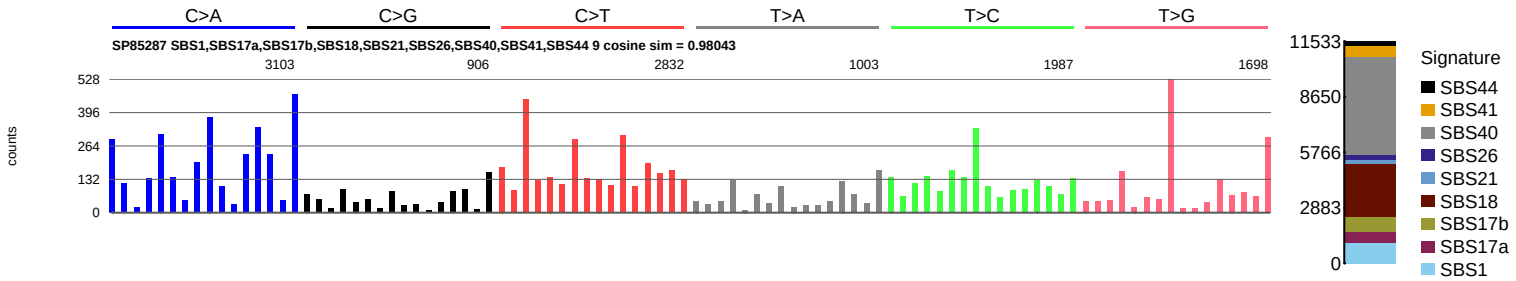
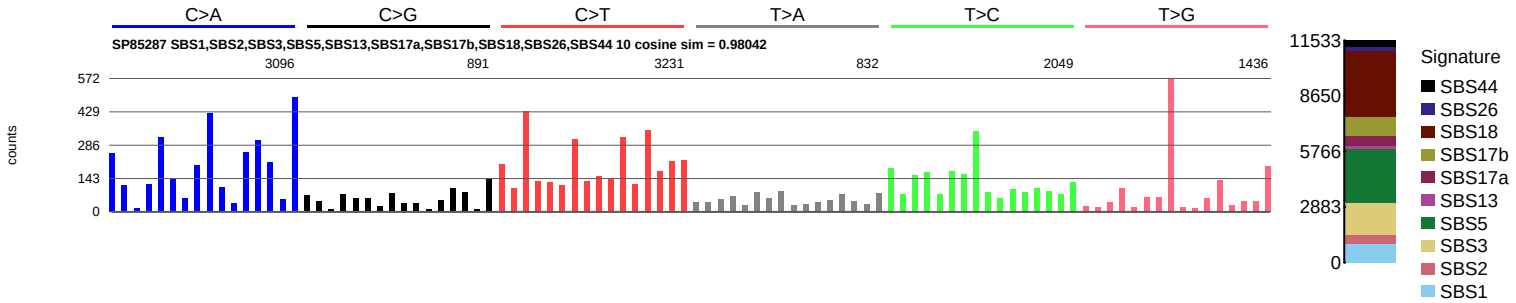
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98043
3103
906
2832
1003
1987
1698
counts
528
396
264
132
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98042
3096
891
3231
832
2049
1436
counts
572
429
286
143
0
11533
8650
5766
2883
0
Signature
SBS44
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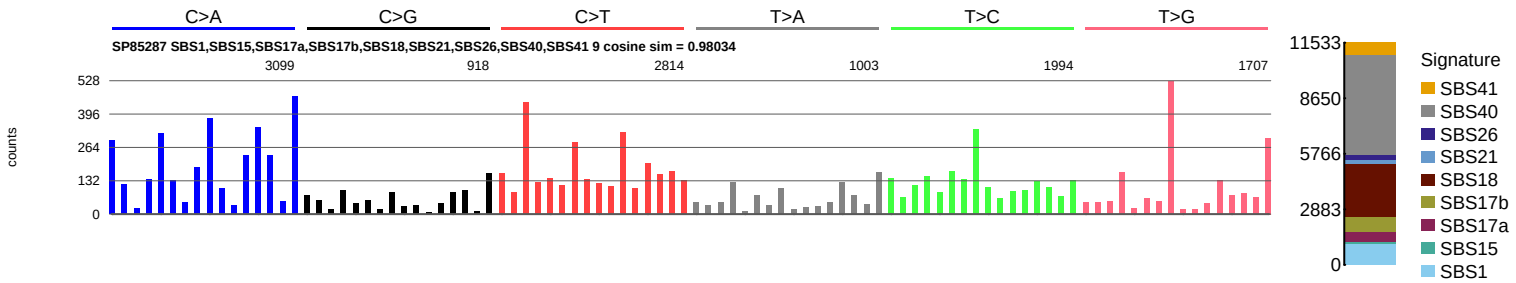
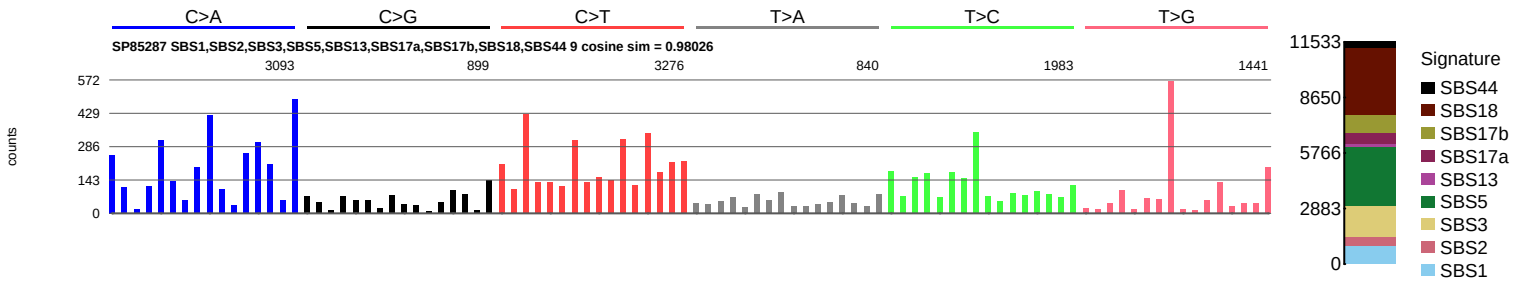
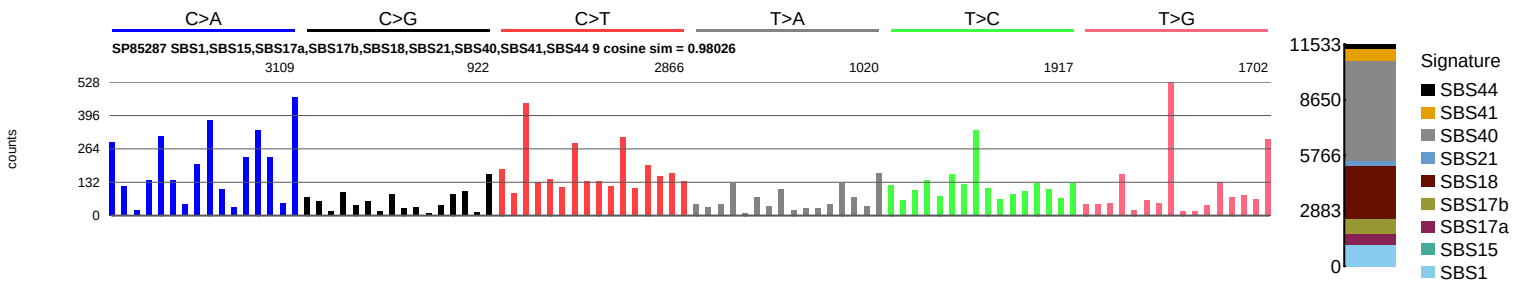
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98034
3099
918
2814
1003
1994
1707
counts
528
396
264
132
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98026
3093
899
3276
840
1983
1441
counts
572
429
286
143
0
11533
8650
5766
2883
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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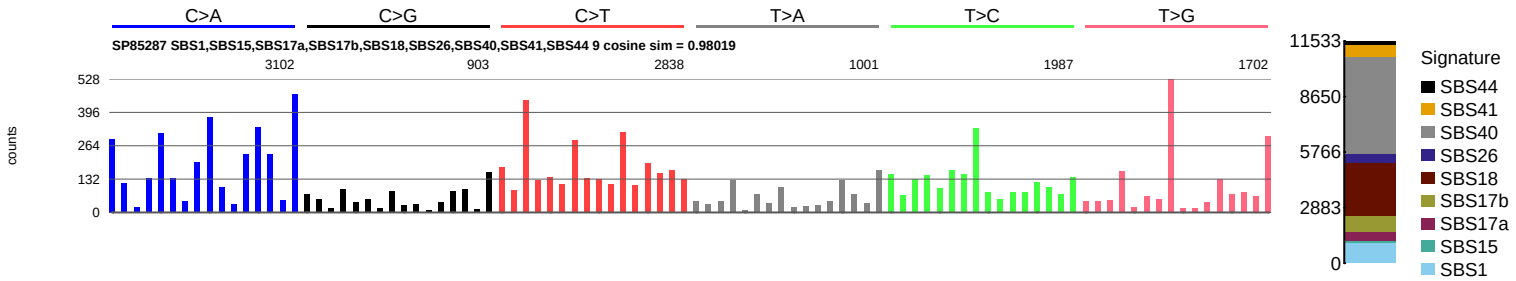
T>C
T>G
SP85287 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98026
3109
922
2866
1020
1917
1702
counts
528
396
264
132
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
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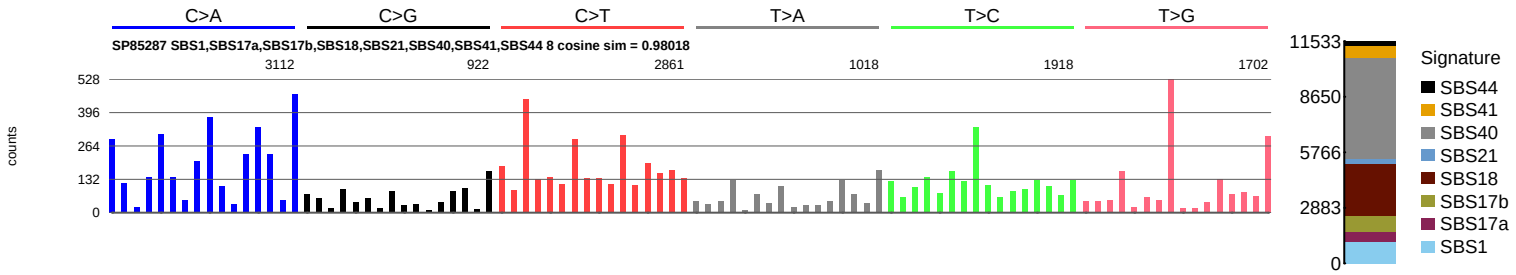
T>G
SP85287 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98019
3102
903
2838
1001
1987
1702
counts
528
396
264
132
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
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T>C
T>G
SP85287 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98018
3112
922
2861
1018
1918
1702
counts
528
396
264
132
0
11533
8650
5766
2883
0
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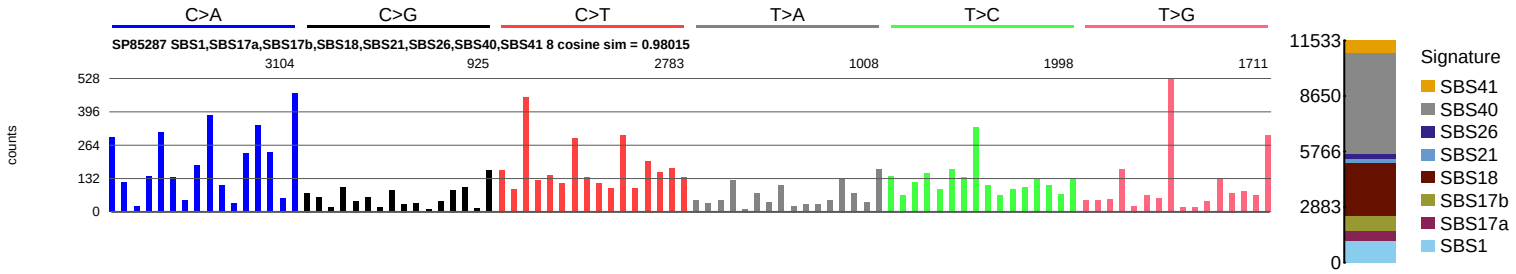
SBS17a
SBS1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C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98015
3104
925
2783
1008
1998
1711
counts
528
396
264
132
0
11533
8650
5766
2883
0
Signature
SBS41
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SBS21
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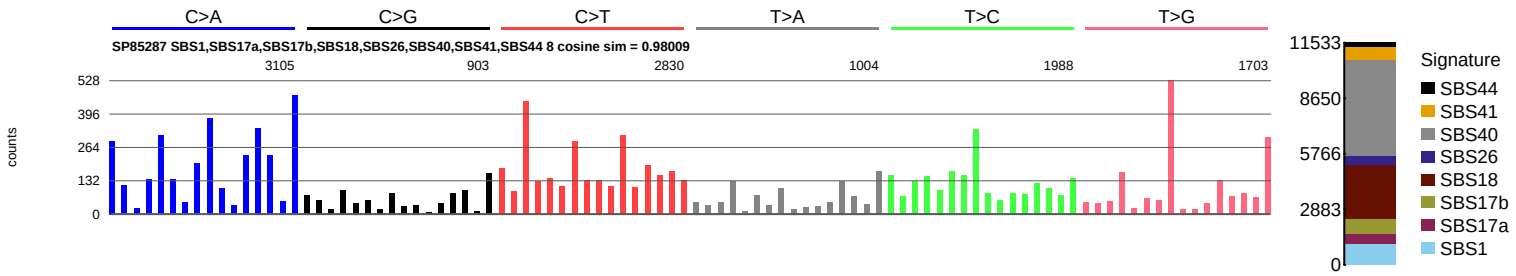
SBS17b
SBS17a
SBS1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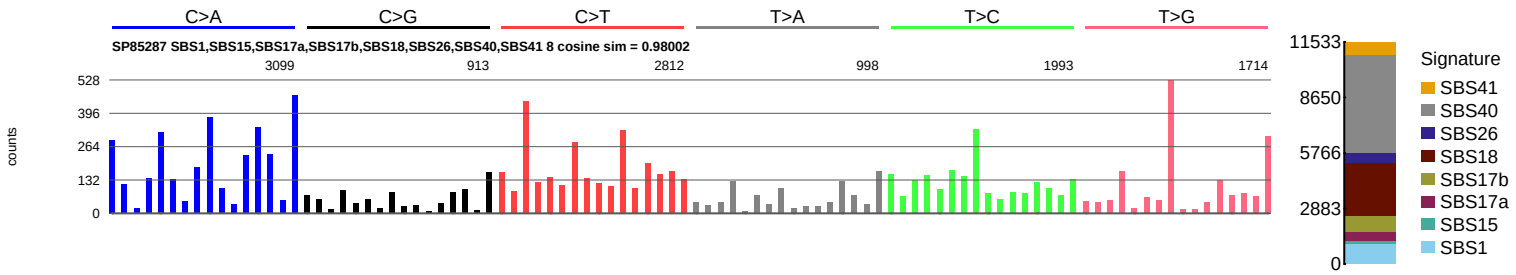
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98009
3105
903
2830
1004
1988
1703
counts
528
396
264
132
0
11533
8650
5766
2883
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85287 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98002
3099
913
2812
998
1993
1714
counts
528
396
264
132
0
11533
8650
5766
2883
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1